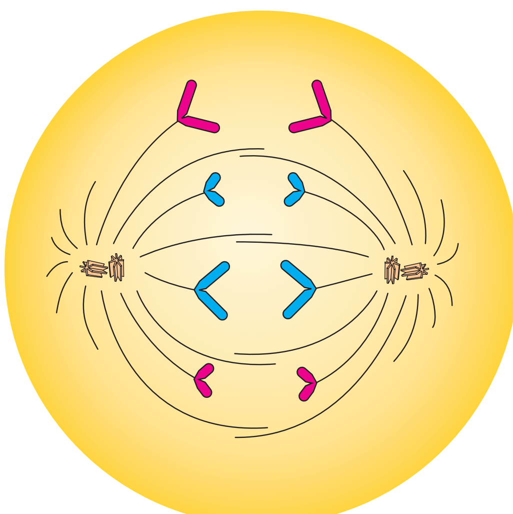
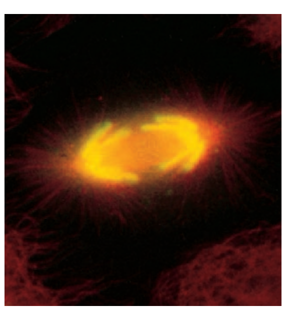
the center of the cell and seems to be involved in the formation of a constricting ring in animal cells or in the growth of a cell plate in plant cells. The cell has now entered the G_1 phase of the cell cycle (see fig. 3.6). Figure 3.18 summarizes mitosis.

The Significance of Mitosis

Cytokinesis and mitosis result in two daughter cells, each with genetic material identical to that of the parent cell. This exact distribution of the genetic material, in the form of chromosomes, to the daughter cells, ensures the stability of cells and the inheritance of traits from one cell generation to the next. Cells have evolved complex life functions; with mitosis, they will produce offspring cells with these same capabilities. With this stability assured, single-celled organisms could thrive and multicellular organisms could evolve.



(a)



(b)

Figure 3.14 (*a*) The mitotic spindle during anaphase. In this cell, $2n = 4$. Maternal chromosomes are *red;* paternal chromosomes are *blue*. (*b*) Fluorescent microscope image of a cultured cell in anaphase. Microtubles are *red;* chromosomes (DNA) are stained *yellow*. ([*b*] John M. Murray, Department of Anatomy, University of Pennsylvania. Cover of *BioTechniques,* volume 7, number 3, March 1989. Reproduced with permission.)

MEIOSIS

Gamete formation presents an entirely new engineering problem to be solved. To form gametes in animals (and, for the most part, to form spores in plants), a diploid organism with two copies of each chromosome must form daughter cells that have only one copy of each chromosome. In other words, the genetic material must be reduced by half so that when gametes recombine to form zygotes, the original number of chromosomes is restored, not doubled.

If we were to try to engineer this task, we would first need to be able to recognize homologous chromosomes. We could then push one member of each pair into one daughter cell and the other into the other daughter cell. If we were unable to recognize homologues, we would not be able to ensure that each daughter cell received one and only one member of each pair. The cell solves this problem by pairing up homologous chromosomes during an extended prophase. The spindle apparatus then separates members of the homologous chromosome pairs, just as it separates sister chromatids during mitosis. But there is one complication. As in mitosis, cells entering meiosis have already replicated their chromosomes. Therefore, two nuclear divisions without an intervening chromosome replication are necessary to produce haploid gametes or

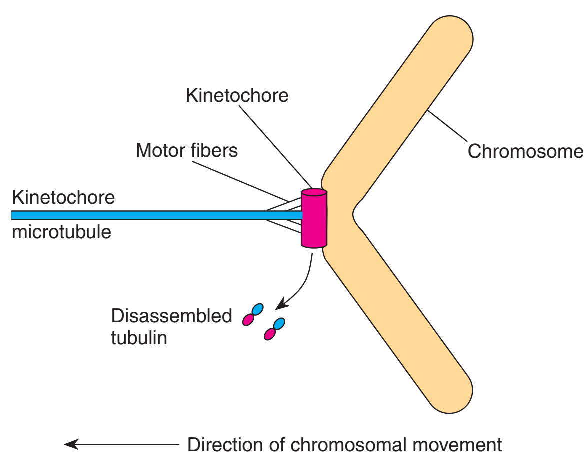


Figure 3.15 The kinetochore acts as a microtubule motor, pulling the chromosome along the kinetochore microtubules toward the pole. One microtubule is shown, although many may be present.

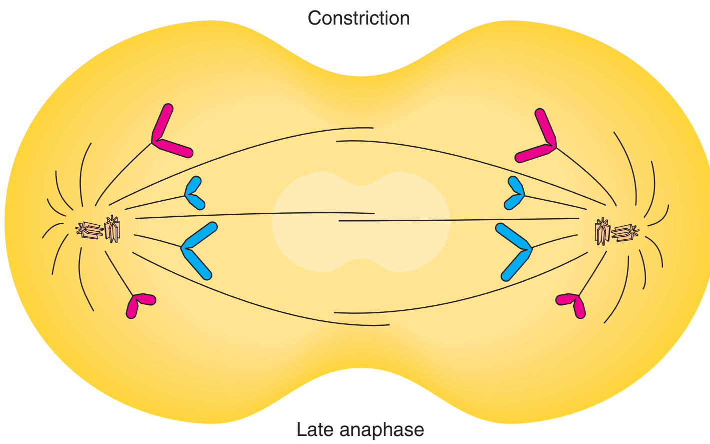


Figure 3.16 Late anaphase of mitosis; $2n = 4$. A constriction begins to form in the middle of the cell (in animals). Maternal chromosomes are *red;* paternal chromosomes are *blue*.

spores. Meiosis is, then, a two-division process that produces four cells from each original parent cell. The two divisions are known as meiosis I and meiosis II.

Unlike mitosis, meiosis occurs only in certain kinds of cells. In animals, meiosis begins in the primary gametocytes; in higher plants, the process takes place only in the spore-mother cells of the sporophyte generation (see fig. 3.1). At the end of this chapter, we review the processes of gamete and spore formation in animals and plants, respectively.

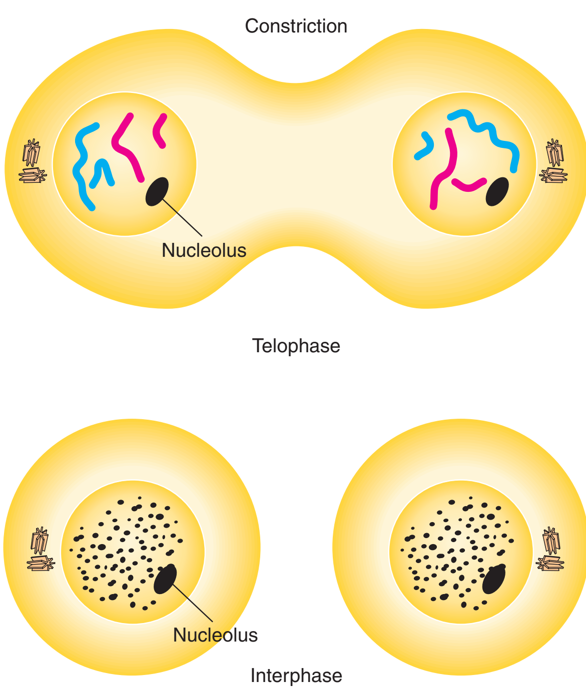


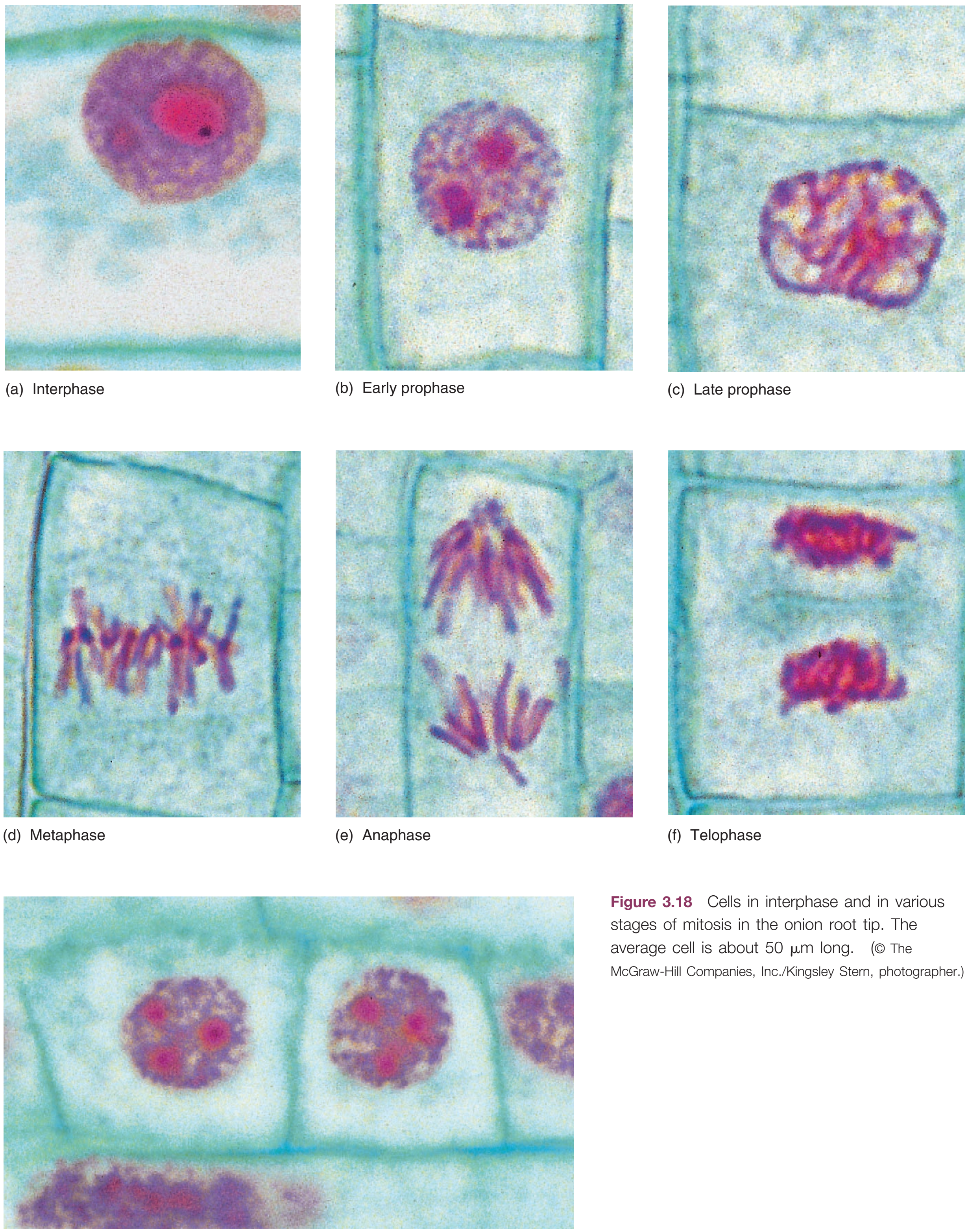
Figure 3.17 Telophase and interphase of mitosis; $2n = 4$. Maternal chromosomes are *red;* paternal chromosomes are *blue*.

Prophase I

Cytogeneticists have divided the prophase of meiosis I into five stages: **leptonema, zygonema, pachynema, diplonema,** and **diakinesis** (Greek: *lepto-,* thin; *zygo-,* yoke-shaped; *pachy-,* thick; *diplo-,* double; *dia-,* across). A cell entering prophase I (leptotene stage) behaves similarly to one entering prophase of mitosis, with the centrosome duplicated and the spindle forming around the intact nucleus. (Note the adjectival forms—*leptotene*—versus the noun forms—*leptonema*—of the stage names.) As the chromosomes coil down in size during leptonema, they are visible as individual threads: sister chromatids are in such close apposition that they are not distinct. The chromosomes are more spread out than they are in mitosis, with dark spheres or bands called chromomeres interspersed.

The tips of the chromosomes are attached to the nuclear membrane in the leptotene stage (fig. 3.19). In the leptotene to zygotene transition, the tips of the chromosomes move until most end up in a limited region near each other. This forms an arrangement called a **bouquet stage.** Presumably, this arrangement helps homologous chromosomes find each other and begin the pairing process without becoming entangled.

The pairing of homologous chromosomes marks the zygotene stage. Initial contact between identical regions of homologous chromosomes leads to a point-for-point pairing along their lengths. This process is referred to as **synapsis.** A proteinaceous complex, referred to as a



(a) Interphase
(b) Early prophase
(c) Late prophase
(d) Metaphase
(e) Anaphase
(f) Telophase
(g) Daughter cells

Figure 3.18 Cells in interphase and in various stages of mitosis in the onion root tip. The average cell is about 50 μm long. (© The McGraw-Hill Companies, Inc./Kingsley Stern, photographer.)

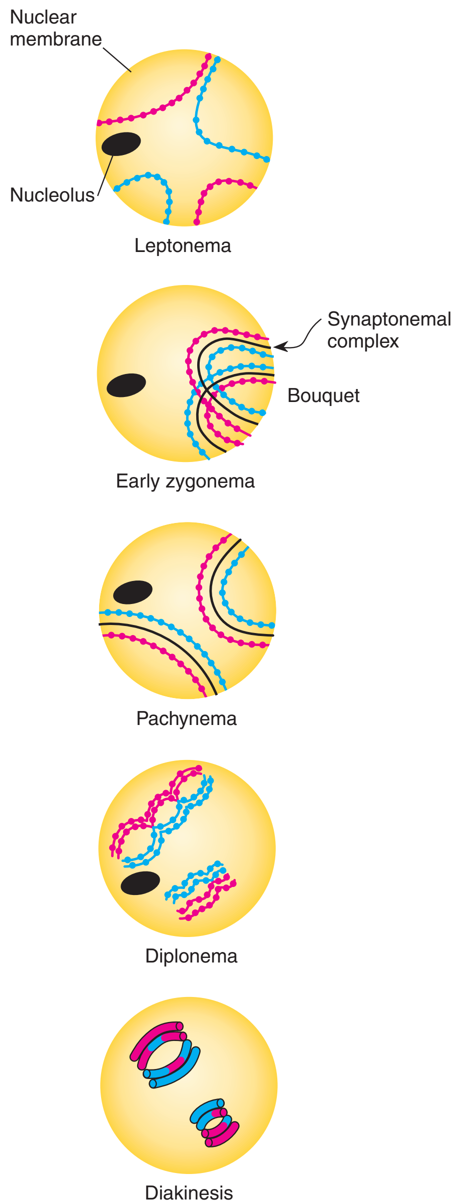


Figure 3.19 Prophase I of meiosis; $2n = 4$ (nuclei shown). Maternal chromosomes are *red;* paternal chromosomes are *blue*. Note that crossing over is evident at diplonema.

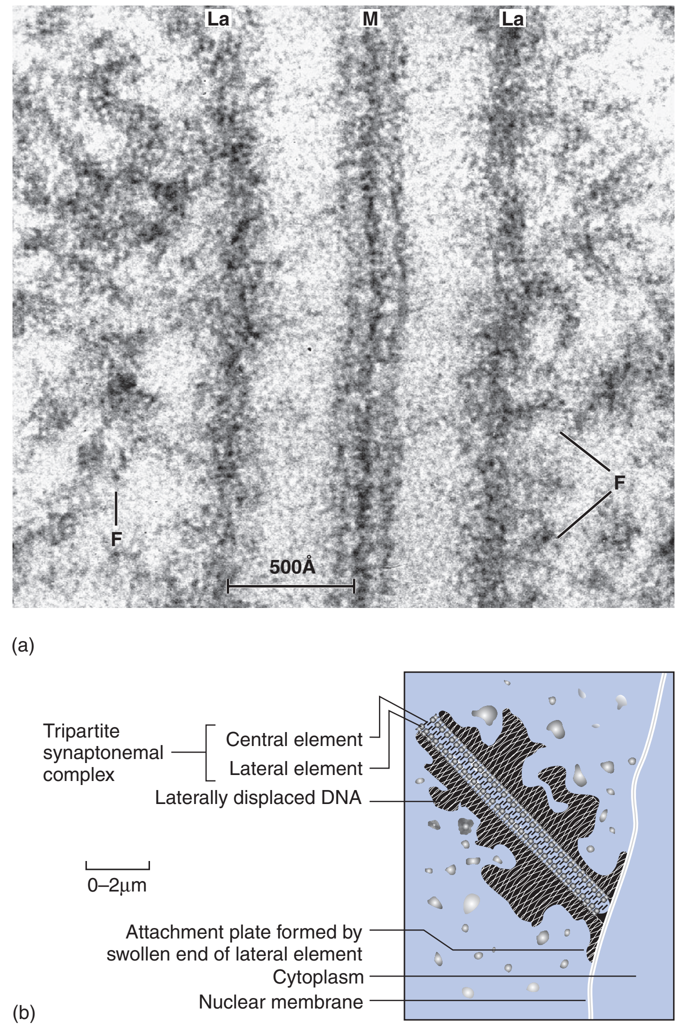


Figure 3.20 The synaptonemal complex. *(a)* In the electron micrograph, *M* is the central element, *La* are lateral elements, and *F* are chromosome fibers. Magnification 400,000×. *(b)* Diagram of the structure. ([*a*] R. Wettstein and J. R. Sotelo, "The molecular architecture of synaptonemal complexes," in E. J. DuPraw, ed., *Advances in Cell and Molecular Biology,* vol. 1 (New York: Academic Press, 1971), p. 118. Reproduced by permission. [*b*] From B. John and K. R. Lewis, *Chromosome Hierarchy*. Copyright © 1975 Oxford University Press, London, England. Reprinted by permission of the Oxford University Press.)

synaptonemal complex (fig. 3.20), appears between the homologous chromosomes and mediates synapsis in an unknown way. At this point, the chromosome figures are referred to as **bivalents,** one bivalent per homologous pair. The synapsis of all chromosomes marks the end of zygonema.

The chromosomes now continue to shorten and thicken, giving pachynema its name. During the entire prophase, **crossing over** takes place. When two chromatids come to lie in close proximity, enzymes can break both chromatid strands and reattach them differently (fig. 3.21). Thus, although genes have a fixed position on a chromosome, alleles that started out attached to a paternal centromere can end up attached to a maternal cen-

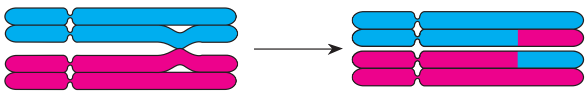


Figure 3.21 Crossing over in a tetrad during prophase of meiosis I. Maternal chromosomes are *red;* paternal chromosomes are *blue.* Note the exchange of chromosome pieces after the process is completed.

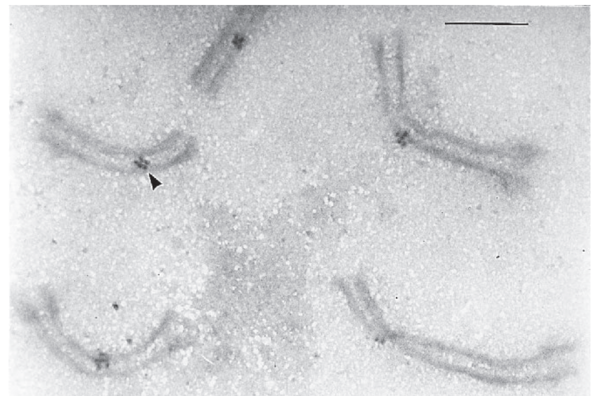
tromere. Crossing over can greatly increase the genetic variability in gametes by associating alleles that were not previously joined. (We examine the molecular mechanism of this process in chapter 12.) Before crossing over takes place, densely staining nodules are visible, first in zygonema and lasting through pachynema. These are called **recombination nodules** (fig. 3.22*a*); they are correlated with crossing over and presumably represent the enzymatic machinery present on the chromosomes.

As the chromosomes shorten and thicken further in diplonema, each chromosome can be seen to be made of two sister chromatids. Now the chromosome figures are referred to as **tetrads** because each is made up of four chromatids (see fig. 3.19). At about this time, the synaptonemal complex disintegrates in all but the areas of the **chiasmata** (singular: chiasma), the X-shaped configurations marking the places of crossing over (fig. 3.22*b*). Virtually all tetrads exhibit chiasmata; in cases in which no crossing over occurs, the tetrads tend to fall apart and segregate randomly. Thus, crossing over not only increases genetic diversity but also ensures the proper separation of homologous chromosomes. A meiosis-specific form of cohesin keeps sister chromatids together.

During the diplotene stage, chromosomes can again uncondense and become active. This is especially obvious in amphibians and birds, which produce a great amount of cytoplasmic nutrient for the future zygote. Recondensation of the chromosomes takes place at the end of diplonema. This stage can be very long; in human females, it begins in the fetus and does not complete until the egg is shed during ovulation, sometimes more than fifty years later. As prophase I moves into diakinesis, the chromosomes become very condensed (see fig. 3.19).

Metaphase I and Anaphase I

Metaphase I is marked by the breakdown of the nuclear membrane and the attachment of kinetochore microtubules to the tetrads. Unlike in mitosis, in which sister chromatids are pulled apart because each sister kinetochore is attached to a different pole, both sister kinetochores become attached to spindle microtubules coming from the same pole in metaphase I (fig. 3.23). During anaphase I, cohesin breaks down every place but at the centromeres, allowing sister chromatids to be pulled to



(a)

(b)

Figure 3.22 (*a*) Recombination nodules (*arrowhead*) in spermatocytes of the pigeon, *Columba livia.* (Bar = 1 μm.) (*b*) A tetrad from the grasshopper, *Chorthippus parallelus,* at diplonema with five chiasmata. ([a] From M. I. Pigozzi and A. J. Solari, "Recombination Nodule Mapping and Chiasma Distribution in Spermatocytes of the Pigeon, *Columba livia,*" in *Genome,* 42: 308–314, 1999. Reprinted by permission. [b] Courtesy of Bernard John.)

the same pole: homologous chromosomes are separated (fig. 3.24). This meiotic division is therefore called a **reductional division** because it reduces the number of chromosomes to half the diploid number in each daughter cell. For every tetrad there is now one chromosome in the form of a chromatid pair, known as a **dyad** or **monovalent,** at each pole of the cell. The initial objective of meiosis, separating homologues into different daughter cells, is accomplished. However, since each dyad consists of two sister chromatids, a second, mitosislike division is required to reduce each chromosome to a single chromatid.

Telophase I and Prophase II

Depending on the organism, telophase I may or may not be greatly shortened in time. In some organisms, all the expected stages take place; chromosomes enter an interphase configuration as cytokinesis takes place. However, no chromosome duplication (DNA replication) occurs during this abbreviated interphase, termed **interkinesis.** Next, in these organisms, prophase II begins and

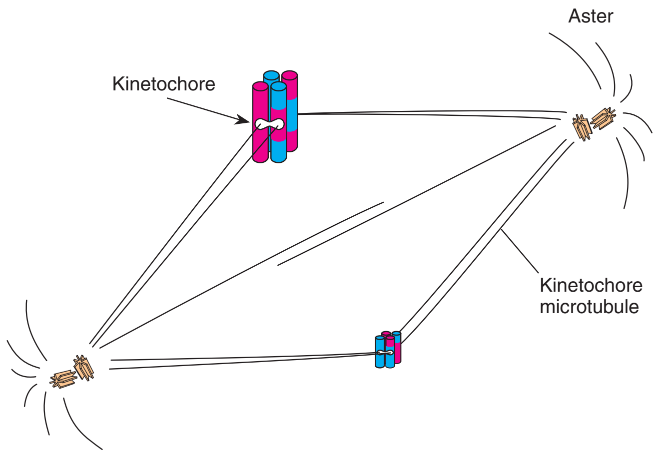
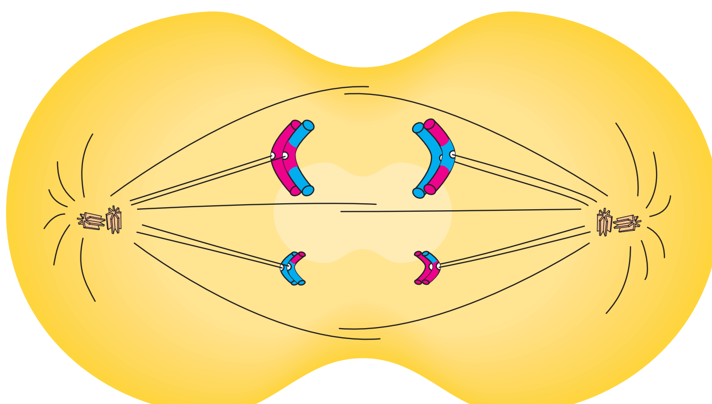


Figure 3.23 Metaphase of meiosis I; $2n = 4$. Maternal chromosomes are *red;* paternal chromosomes are *blue.* Sister kinetochores (effectively single, merged kinetochores) are attached to microtubules from the same pole.



Anaphase I

Figure 3.24 Anaphase of meiosis I; $2n = 4$. Maternal chromosomes are *red;* paternal chromosomes are *blue.* Homologous chromosomes separate and move to opposite poles.

meiosis II proceeds. In still other organisms, the late anaphase I chromosomes go almost directly into metaphase II, virtually skipping telophase I, interphase, and prophase II.

Meiosis II

Meiosis II is basically a mitotic division in which the chromatids of each chromosome are pulled to opposite poles. For each original cell entering meiosis I, four cells emerge at telophase II. Meiosis II is an **equational division;** although it reduces the amount of genetic material per cell by half, it does not further reduce the chromosome number per cell (fig. 3.25). (Sometimes it is simpler to concentrate on the behavior of centromeres during meiosis than on the chromosomes and chromatids. Meiosis I separates maternal from paternal centromeres, and meiosis II separates sister centromeres.) Figure 3.26 summarizes meiosis in corn (*Zea mays*).

In terms of chromosomes, meiosis begins with a diploid cell and produces four haploid cells. In terms of DNA, the process is a bit more complex but has the same

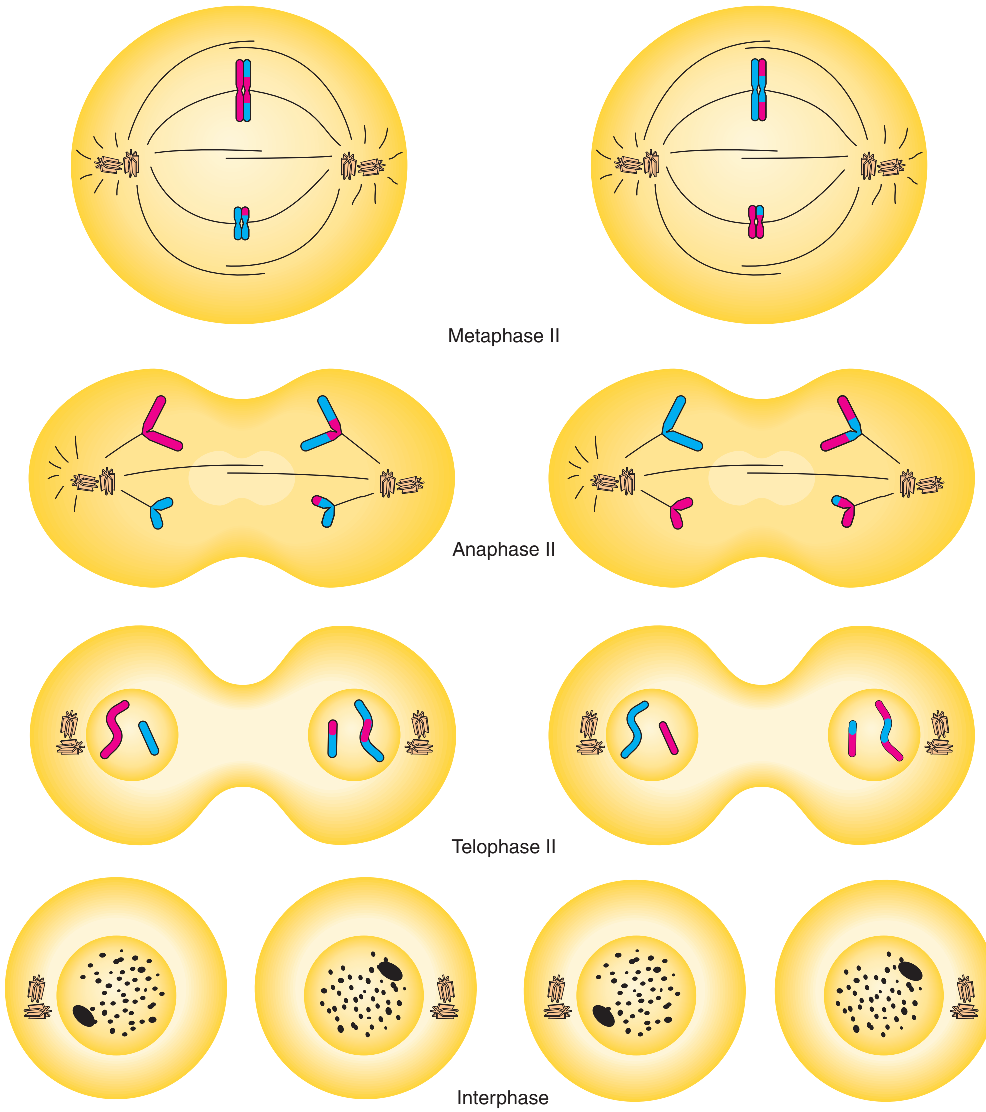


Figure 3.25 Meiosis II; $2n = 4$. Maternal chromosomes are *red;* paternal chromosomes are *blue.*

result. Let us call the quantity of DNA in a gamete "C." A diploid cell before S phase has 2C DNA, and the same cell after S phase, but before mitosis, has 4C DNA. Mitosis reduces the quantity of DNA to 2C. A cell entering meiosis also has 4C DNA. After the first meiotic division, each daughter cell has 2C DNA, and after the second meiotic division, each daughter cell has C DNA, the quantity appropriate for a gamete.

The Significance of Meiosis

Meiosis is significant for several reasons. First, it reduces the diploid number of chromosomes so that each of four daughter cells has one complete haploid chromosome set. Second, because of the randomness of the process of chromosomal separation, a very large number of different chromosomal combinations can form in the gametes. For example, in human beings, if each gamete could get either the maternal or paternal chromosome, and we have twenty-three chromosomal pairs, 2^{23} or 8,388,608 different combinations can occur. Third, because of crossing over, even more allelic combinations are possible. The process of creating new arrangements, either by crossing over or by independent segregation of homologous pairs of chromosomes, is called **recombination.** Assuming one hundred thousand genes in a human

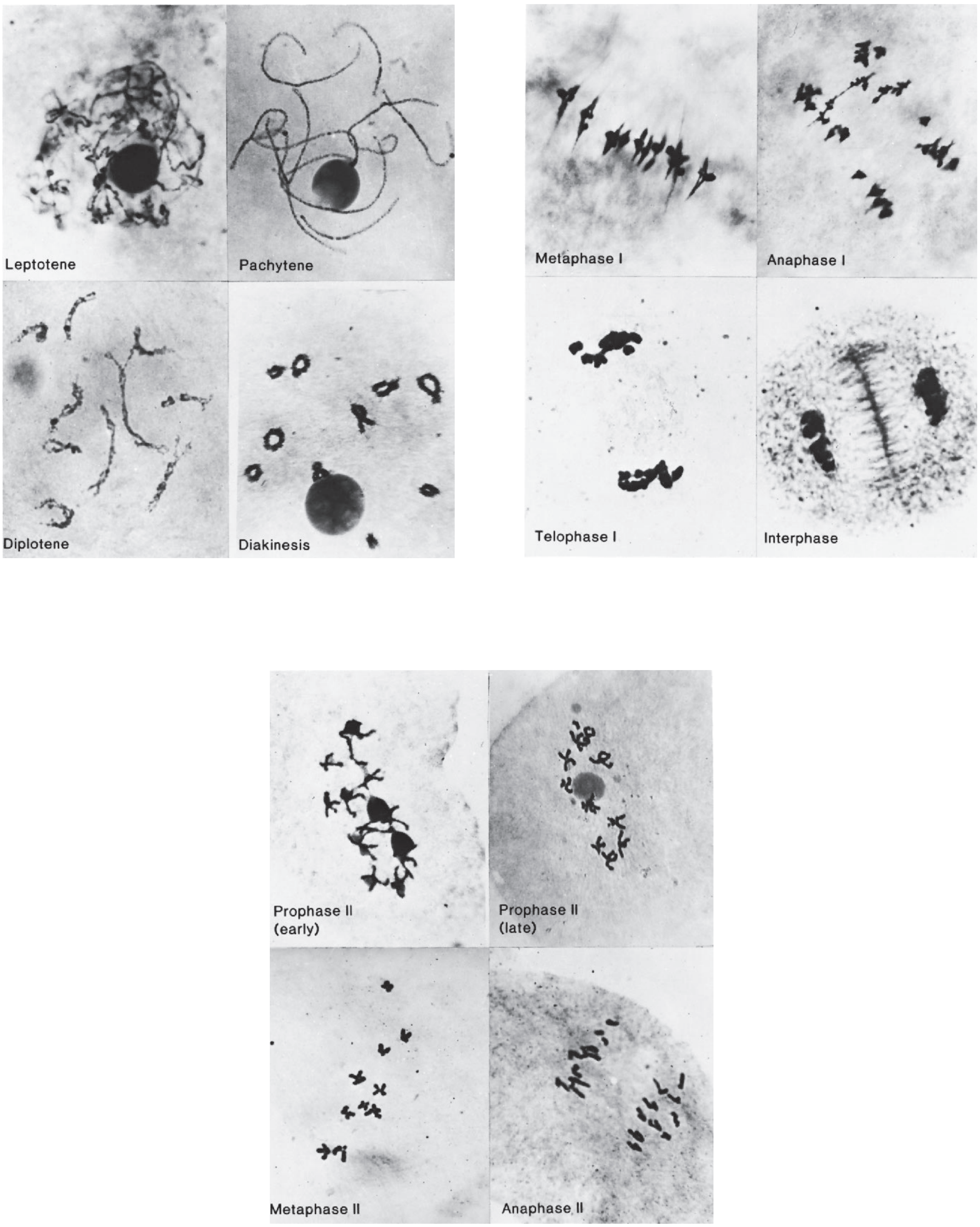


Figure 3.26 Meiosis in corn (*Zea mays*). (Courtesy of Dr. M. M. Rhoades. "Meiosis in maize," *Journal of Heredity*, 41: 59–67, 1950. Reproduced by permission.)

being with two alleles each, $2^{100,000}$ different gametes could potentially arise by meiosis.

The behavior of any tetrad follows the pattern of Mendel's rule of segregation. At spore or gamete formation (meiosis), the diploid number of chromosomes is halved; each gamete receives only one chromosome from a homologous pair. This process, of course, explains Mendel's rule of segregation. Chromosomal behavior at meiosis also explains independent assortment (fig. 3.27). In anaphase I, the direction of separation is independent in different tetrads. Whereas one pole may get the maternal centromere from chromosomal pair number 1, it could get either the maternal or the paternal centromere from chromosomal pair number 2, and so on (see fig. 3.27). Alleles of one gene segregate independently of alleles of other genes. Very shortly after the rediscovery of Mendel's principles in 1900, geneticists were quick to realize this.

MEIOSIS IN ANIMALS

In male animals, each meiosis produces four equal-sized **sperm cells** in a process called **spermatogenesis** (fig. 3.28). In vertebrates, a cell type in the testes known as a **spermatogonium** produces **primary spermatocytes,** as well as additional spermatogonia, by mitosis.

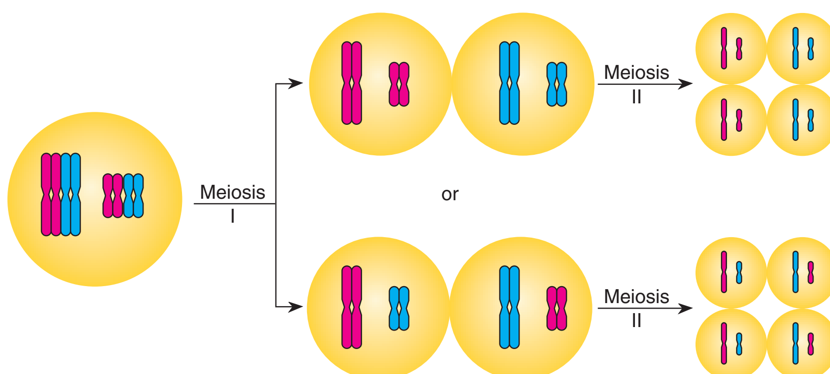


Figure 3.27 Relationship of meiosis to the rule of independent assortment. Maternal (*red*) and paternal (*blue*) chromosomes separate independently in different tetrads.

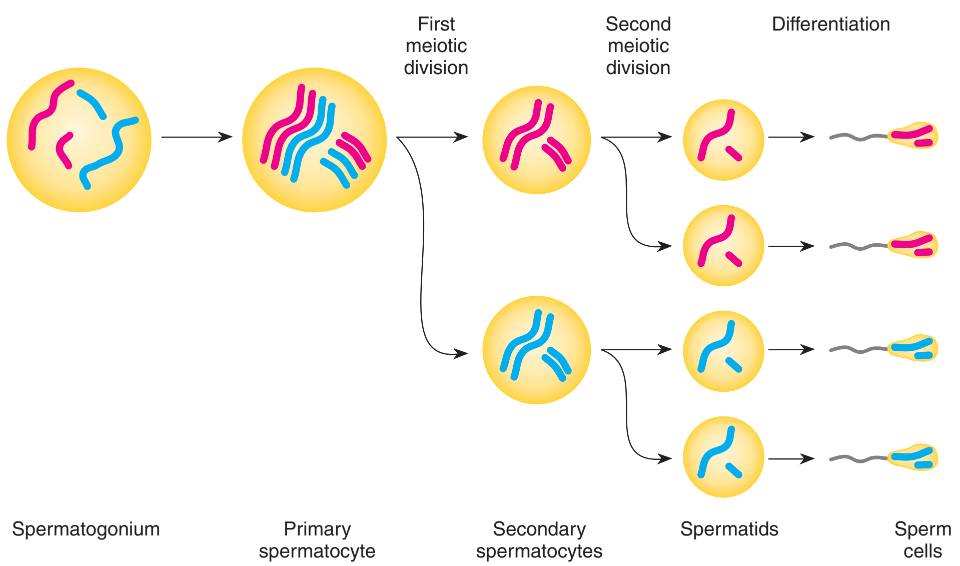


Figure 3.28 Spermatogenesis; $2n = 4$. Maternal chromosomes are *red;* paternal chromosomes are *blue.*

The primary spermatocytes then undergo meiosis. After the first meiotic division, these cells are known as **secondary spermatocytes;** after the second meiotic division, they are known as **spermatids.** The spermatids mature into spermatozoa by a process called **spermiogenesis**—with four sperm cells resulting from each primary spermatocyte. In human beings and other vertebrates without a specific mating season, the process of spermatogenesis is continuous throughout adult life. A normal human male may produce several hundred million sperm cells per day.

During embryonic development in human females, cells in the ovary, known as **oogonia,** proliferate by numerous mitotic divisions to form **primary oocytes.** About one million form per ovary. These begin the first meiotic division and then stop before the birth of the female in a prolonged diplonema, called the **dictyotene** stage. A primary oocyte does not resume meiosis until the female is past puberty, when, under hormonal control, ovulation takes place. This process usually occurs for only one oocyte per month during the female's reproductive life span (from about twelve to fifty years of age). Meiosis only then proceeds in the ovulated oocyte. In the female, the two cells formed by meiosis I are of unequal size. One, termed the **secondary oocyte,** contains almost all the nutrient-rich cytoplasm; the other, a **polar body,** receives very little cytoplasm. The second meiotic division in the larger cell yields another polar body and an **ovum.** The first polar body may or may not divide to form two other polar bodies, which ultimately disintegrate. Thus, **oogenesis** produces cells of unequal size—an ovum and two or three polar bodies (fig. 3.29). Cells of unequal size are produced because the oocyte nucleus and meiotic spindle reside very close to the surface of this large cell.

LIFE CYCLES

For eukaryotes, the basic pattern of the life cycle alternates between a diploid and a haploid state (see fig. 3.1). With the exception of the life cycles of bacteria and viruses, all life cycles are modifications of this general pattern. Bacteria, including blue-green algae, have a single circular chromosome; with exceptions described later, they are always in the haploid state. They divide by replicating their DNA and having the two copies separate into two daughter cells by simple cell division (see chapter 7). Viruses, on the border of being called alive, insert their genetic material into the cells of other organisms, and then manufacture new copies of themselves (see chapter 7).

Most animals are diploids that form gametes by meiosis, then restore the diploid number by fertilization. Exceptions, however, are numerous. For example, in the bees, wasps, and ants (hymenoptera), males are haploid and produce gametes by mitosis; females are diploid. Some fishes exist by **parthenogenesis,** in which the offspring come from unfertilized eggs that do not undergo meiosis. And, in some copepods, the sexual and parthenogenetic stages of their life cycles alternate.

The general pattern of the life cycle of plants alternates between two distinct generations, each of which, depending on the species, may exist independently. In lower plants, the haploid generation predominates, whereas in higher plants, the diploid generation is dominant. In flowering plants **(angiosperms),** the plant you see is the diploid **sporophyte** (see fig. 3.1). It is referred to as a sporophyte because, through meiosis, it will give rise to spores. The spores germinate into the alternate generation, the haploid **gametophyte,** which produces gametes by mitosis. Fertilization then produces the next generation

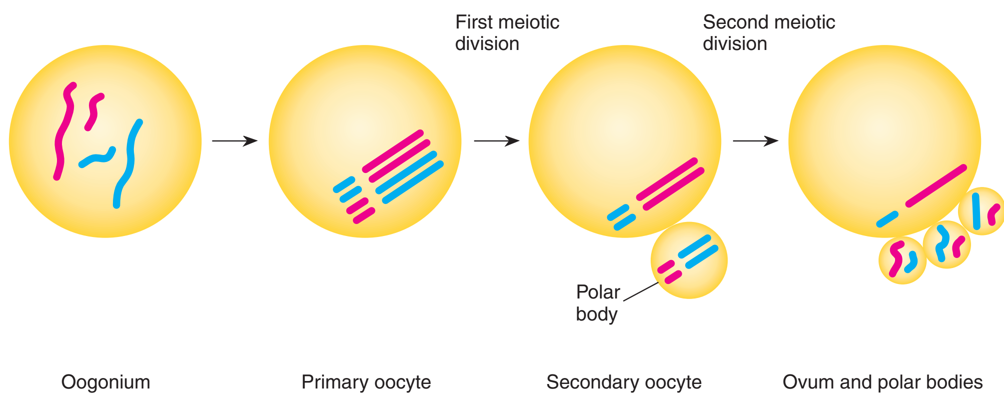


Figure 3.29 Oogenesis; $2n = 4$. Maternal chromosomes are *red;* paternal chromosomes are *blue.*

of diploid sporophytes. In lower plants, the gametophyte has an independent existence; in angiosperms, this generation is radically reduced. For example, in corn (fig. 3.30), an angiosperm, the mature corn plant is the sporophyte. The male flowers produce microspores by meiosis. After mitosis, three cells exist in each spore, a structure that we call a **pollen grain,** the male gametophyte. In female flowers, meiosis produces megaspores. Mitosis within a megaspore produces an embryo sac of seven cells with eight nuclei. This is the female gametophyte. A sperm cell fertilizes the egg cell. The two polar nuclei of the embryo sac are fertilized by a second sperm cell, producing triploid ($3n$) nutritive endosperm tissue. The sporophyte grows from the diploid fertilized egg.

Many fungi and protista are haploid. Fertilization produces a diploid stage, which almost immediately undergoes meiosis to form haploid cells. These cells, in turn, increase in number by mitosis. We will analyze organisms such as *Neurospora,* the pink bread mold, in more detail later (see chapter 6).

Much of our knowledge of genetics derives from the study of specific organisms with unique properties. Mendel found pea plants useful because he could control matings carefully, their generation time was only a year, he could easily grow them in his garden, and they had the discrete traits that he was seeking. Our interest in human beings is obvious. However, we are members of a very difficult species to study experimentally. We have a long generation time and a small number of offspring from matings that we cannot tailor for research purposes. The fruit fly, ***Drosophila melanogaster,*** is one of the organisms geneticists have studied most extensively. Fruit flies have a short generation time (twelve to fourteen days), which means that many matings can be carried out in a reasonable amount of time. In addition, they do exceptionally well in the laboratory, they have many easily observable mutants, and in several organs they have giant banded chromosomes of great interest to cytogeneticists.

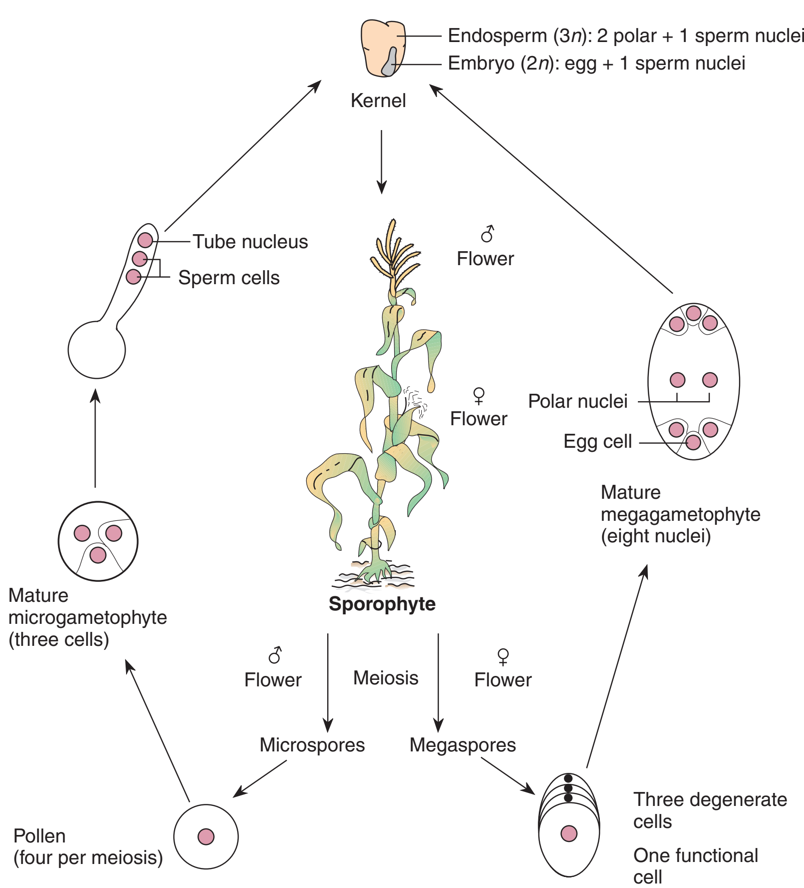


Figure 3.30 Life cycle of the corn plant.

Note that species used in food production tend to be intermediate in their life cycles. That is, many crop plants, such as peas and corn, have only one generation interval per year under normal circumstances. (We use the term *generation interval* here in the broadest sense, as the time it takes to complete an entire life cycle; see also chapter 19.) Crop plants are easier to work with from a genetic standpoint than people, but much more difficult than, say, *Drosophila* or bacteria (table 3.4). Because of their relatively long generation interval, crop plants are limited in their utility for studying basic genetic concepts or applying genetic technology to agriculture.

As you make your way through this book and through other readings on genetics, and as you come across studies involving new organisms, ask yourself the question, What are the properties of this organism that make it ideal for this type of research?

CHROMOSOMAL THEORY OF HEREDITY

In a paper in 1903, cytologist Walter Sutton firmly stated the concepts we have developed here: The behavior of chromosomes during meiosis explains Mendel's principles. Genes, then, must be located on chromosomes. This idea, which several other biologists were also developing at the time, was immediately accepted, ushering in the era of the **chromosomal theory of inheritance.** During this era, intensive effort was devoted to studying the relationships between genes and chromosomes. The major portion of the first section of this book is devoted to classical studies of **linkage** and **mapping.** Linkage deals with the association of genes to each other and to specific chromosomes. Mapping deals with the sequence of genes on a chromosome and the distances between genes on the same chromosome. This is basic information for a study of the structure and function of genes. Here we introduce a new term for the gene. The term **locus** (plural: *loci*), meaning "place" in Latin, refers to the location of a gene on the chromosome.

Table 3.4 Approximate Generation Intervals of Some Organisms of Genetic Interest

Organism	Approximate Generation Interval
Intestinal bacterium (*Escherichia coli*)	20 minutes
Bacterial virus (*lambda*)	1 hour
Pink bread mold (*Neurospora crassa*)	2 weeks
Fruit fly (*Drosophila melanogaster*)	2 weeks
House mouse (*Mus musculus*)	2 months
Corn (*Zea mays*)	6 months
Sheep (*Ovus aries*)	1 year
Cattle (*Bos taurus*)	2 years
Human being (*Homo sapiens*)	14 years

SUMMARY

STUDY OBJECTIVE 1: To observe the morphology of chromosomes 48–50

Chromosomes are made of chromatin and divided by centromeres. Within centromeres are kinetochores, attachment points for spindle fibers. Structure within the chromosomes is evident from bands on chromosomes called chromomeres.

STUDY OBJECTIVE 2: To understand the processes of mitosis and meiosis 50–61

During eukaryotic cell division, the processes of mitosis and meiosis apportion the chromosomes to daughter cells. Both processes are preceded by chromosome replication during the S phase of the cell cycle, which is under genetic control. In mitosis, the two sister chromatids making up each replicated chromosome separate into two daughter cells. Sex cells—gametes in animals and spores in plants—are produced by the two-stage process of meiosis. In meiosis, homologous chromosomes are first separated into two daughter cells, and then the sister chromatids making up each chromosome are distributed to two new daughter cells. We end up with four cells, each with the haploid chromosomal complement. The spindle is the apparatus that separates chromosomes in both mitosis and meiosis.

STUDY OBJECTIVE 3: To analyze the relationships between meiosis and Mendel's rules 61–65

The behavior of chromosomes during meiosis explains Mendel's two principles, segregation and independent assortment.

At the end of this chapter, we define the chromosomal theory of inheritance, the concept that shapes the first section of this book. This theory states that genes are located on chromosomes; their positions and order on the chromosomes can be discovered by mapping techniques described in later chapters.

SOLVED PROBLEMS

PROBLEM 1: What are the differences between chromosomes and chromatids?

Answer: In higher organisms, a chromosome is a linear DNA molecule complexed with protein and, generally, with a centromere somewhere along its length. During the cell cycle, in the S phase, the DNA replicates and each chromosome is duplicated. The duplication is visible in the early stages of mitosis and meiosis when chromosomes shorten. At this point, each duplicated chromosome is made up of two chromatids. The chromatids are called chromosomes when their centromeres are pulled to opposite poles of the spindle and each chromatid becomes independent.

PROBLEM 2: What are the relationships between mitosis and meiosis and Mendel's rules of segregation and independent assortment?

Answer: The process of mitosis does not relate directly to Mendel's rules. The behavior of chromosomes during meiosis, however, explains both segregation and independent assortment. Segregation is explained by the fact that only one chromosome from each homologous pair goes into a gamete; this is also true for the maternal and paternal alleles of a given gene. Independent assortment is explained by the independent behavior of each tetrad at meiosis. That is, the separation of maternal and paternal alleles in one tetrad is independent of the separation of maternal and paternal alleles in any other tetrad.

PROBLEM 3: A hypothetical organism has six chromosomes ($2n = 6$). How many different combinations of maternal and paternal chromosomes can appear in the gametes?

Answer: You could figure this empirically by listing all combinations. For example, let A, B, and C = maternal chromosomes and A′, B′, and C′ = paternal chromosomes. Two combinations in the gametes could be A B C′ and A′ B′ C; obviously, several other combinations exist. It is easier to recall that 2^n = number of combinations, where n = the number of chromosome pairs. In this case, $n = 3$, so we expect $2^3 = 8$ different combinations.

EXERCISES AND PROBLEMS*

CHROMOSOMES

1. What are the major differences between prokaryotes and eukaryotes?
2. What is the difference between a centromere and a kinetochore?
3. What is the difference between sister and nonsister chromatids? Between homologous and nonhomologous chromosomes?
4. In human beings, $2n = 46$. How many chromosomes would you find in a
   a. brain cell?
   b. red blood cell?
   c. polar body?
   d. sperm cell?
   e. secondary oocyte?

   (See also MEIOSIS IN ANIMALS)

MITOSIS

5. You are working with a species with $2n = 6$, in which one pair of chromosomes is telocentric, one pair subtelocentric, and one pair metacentric. The *A, B,* and *C* loci, each segregating a dominant and recessive allele (*A* and *a, B* and *b, C* and *c*), are each located on different chromosome pairs. Draw the stages of mitosis.
6. Identify stages *a–f* in the nuclear division shown in figure 1 (on the next page). Include the process, stage, and diploid number (e.g., meiosis I, prophase, $2n = 10$). Keep in mind that one picture could represent more than one process and stage. Chromosomes are drawn as threads, with circles representing kinetochores. (See also MEIOSIS)
7. When during the cell cycle does chromosome replication take place?
8. A mature human sperm cell has *c* amount of DNA. How much DNA (*c,* 2*c,* 4*c,* etc.) will a somatic cell have if it is in
   a. G_1?
   b. G_2?
   How much DNA will be in a cell at the end of meiosis I?

*Answers to selected exercises and problems are on page A-3.

MEIOSIS

9. Given the same information as in problem 5, diagram one of the possible meioses. How many different gametes can arise, absent crossing over? What variation in gamete genotype is introduced by a crossover between the *A* locus and its centromere?

10. How many bivalents, tetrads, and dyads would you find during meiosis in human beings? in fruit flies? in the other species of table 3.3?

11. Can you devise a method of chromosome partitioning during gamete formation that would not involve synapsis—that is, can you reengineer meiosis without passing through a synapsis stage?

12. What are the differences between a reductional and an equational division? What do these terms refer to?

13. How does the process of meiosis explain Mendel's two rules of inheritance?

14. *Drosophila* has four pairs of chromosomes. Let chromosomes from the male parent be A, B, C, and D, and those from the female parent be A′, B′, C′, and D′. What fraction of the gametes from an AA′ BB′ CC′ DD′ individual will be

a. all of paternal origin?
b. all of maternal origin?
c. half of maternal origin and half of paternal origin?

15. Wheat has $2n = 42$ and rye has $2n = 14$ chromosomes. Explain why a wheat-rye hybrid is usually sterile.

16. The arctic fox has fifty small chromosomes, and the red fox has thirty-eight larger chromosomes. Hybrids of these two species are sterile, but cytological studies during meiosis in these hybrids reveal both paired and unpaired chromosomes.

a. Account for the sterility of the hybrids.
b. How can you explain the paired chromosomes?

17. An organism has six pairs of chromosomes. In the absence of crossing over, how many different chromosomal combinations are possible in the gametes?

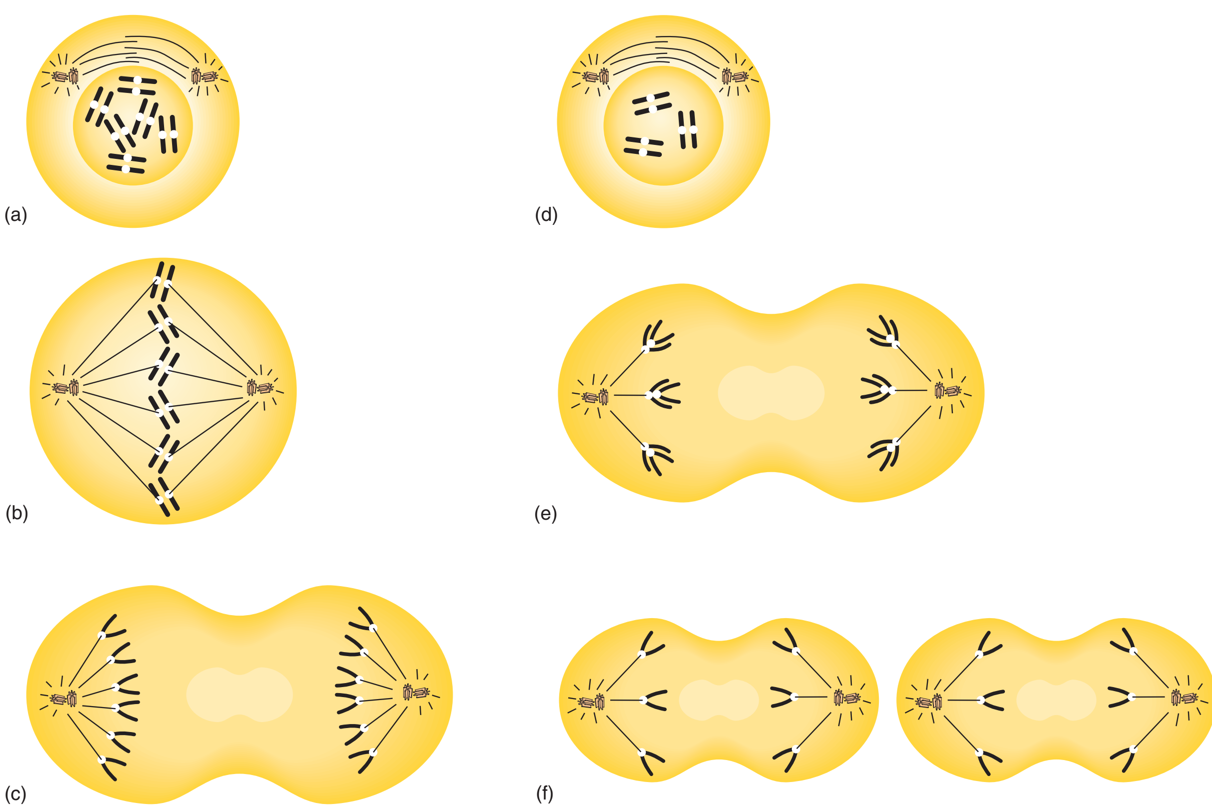


Figure 1 Stages in nuclear division.

MEIOSIS IN ANIMALS

18. How many sperm come from ten primary spermatocytes? How many ova from ten primary oocytes?

19. How do the quantity of genetic material and the ploidy change from stage to stage of spermatogenesis and oogenesis (see figs. 3.28 and 3.29)? (Consider the spermatogonium and the oogonium to be diploid, with the chromosome number arbitrarily set at two.)

20. How many sperm cells will form from
- **a.** fifty primary spermatocytes?
- **b.** fifty secondary spermatocytes?
- **c.** fifty spermatids?

21. In human beings, how many eggs will form from
- **a.** fifty primary oocytes?
- **b.** fifty secondary oocytes?

LIFE CYCLES

22. In corn (see fig. 3.30), the diploid number is twenty. How many chromosomes would you find in a(n)
- **a.** sporophyte leaf cell?
- **b.** embryo cell?
- **c.** endosperm cell?
- **d.** pollen grain?
- **e.** polar nucleus?

23. If a dihybrid corn plant is self-fertilized, what genotypes of the triploid endosperm can result? If you know the endosperm genotype, can you determine the genotype of the embryo?

24. Change the generalized life cycles of figure 3.1 so they describe the life cycles of human beings, peas, and *Neurospora.*

25. If the cytoplasm rather than nuclear genes controlled inheritance, what might be the relationship in phenotype and genotype between an organism and its parents in
- **a.** *Drosophila*?
- **b.** corn?
- **c.** *Neurospora*?

26. A drone (male) honeybee is haploid (arising from unfertilized eggs), and a queen (female) is diploid. Draw a testcross between a dihybrid queen and a drone. How many different kinds of sons and daughters might result from this cross?

27. The plant *Arabidopsis thaliana* has five pairs of chromosomes: AA, BB, CC, DD, and EE. If this plant is self-fertilized, what chromosome complement would be found in a root cell of the offspring?
- **a.** A B C D E
- **b.** AA BB CC DD EE
- **c.** AAA BBB CCC DDD EEE
- **d.** AAAA BBBB CCCC DDDD EEEE

28. In wheat, the haploid number is twenty-one. How many chromosomes would you expect to find in
- **a.** the tube nucleus?
- **b.** a leaf cell?
- **c.** the endosperm?

CHROMOSOMAL THEORY OF HEREDITY

29. A hypothetical organism has two distinct chromosomes ($2n = 4$) and fifty known genes, each with two alleles. If an individual is heterozygous at all known loci, how many gametes can be produced if
- **a.** all genes behave independently?
- **b.** all genes are completely linked?

CRITICAL THINKING QUESTIONS

1. Can meiosis occur in a haploid cell? can mitosis?

2. What is the minimum number of chromosomes that an organism can have? the maximum number?

Suggested Readings for chapter 3 are on page B-1.

4

PROBABILITY AND STATISTICS

STUDY OBJECTIVES

STUDY OUTLINE

An agricultural worker studies variability in plants in a greenhouse. Probability influences the differences among organisms. (© David Joel/Tony Stone Images.)

In an experimental science, such as genetics, scientists make decisions about hypotheses on the basis of data gathered during experiments. Geneticists must therefore have an understanding of probability theory and statistical tests of hypotheses. Probability theory allows geneticists to construct accurate predictions of what to expect from an experiment. Statistical testing of hypotheses, particularly with the chi-square test, allows geneticists to have confidence in their interpretations of experimental data.

PROBABILITY

Part of Gregor Mendel's success was due to his ability to work with simple mathematics. He was capable of turning numbers into ratios and deducing the mechanisms of inheritance from them. Taking numbers that did not exactly fit a ratio and rounding them off to fit lay at the heart of Mendel's deductive powers. The underlying rules that make the act of "rounding to a ratio" reasonable are the rules of probability.

In the **scientific method,** scientists make predictions, perform experiments, and gather data that they then compare with their original predictions (see chapter 1). However, even if the bases for the predictions are correct, the data almost never exactly fit the predicted outcome. The problem is that we live in a world permeated by random, or **stochastic,** events. A bright new penny when flipped in the air twice in a row will not always give one head and one tail. In fact, that penny, if flipped one hundred times, could conceivably give one hundred heads. In a stochastic world, we can guess how often a coin should land heads up, but we cannot know for certain what the next toss will bring. We can guess how often a pea should be yellow from a given cross, but we cannot know with certainty what the next pod will contain. Thus, we need **probability theory** to tell us what to expect from data. This chapter closes with some thoughts on statistics, a branch of mathematics that helps us with criteria for supporting or rejecting our hypotheses.

Types of Probabilities

The **probability** (P) that an event will occur is the number of favorable cases (a) divided by the total number of possible cases (n):

$$P = a/n$$

The probability can be determined either by observation (empirical) or by the nature of the event (theoretical). For example, we observe that about one child in ten thousand is born with phenylketonuria. Therefore, the probability that the next child born will have phenylketonuria is 1/10,000. The odds based on the geometry of an event are, for example, like the familiar toss of dice. A die (singular of dice) has six faces. When that die is tossed, there is no reason one face should land up more often than any other. Thus, the probability of any one of the faces being up (e.g., a four) is one-sixth:

$$P = a/n = 1/6$$

Similarly, the probability of drawing the seven of clubs from a deck of cards is

$$P = 1/52$$

The probability of drawing a spade from a deck of cards is

$$P = 13/52 = 1/4$$

The probability (assuming a 1:1 sex ratio, though the actual ratio is about 1.06 males per female born in the United States) of having a daughter in any given pregnancy is

$$P = 1/2$$

And the probability that an offspring from a self-fertilized dihybrid will show the dominant phenotype is

$$P = 9/16$$

From the probability formula, we can say that an event with certainty has a probability of one, and an event that is an impossibility has a probability of zero. If an event has the probability of P, all the other alternatives combined will have a probability of $Q = 1 - P$; thus $P + Q = 1$. That is, the probability of the completely dominant phenotype in the F_2 of a selfed dihybrid is 9/16. The probability of any other phenotype is 7/16, and when the two are added together, they equal 16/16, or 1.

Combining Probabilities

The basic principle of probability can be stated as follows: If one event has c possible outcomes and a second event has d possible outcomes, then there are cd possible outcomes of the two events. From this principle, we obtain three rules that concern us as geneticists.

To understand these rules of probability requires a few definitions. *Mutually exclusive events* are events in which the occurrence of one possibility excludes the occurrence of the other possibilities. In the throwing of a die, for example, only one face can land up. Thus, if it comes up a four, it precludes the possibility of any of the other faces. Similarly, a blue-eyed daughter is mutually exclusive of a brown-eyed son or any other combination of gender and eye color. *Independent events,* however, are events whose outcomes do not influence one another. For example, if two dice are thrown, the face value of one die is not able to affect the face value of the other; they are thus independent of each other. Similarly, the gender of one child in a family is generally independent of the gender of the children who have come before or might come after. Finally, *unordered events* are events whose probability of outcome does not depend on the order in which the events occur; the probabilities combine both mutual exclusivity and independence. For example, when two dice (one red, one green) are thrown at the same time, we generally do not specify which die has which value; a seven can occur whether the green die is the four or the red die is the four. Similarly, the probability that a family of several children will have two boys and one girl is the same irrespective of their birth order. If the family has two boys and one girl, it does not matter whether the daughter is born first, second, or third. In general, probabilities differ depending on whether order is specified. With these definitions in mind, let us look at three rules of probability that affect genetics.

1. Sum Rule

When events are mutually exclusive, the **sum rule** is used: The probability that one of several mutually exclusive events will occur is the sum of the probabilities of the individual events. This is also known as the *either-or rule.* For example, what is the probability, when we throw a die, of its showing *either* a four *or* a six? According to the sum rule,

$$P = 1/6 + 1/6 = 2/6 = 1/3$$

2. Product Rule

When the occurrence of one event is independent of the occurrence of other events, the **product rule** is used: The probability that two independent events will both occur is the product of their separate probabilities. This is known as the *and rule.* For example, the probability of throwing a die two times and getting a four *and* then a six, in that order, is

$$P = 1/6 \times 1/6 = 1/36$$

3. Binomial Theorem

The **binomial theorem** is used for unordered events: The probability that some arrangement will occur in which the final order is not specified is defined by the binomial theorem. For example, what is the probability when tossing two pennies simultaneously of getting a head and a tail? We will look more closely at how to use the rules of probability to answer this question.

USE OF RULES

There are several ways to calculate the probability just asked for. To put the problem in the form for rule 3 is the quickest method, but this problem can also be solved by using a combination of rules 1 and 2. For each penny, the probability of getting a head (H) *or* a tail (T) is

$$\text{for H: } P = 1/2$$
$$\text{for T: } Q = 1/2$$

Tossing the pennies one at a time, it is possible to get a head *and* a tail in two ways:

first head, then tail (HT)
or
first tail, then head (TH)

Within a sequence (HT or TH), the probabilities apply to independent events. Thus, the probability for any one of the two sequences involves the product rule (rule 2):

$$1/2 \times 1/2 = 1/4 \text{ for HT or TH}$$

The two sequences (HT or TH) are mutually exclusive. Thus, the probability of getting either of the two sequences involves the sum rule (rule 1):

$$1/4 + 1/4 = 1/2$$

Thus, for unordered events, we can obtain the probability by combining rules 1 and 2. The binomial theorem (rule 3) provides the shorthand method.

To use rule 3, we must state the theorem as follows: If the probability of an event (X) is p and an alternative (Y) is q, then the probability in n trials that event X will occur s times and Y will occur t times is

$$P = \frac{n!}{s!t!}p^s q^t$$

In this equation, $s + t = n$, and $p + q = 1$. The symbol !, as in n!, is called **factorial,** as in "n factorial," and is the product of all integers from n down to one. For example, $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. Zero factorial equals one, as does anything to the power of zero ($0! = n^0 = 1$).

Now, what is the probability of tossing two pennies at once and getting one head and one tail? In this case, $n = 2$, s and $t = 1$, and p and $q = 1/2$. Thus,

$$P = \frac{2!}{1!1!}(1/2)^1(1/2)^1 = 2(1/2)^2 = 1/2$$

This is, of course, our original answer. Now on to a few more genetically relevant problems. What is the probability that a family with six children will have five girls and one boy? (We assume that the probability of either a son or a daughter equals 1/2.) Since the order is not specified, we use rule 3:

$$P = \frac{6!}{5!1!}(1/2)^5(1/2)^1 = 6(1/2)^6 = 6/64 = 3/32$$

What would happen if we asked for a specific family order, in which four girls were born, then one boy, and then one girl? This would entail rule 2; for a sequence of six independent events:

$$P = 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 = 1/64$$

When no order is specified, the probability is six times larger than when the order is specified; the reason is that there are six ways of getting five girls and one boy, and the sequence 4-1-1 is only one of them. Rule 3 tells us that there are six ways. These are (letting B stand for boy and G for girl) as follows:

Birth Order

1	2	3	4	5	6
B	G	G	G	G	G
G	B	G	G	G	G
G	G	B	G	G	G
G	G	G	B	G	G
G	G	G	G	B	G
G	G	G	G	G	B

Let us look at yet another problem. If two persons, heterozygous for albinism (a recessive condition), have four children, what is the probability that all four children will be normal? The answer is simply $(3/4)^4$ by rule 2. What is the probability that three will be normal and one albino? If we specify which of the four children will be albino (e.g., the fourth), then the probability is $(3/4)^3(1/4)^1 = 27/256$. If, however, we do not specify order,

$$P = \frac{4!}{3!1!}(3/4)^3(1/4)^1 = 4(3/4)^3(1/4)^1$$
$$= 4(27/256) = 108/256$$

This is precisely four times the ordered probability because the albino child could have been born first, second, third, or last.

The formula for rule 3 is the formula for the terms of the **binomial expansion.** That is, if $(p + q)^n$ is expanded (multiplied out), the formula $(n!/s!t!)p^sq^t$ gives the probability for any one of these terms, given that $p + q = 1$ and that $s + t = n$. Since there are $(n + 1)$ terms in the binomial, the formula gives the probability for the term numbered $(t + 1)$. Two bits of useful information come from recalling that rule 3 is in reality the binomial expansion formula. First, if you have difficulty calculating the term, you can use **Pascal's triangle** to get the coefficients:

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
1   5  10  10   5   1
```

Pascal's triangle is a triangular array made up of coefficients in the binomial expansion. It is calculated by starting any row with a 1, proceeding by adding two adjacent terms from the row above, and then ending with a 1. For example, the next row would be

$$1, (1 + 5), (5 + 10), (10 + 10), (10 + 5), (5 + 1), 1$$
$$\text{or } 1, 6, 15, 20, 15, 6, 1$$

These numbers give us the combinations for any p^sq^t term. That is, in our previous example, $n = 4$; so we use the $(n + 1)$, or fifth, row of Pascal's triangle. (The second number in any row of the triangle gives the power of the expansion, or n. Here, 4 is the second number in the row.) We were interested in the case of one albino child in a family of four children, or p^3q^1, where p is the probability of the normal child (3/4) and q is the probability of an albino child (1/4). Hence, we are interested in the $(t + 1)$—that is, the $(1 + 1)$—or the second term of the fifth row of Pascal's triangle, which will tell us the number of ways of getting a four-child family with one albino child. That number is 4. Thus, using Pascal's triangle, we see that the solution to the problem is

$$4(3/4)^3(1/4)^1 = 108/256$$

This is the same as the answer we obtained the conventional way.

The second advantage from knowing that rule 3 is the binomial expansion formula is that we can now generalize to more than two outcomes. The general form for the **multinomial expansion** is $(p + q + r + \ldots)^n$ and the general formula for the probability is

$$P = \frac{n!}{s!t!u! \ldots} p^sq^tr^u \ldots$$

where $s + t + u + \ldots = n$ and $p + q + r + \ldots = 1$ For example, our albino-carrying heterozygous parents may want an answer to the following question: If we have five children, what is the probability that we will have two normal sons, two normal daughters, and one albino son? (This family will have no albino daughters.) By rule 2, the probability of

a normal son = (3/4)(1/2) = 3/8
a normal daughter = (3/4)(1/2) = 3/8
an albino son = (1/4)(1/2) = 1/8
an albino daughter = (1/4)(1/2) = 1/8

Thus:

$$P = \frac{5!}{2!2!1!0!}(3/8)^2(3/8)^2(1/8)^1(1/8)^0$$
$$= 30(3/8)^4(1/8)^1 = 30(3)^4/(8)^5 = 2{,}430/32{,}768$$
$$= 0.074$$

STATISTICS

In one of Mendel's experiments, F_1 heterozygous pea plants, all tall, were self-fertilized. In the next generation (F_2), he recorded 787 tall offspring and 277 dwarf offspring for a ratio of 2.84:1. Mendel saw this as a 3:1 ratio, which supported his proposed rule of inheritance. In fact, is 787:277 "roundable" to a 3:1 ratio? From a brief discussion of probability, we expect some deviation from an exact 3:1 ratio (798:266), but how much of a deviation is acceptable? Would 786:278 still support Mendel's rule? Would 785:279 support it? Would 709:355 (a 2:1 ratio) or 532:532 (a 1:1 ratio)? Where do we draw the line? It is at this point that the discipline of statistics provides help.

We can never speak with certainty about stochastic events. For example, take Mendel's cross. Although a 3:1 ratio is expected on the basis of Mendel's hypothesis, chance could mean that the data yield a 1:1 ratio (532:532), yet the mechanism could be the one that Mendel suggested. In other words, we could flip an honest coin and get ten heads in a row. Conversely, Mendel could have gotten exactly a 3:1 ratio (798:266) in his F_2 generation, yet his hypothesis of segregation could have been wrong. The point is that any time we deal with probabilistic events there is some chance that the data will lead us to support a bad hypothesis or reject a good one. Statistics quantifies these chances. We cannot say with certainty that a 2.84:1 ratio represents a 3:1 ratio; we can say, however, that we have a certain degree of confidence in the ratio. Statistics helps us ascertain these **confidence limits.**

Statistics is a branch of probability theory that helps the experimental geneticist in three ways. First, part of statistics deals with **experimental design**. A bit of thought applied before an experiment may help the investigator design the experiment in the most efficient way. Although he did not know statistics, Mendel's experimental design was very good. The second way in which statistics is helpful is in summarizing data. Familiar terms such as *mean* and *standard deviation* are part of the body of descriptive statistics that takes large masses of data and reduces them to one or two meaningful values. We examine further some of these terms and concepts in the chapter on quantitative inheritance (chapter 18).

Hypothesis Testing

The third way that statistics is valuable to geneticists is in the **testing of hypotheses:** determining whether to support or reject a hypothesis by comparing the data to the predictions of the hypothesis. This area is the most germane to our current discussion. For example, was the ratio of 787:277 really indicative of a 3:1 ratio? Since we know now that we cannot answer with an absolute yes, how can we decide to what level the data support the predicted 3:1 ratio?

Statisticians would have us proceed as follows. To begin, we need to establish how much variation to expect. We can determine this by calculating a **sampling distribution:** the frequencies with which various possible events could occur in a particular experiment. For example, if we self-fertilized a heterozygous tall plant, we would expect a 3:1 ratio of tall to dwarf plants among the progeny. (The 3:1 ratio is our hypothesis based on the assumption that height is genetically controlled by one locus with two alleles.) If we looked at the first four offspring, what is the probability we would see three tall and one dwarf plant? We can calculate the answer using the formula for the terms of the binomial expansion:

$$P = \frac{4!}{3!1!}(3/4)^3(1/4)^1 = 108/256 = 0.42$$

Similarly, we can calculate the probability of getting all tall (81/256 = 0.32), two tall and two dwarf (54/256 = 0.21), one tall and three dwarf (12/256 = 0.05), and all dwarf (1/256 = 0.004) in this first set of four. Table 4.1 shows this distribution, as well as the distributions for samples of eight and forty progeny. Figure 4.1 shows these distributions in graph form.

As sample sizes increase (from four to eight to forty in fig. 4.1), the sampling distribution takes on the shape of a smooth curve with a peak at the true ratio of 3:1 (75% tall progeny)—that is, there is a high probability of getting very close to the true ratio. However, there is some chance the ratio will be fairly far off, and a very small part of the time our ratio will be very far off. It is important to see that any ratio could arise in a given experiment even though the true ratio is 3:1. At what point do we decide that an experimental result is not indicative of a 3:1 ratio?

Statisticians have agreed on a convention. When all the frequencies are plotted, as in figure 4.1, we can treat the area under the curve as one unit, and we can draw lines to mark 95% of this area (fig. 4.2). Any ratios in-

Table 4.1 **Sampling Distribution for Sample Sizes of Four, Eight, and Forty, Given a 3:1 Ratio of Tall and Dwarf Plants**

n = 4		*n* = 8		*n* = 40	
No. Tall Plants	**Probability***	**No. Tall Plants**	**Probability***	**No. Tall Plants**	**Probability***
4	$\frac{81}{256} = 0.32$	8	0.10	40	0.00001
		7	0.27	39	0.0001
3	$\frac{108}{256} = 0.42$	6	0.31	38	0.0009
		5	0.21		. . .
2	$\frac{54}{256} = 0.21$	4	0.09	30	0.14
		3	0.02		. . .
1	$\frac{12}{256} = 0.05$	2	0.004	2	0.59×10^{-20}
		1	0.0004	1	0.10×10^{-21}
0	$\frac{1}{256} = 0.004$	0	0.00002	0	0.83×10^{-24}

* Probabilities are calculated from the binomial theorem.
probability = $(n!/s!t!)p^s q^t$
where n = number of progeny observed
s = number of progeny that are tall
t = number of progeny that are dwarf
p = probability of a progeny plant being tall (3/4)
q = probability of a progeny plant being dwarf (1/4)

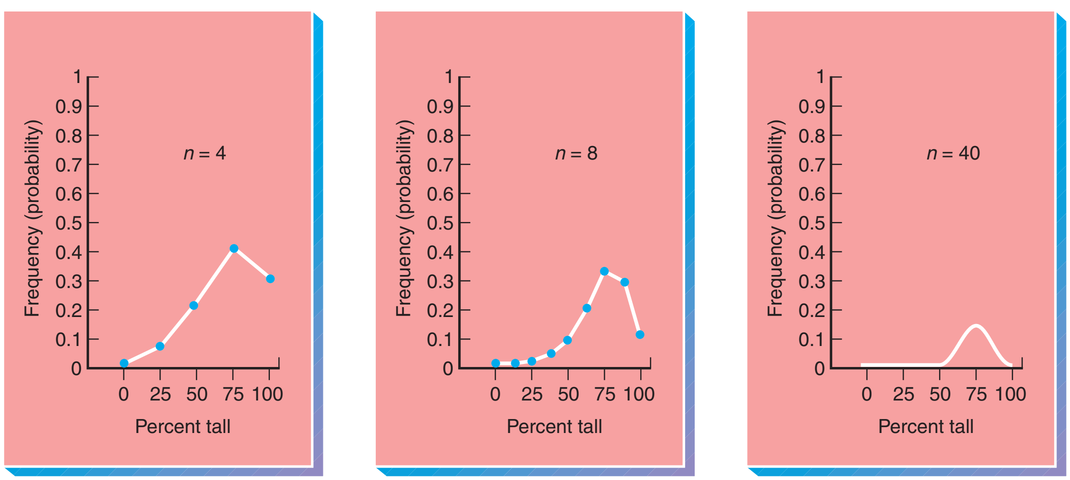


Figure 4.1 Sampling distributions from an experiment with an expected ratio of three tall to one dwarf plant. As the sample size, *n,* gets larger, the distribution curve becomes smoother. These distributions are plotted terms of the binomial expansion (see table 4.1). Note also that as *n* gets larger, the peak of the curve gets lower because as more points (possible ratios) are squeezed in along the x-axis, the probability of producing any one ratio decreases.

cluded within the 95% limits are considered supportive of (failing to reject) the hypothesis of a 3:1 ratio. Any ratio in the remaining 5% area is considered unacceptable. (Other conventions also exist, such as rejection within the outer 10% or 1% limits; we consider these at the end of the chapter.) Thus, it is possible to see whether the experimental data support our hypothesis (in this case, the hypothesis of 3:1). One in twenty times (5%) we will make a **type I error:** We will reject a true hypothesis. (A **type II error** is failing to reject a false hypothesis.)

To determine whether to reject a hypothesis, we must derive a frequency distribution for each type of experiment. Mendel could have used the distribution shown in figure 4.1 for seed coat or seed color, as long as he was expecting a 3:1 ratio and had a similar sample size. What about independent assortment, which predicts a 9:3:3:1 ratio? A geneticist would have to calculate a new sampling distribution based on a 9:3:3:1 ratio and a particular sample size. Statisticians have devised shortcut methods by using standardized probability distributions. Many are in use, such as the *t*-distribution, binomial distribution, and chi-square distribution. Each is useful for particular kinds of data; geneticists usually use the chi-square distribution to test hypotheses regarding breeding data.

Chi-Square

When sample subjects are distributed among discrete categories such as tall and dwarf plants, geneticists frequently use the **chi-square distribution** to evaluate data. The formula for converting categorical experimental data to a chi-square value is

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where χ is the Greek letter chi, O is the observed number for a category, E is the expected number for that category, and Σ means to sum the calculations for all categories.

A chi-square (χ^2) value of 0.60 is calculated in table 4.2 for Mendel's data on the basis of a 3:1 ratio. If Mendel had originally expected a 1:1 ratio, he would have calculated a chi-square of 244.45 (table 4.3). However, these χ^2 values have little meaning in themselves: they are not probabilities. We can convert them to probabilities by determining where the chi-square value falls in relation to the area under the chi-square distribution curve. We usually use a chi-square table that contains probabilities that have already been calculated (table 4.4). Before we can use this table, however, we must define the concept of **degrees of freedom.**

Reexamination of the chi-square formula and tables 4.2 and 4.3 reveals that each category of data contributes to the total chi-square value, because chi-square is a summed value. We therefore expect the chi-square value to increase as the total number of categories increases. That is, the more categories involved, the larger the chi-square value, even if the sample fits relatively well against the hypothesized ratio. Hence, we need some way of keeping track of categories. We can do this with degrees of freedom, which is basically a count of independent categories. With Mendel's data, the total number of offspring is 1,064, of which 787 had tall stems. Therefore, the short-stem group had to consist of 277 plants (1,064 − 787) and isn't an independent category. For our purposes here, degrees of freedom equal the number of categories minus one. Thus, with two phenotypic categories, there is only one degree of freedom.

Table 4.4, the table of chi-square probabilities, is read as follows. Degrees of freedom appear in the left column. We are interested in the first row, where there is one degree of freedom. The numbers across the top of the table are the probabilities. We are interested in the next-to-the-last column, headed by the 0.05. We thus gain the following information from the table: *The probability is 0.05 of getting a chi-square value of 3.841 or larger by chance alone, given that the hypothesis is correct.* This statement formalizes the information in our discussion of frequency distributions. Hence, we are interested in how large a chi-square value will be found in the 5% unacceptable area of the curve. For Mendel's plant experiment, the **critical chi-square** (at $p = 0.05$, one degree of freedom) is 3.841. This is the value to which we compare the calculated χ^2 values (0.60 and 244.45). Since the chi-square value for the 3:1 ratio is 0.60 (table 4.2), which is less than the critical value of 3.841, we do not reject the hypothesis of a 3:1 ratio. But since χ^2 for the 1:1 ratio (table 4.3) is 244.45, which is greater than the critical value, we reject the hypothesis of a 1:1 ratio. Notice that once we did the chi-square test for the 3:1 ratio and failed to reject the hypothesis, no other statistical tests were needed: Mendel's data are consistent with a 3:1 ratio.

A word of warning when using the chi-square: If the expected number in any category is less than five, the conclusions are not reliable. In that case, you can repeat the experiment to obtain a larger sample size, or you can

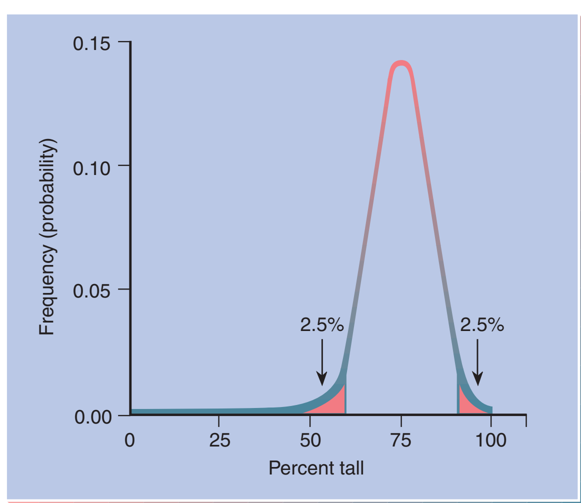


Figure 4.2 Sampling distribution of figure 4.1, $n = 40$. By convention, 5% of the area is marked off (2.5% at each end).

combine categories. Note also that chi-square tests are always done on whole numbers, not on ratios or percentages.

Failing to Reject Hypotheses

Hypothesis testing, in general, involves testing the assumption that there is no difference between the observed and the expected samples. Therefore, the hypothesis against which the data are tested is referred to as the **null hypothesis.** If the null hypothesis is not rejected, then we say that the data are consistent with it, not that the hypothesis has been proved. (As previously discussed, it is always possible we are not rejecting a false hypothesis or are rejecting the true one.) If, however, the hypothesis is rejected, as we rejected a 1:1 ratio for Mendel's data, we fail to reject the alternative hypothesis: that there is a difference between the observed and the expected values. We may then retest the data against some other hypothesis. (We don't say "accept the hypothesis" but rather "fail to reject the hypothesis," because supportive numbers could arise for many reasons. Our failure to reject is tentative acceptance of a hypothesis. However, we are on stronger ground when we reject a hypothesis.)

The use of the 0.05 probability level as a cutoff for rejecting a hypothesis is a convention called the **level of significance.** When a hypothesis is rejected at that level, statisticians say that the data depart *significantly* from the expected ratio. Other levels of significance are also

Table 4.2 Chi-Square Analysis of One of Mendel's Experiments, Assuming a 3:1 Ratio

	Tall Plants	Dwarf Plants	Total
Observed numbers (O)	787	277	1,064
Expected ratio	3/4	1/4	
Expected numbers (E)	798	266	1,064
$O - E$	−11	11	
$(O - E)^2$	121	121	
$(O - E)^2/E$	0.15	0.45	$0.60 = \chi^2$

Table 4.3 Chi-Square Analysis of One of Mendel's Experiments, Assuming a 1:1 Ratio

	Tall Plants	Dwarf Plants	Total
Observed numbers (O)	787	277	1,064
Expected ratio	1/2	1/2	
Expected numbers (E)	532	532	1,064
$O - E$	255	−255	
$(O - E)^2$	65,025	65,025	
$(O - E)^2/E$	122.23	122.23	$244.45 = \chi^2$

Table 4.4 Chi-Square Values

Degrees of Freedom	Probabilities						
	0.99	0.95	0.80	0.50	0.20	0.05	0.01
1	0.000	0.004	0.064	0.455	1.642	3.841	6.635
2	0.020	0.103	0.446	1.386	3.219	5.991	9.210
3	0.115	0.352	1.005	2.366	4.642	7.815	11.345
4	0.297	0.711	1.649	3.357	5.989	9.488	13.277
5	0.554	1.145	2.343	4.351	7.289	11.070	15.086
6	0.872	1.635	3.070	5.348	8.558	12.592	16.812
7	1.239	2.167	3.822	6.346	9.803	14.067	18.475
8	1.646	2.733	4.594	7.344	11.030	15.507	20.090
9	2.088	3.325	5.380	8.343	12.242	16.919	21.666
10	2.558	3.940	6.179	9.342	13.442	18.307	23.209
15	5.229	7.261	10.307	14.339	19.311	24.996	30.578
20	8.260	10.851	14.578	19.337	25.038	31.410	37.566
25	11.524	14.611	18.940	24.337	30.675	37.652	44.314
30	14.953	18.493	23.364	29.336	36.250	43.773	50.892

used, such as 0.01. If a calculated chi-square is greater than the critical value in the table at the 0.01 level, we say that the data depart in a *highly significant* manner from the null hypothesis. Since the chi-square value at the 0.01 level is larger than the value at the 0.05 level, it is more difficult to reject a hypothesis at this level and hence more convincing when it is rejected. Other levels of rejection are also set. In clinical trials of medication, for example, experimenters attempt to make it very easy to reject the null hypothesis: a level of significance of 0.10 or higher is set. The rationale is that it is not desirable to discard a drug or treatment that may well be beneficial. Since the null hypothesis states that the drug has no effect—that is, the control and drug groups show the same response—clinicians would rather be overly conservative. Not rejecting the hypothesis means concluding that the drug has no effect. Rejecting the hypothesis means that the drug has some effect and should be tested further. It is much better to have to retest some drugs that are actually worthless than to discard drugs that have potential value.

SUMMARY

STUDY OBJECTIVE 1: To understand the rules of probability and how they apply to genetics 71–74

We have examined the rules of probability theory relevant to genetic experiments. Probability theory allows us to predict the outcomes of experiments. The probability (P) of independent events occurring is calculated by multiplying their separate probabilities. The probability of mutually exclusive events occurring is calculated by adding their individual probabilities. And the probability of unordered events is defined by the polynomial expansion $(p + q + r + \ldots)^n$:

$$P = \frac{n!}{s!t!u!\ldots} p^s q^t r^u \ldots$$

STUDY OBJECTIVE 2: To understand the use of the chi-square statistical test in genetics 74–78

To assess whether data gathered during an experiment actually support a particular hypothesis, it is necessary to determine what the probability is of getting a particular data set when the null hypothesis is correct. One way to do this is through the chi-square test:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

This test gives us a method of quantifying the confidence we can place in the results obtained from typical genetic experiments. The rules of probability and statistics allow us to devise hypotheses about inheritance and to test these hypotheses with experimental data.

SOLVED PROBLEMS

PROBLEM 1: Mendel self-fertilized a dihybrid plant that had round, yellow peas. In the offspring generation: What is the probability that a pea picked at random will be round and yellow? What is the probability that five peas picked at random will be round and yellow? What is the probability that of five peas picked at random, four will be round and yellow, and one will be wrinkled and green?

Answer: The offspring peas will be round and yellow, round and green, wrinkled and yellow, and wrinkled and green in a ratio of 9:3:3:1. Thus, the probabilities that a pea picked at random will be one of these four categories are 9/16, 3/16, 3/16, and 1/16, respectively. Thus, the probability that a pea picked at random will be round and yellow is 9/16, or 0.563. The probability of picking five of these peas in a row is $(9/16)^5$ or 0.056. The probability that of five peas picked at random, four will be round and yellow, and one will be wrinkled and green, is (substituting into the binomial equation): $(5!/4!1!)(9/16)^4 (1/16)^1 = 5(9^4)/(16^5) = 5(0.006) = 0.031$.

PROBLEM 2: On a chicken farm, walnut-combed fowl that were crossed with each other produced the following offspring: walnut-combed, 87; rose-combed, 31; pea-combed, 30; and single-combed, 12. What hypothesis might you have (based on chapter 2) about the control of comb shape in fowl? Do the data support that hypothesis?

Answer: The numbers 87, 31, 30, and 12 are very similar to 90, 30, 30, and 10, which would be a perfect fit to a 9:3:3:1 ratio. We might expect that ratio, having previously learned something about how comb type is inherited in fowl. Thus, we hypothesize that inheritance of comb type is by two loci, and that dominant alleles at both result in walnut combs, a dominant allele at one locus and recessives at the other result in rose or pea combs, and the recessive alleles at both loci result in a single comb. The results of the cross of dihybrids should produce fowl with the four comb types in a 9:3:3:1 ratio of walnut-, rose-, pea-, and single-combed fowl, respectively. Therefore, our observed numbers are 87, 31, 30, and 10 (sum = 160). Our expected ratio is 9:3:3:1, or 90, 30, 30, and 10 fowl, which are 9/16, 3/16, 3/16, and 1/16, respectively, of the sum of 160. We therefore set up the following chi-square table:

	Comb Type				
	Walnut	**Rose**	**Pea**	**Single**	**Total**
Observed Numbers (*O*)	87	31	30	12	160
Expected Ratio	9/16	3/16	3/16	1/16	
Expected Numbers (*E*)	90	30	30	10	160
$O - E$	−3	1	0	2	
$(O - E)^2$	9	1	0	4	
$(O - E)^2/E$	0.1	0.033	0	0.4	$0.533 = \chi^2$

There are three degrees of freedom since there are four categories of combs. $(4 - 1 = 3)$. The critical chi-square value with three degrees of freedom and probability of 0.05 = 7.815 (table 4.4). Since our calculated chi-square value (0.533) is less than this critical value, we cannot reject our hypothesis. In other words, our data are consistent with the hypothesis of a 9:3:3:1 phenotypic ratio, indicative of a two-locus genetic model with dominance at each locus.

EXERCISES AND PROBLEMS*

PROBABILITY

1. Assuming a 1:1 sex ratio, what is the probability that five children produced by the same parents will consist of
   a. three daughters and two sons?
   b. alternating sexes, starting with a son?
   c. alternating sexes?
   d. all daughters?
   e. all the same sex?
   f. at least four daughters?
   g. a daughter as the eldest child and a son as the youngest?

   (*See also* USE OF RULES)

2. Phenylthiocarbamide (PTC) tasting is dominant (*T*) to nontasting (*t*). If a taster woman with a nontaster father produces children with a taster man, and the man previously had a nontaster daughter, what would be the probability that
   a. their first child would be a nontaster?
   b. their first child would be a nontaster girl?
   c. if they had six children, they would have two nontaster sons, two nontaster daughters, and two taster sons?
   d. their fourth child would be a taster daughter?

   (*See also* USE OF RULES)

3. Albinism is recessive; assume for this problem that blue eyes are also recessive (albinos have blue eyes). What is the probability that two brown-eyed persons, heterozygous for both traits, would produce (remembering epistasis)
   a. five albino children?
   b. five albino sons?
   c. four blue-eyed daughters and a brown-eyed son?
   d. two sons genotypically like their father and two daughters genotypically like their mother?

   (*See also* USE OF RULES)

4. On the average, about one child in every ten thousand live births in the United States has phenylketonuria (PKU). What is the probability that
   a. the next child born in a Boston hospital will have PKU?
   b. after that child with PKU is born, the next child born will have PKU?
   c. two children born in a row will have PKU?

5. In fruit flies, the diploid chromosome number is eight.
   a. What is the probability that a male gamete will contain only paternal centromeres or only maternal centromeres?

*Answers to selected exercises and problems are on page A-4.

b. What is the probability that a zygote will contain only centromeres from male grandparents? (Disregard the problems that the sex chromosomes may introduce.)

6. How many seeds should Mendel have tested to determine with complete certainty that a plant with a dominant phenotype was heterozygous? With 99% certainty? With 95% certainty? With "pretty reliable" certainty?

7. What chance do a man and a woman have of producing one son and one daughter?

8. PKU and albinism are two autosomal recessive disorders, unlinked in human beings. If two people, each heterozygous for both traits, produce a child, what is the chance of their having a child with
a. PKU?
b. either PKU or albinism?
c. both traits?

9. In human beings, the absence of molars is inherited as a dominant trait. If two heterozygotes have four children, what is the probability that
a. all will have no molars?
b. three will have no molars and one will have molars?
c. the first two will have molars and the second two will have no molars?

(*See also* USE OF RULES)

10. Galactosemia is inherited as a recessive trait. If two normal heterozygotes produce children, what is the chance that
a. one of four children will be affected?
b. three children will be born in this order: normal boy, affected girl, affected boy?

(*See also* USE OF RULES)

11. A normal man (A) whose grandfather had galactosemia and a normal woman (B) whose mother was galactosemic want to produce a child. What is the probability that their first child will be galactosemic?

12. A city had nine hundred deaths during the year, and of these, three hundred were from cancer and two hundred from heart disease. What is the probability that the next death will be from
a. cancer?
b. either cancer or heart disease?

13. A plant that has the genotype *AA bb cc DD EE* is mated with one that is *aa BB CC dd ee.* F_1 individuals are selfed. What is the chance of getting an F_2 plant whose genotype exactly matches the genotype of one of the parents?

USE OF RULES

14. The ability to taste phenylthiocarbamide is dominant in human beings. If a heterozygous taster mates with a nontaster, what is the probability that of their five children, only one will be a taster?

15. In mice, coat color is determined by two independent genes, A and C, as indicated here: *A-C-,* agouti; *aaC-,* black; $A\text{-}c^ac^a$ and aac^ac^a, albino. If the following two mice are crossed $AaCc^a \times Aac^ac^a$, what is the probability that among the first six offspring, two will be agouti, two will be black, and two will be albino?

STATISTICS

16. The following data are from Mendel's original experiments. Suggest a hypothesis for each set and test this hypothesis with the chi-square test. Do you reach different conclusions with different levels of significance?
a. Self-fertilization of round-seeded hybrids produced 5,474 round seeds and 1,850 wrinkled ones.
b. One particular plant from *a* yielded 45 round seeds and 12 wrinkled ones.
c. Of the 565 plants raised from F_2 round-seeded plants, 372, when self-fertilized, gave both round and wrinkled seeds in a 3:1 proportion, whereas 193 yielded only round seeds.
d. A violet-flowered, long-stemmed plant was crossed with a white-flowered, short-stemmed plant, producing the following offspring:

47 violet, long-stemmed plants
40 white, long-stemmed plants
38 violet, short-stemmed plants
41 white, short-stemmed plants

17. Mendel self-fertilized pea plants with round and yellow peas. In the next generation he recovered the following numbers of peas:

315 round and yellow peas
108 round and green peas
101 wrinkled and yellow peas
32 wrinkled and green peas

What is your hypothesis about the genetic control of the phenotype? Do the data support this hypothesis?

18. Two agouti mice are crossed, and over a period of a year they produce 48 offspring with the following phenotypes:

28 agouti mice
7 black mice
13 albino mice

What is your hypothesis about the genetic control of coat color in these mice? Do the data support that hypothesis?

19. Two curly-winged flies, when mated, produce sixty-one curly and thirty-five straight-winged progeny. Use a chi-square test to determine whether these numbers fit a 3:1 ratio.

20. A short-winged, dark-bodied fly is crossed with a long-winged, tan-bodied fly. All the F_1 progeny are long-winged and tan-bodied. F_1 flies are crossed among themselves to yield 84 long-winged, tan-bodied flies; 27 long-winged dark-bodied flies; 35 short-winged, tan-bodied flies; and 14 short-winged, dark-bodied flies.

a. What ratio do you expect in the progeny?

b. Use the chi-square test to evaluate your hypothesis. Is the observed ratio within the expected range?

CRITICAL THINKING QUESTIONS

1. A friend shows you three closed boxes, one of which contains a prize, and asks you to choose one. Your friend then opens one of the two remaining boxes, a box she knows is empty. At that point, she gives you the opportunity to change your choice to the last remaining box. Should you?

2. If all couples wanted at least one child of each sex, approximately what would the average family size be?

Suggested Readings for chapter 4 are on page B-2.

5

SEX DETERMINATION, SEX LINKAGE, AND PEDIGREE ANALYSIS

Three generations of a family.

(© Frank Siteman/Tony Stone Images.)

STUDY OBJECTIVES

STUDY OUTLINE

We ended chapter 3 with a discussion of the chromosomal theory of heredity, stated lucidly in 1903 by Walter Sutton, that genes are located on chromosomes. In 1910, T. H. Morgan, a 1933 Nobel laureate, published a paper on the inheritance of white eyes in fruit flies. The mode of inheritance for this trait, discussed later in this chapter, led inevitably to the conclusion that the locus for this gene is on a chromosome that determines the sex of the flies: when a white-eyed male was mated with a red-eyed female, half of the F_2 sons were white-eyed and half were red-eyed; all F_2 daughters were red-eyed. Not only was this the first evidence that localized a particular gene to a particular chromosome, but this study also laid the foundation for our understanding of the genetic control of sex determination.

SEX DETERMINATION

Patterns

At the outset, we should note that the sex of an organism usually depends on a very complicated series of developmental changes under genetic and hormonal control. However, often one or a few genes can determine which pathway of development an organism takes. Those switch genes are located on the **sex chromosomes,** a heteromorphic pair of chromosomes, when those chromosomes exist.

However, sex chromosomes are not the only determinants of an organism's sex. The ploidy of an individual, as in many hymenoptera (bees, ants, wasps), can determine sex; males are haploid and females are diploid. Allelic mechanisms may determine sex by a single allele or multiple alleles not associated with heteromorphic chromosomes; even environmental factors may control sex. For example, temperature determines the sex of some geckos, and the sex of some marine worms and gastropods depends on the substrate on which they land. In this chapter, however, we concentrate on chromosomal sex-determining mechanisms.

Sex Chromosomes

Basically, four types of chromosomal sex-determining mechanisms exist: the XY, ZW, X0, and compound chromosomal mechanisms. In the XY case, as in human beings or fruit flies, the females have a homomorphic pair of chromosomes (XX) and males are heteromorphic (XY). In the ZW case, males are homomorphic (ZZ), and females are heteromorphic (ZW). (XY and ZW are chromosome notations and imply nothing about the sizes or shapes of these chromosomes.) In the X0 case, the organism has only one sex chromosome, as in some grasshoppers and beetles; females are usually XX and males X0. And in the compound chromosome case, several X and Y chromosomes combine to determine sex, as in bedbugs and some beetles. We need to emphasize that the chromosomes themselves do not determine sex, but the genes they carry do. In general, the genotype determines the type of gonad, which then determines the phenotype of the organism through male or female hormonal production.

The XY System

The XY situation occurs in human beings, in which females have forty-six chromosomes arranged in twenty-three homologous, homomorphic pairs. Males, with the same number of chromosomes, have twenty-two homomorphic pairs and one heteromorphic pair, the XY pair (fig. 5.1). During meiosis, females produce gametes that contain only the X chromosome, whereas males produce two kinds of gametes, X- and Y-bearing (fig. 5.2). For this reason, females are referred to as **homogametic** and males as **heterogametic.** As you can see from figure 5.2, in people, fertilization has an equal chance of producing either male or female offspring. In *Drosophila,* the system is the same, but the Y chromosome is almost 20% larger than the X chromosome (fig. 5.3).

Since both human and *Drosophila* females normally have two X chromosomes, and males have an X and a Y chromosome, it seems impossible to know whether maleness is determined by the presence of a Y chromosome or the absence of a second X chromosome. One way to resolve this problem would be to isolate individuals with odd numbers of chromosomes. In chapter 8, we examine the causes and outcomes of anomalous chromosome numbers. Here, we consider two facts from that chapter. First, in rare instances, individuals form, although they are not necessarily viable, with extra sets of chromosomes. These individuals are referred to as **polyploids** (*triploids* with 3*n, tetraploids* with 4*n,* etc.). Second, also infrequently, individuals form that have more or fewer than the normal number of any one chromosome. These **aneuploids** usually come about when a pair of chromosomes fails to separate properly during meiosis, an occurrence called **nondisjunction.** The existence of polyploid and aneuploid individuals makes it possible to test whether the Y chromosome is male determining. For example, a person or a fruit fly that has all the proper nonsex chromosomes, or **autosomes** (forty-four in human beings, six in *Drosophila*), but only a single X without a Y would answer our question. If the Y were absolutely male determining, then this X0 individual should be female. However, if the sex-determining mechanism is a result of the number of X chromosomes, this individual should be a male. As it turns out, an X0 individual is a *Drosophila* male and a human female.

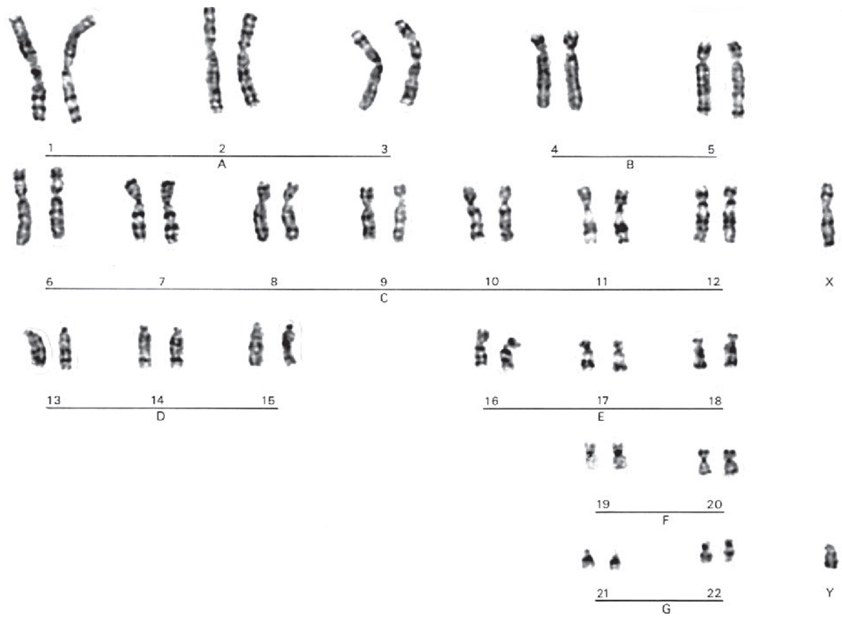
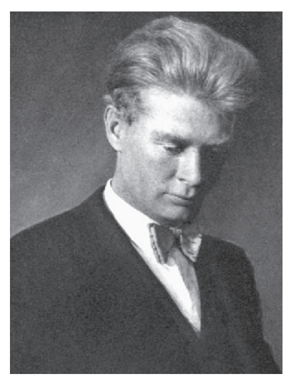


Figure 5.1 Human male karyotype. Note the X and Y chromosomes. A female would have a second X chromosome in place of the Y. (Reproduced courtesy of Dr. Thomas G. Brewster, Foundation for Blood Research, Scarborough, Maine.)

Genic Balance in Drosophila

When geneticist Calvin Bridges, working with *Drosophila,* crossed a triploid (3*n*) female with a normal male, he observed many combinations of autosomes and sex chromosomes in the offspring. From his results, Bridges suggested in 1921 that sex in *Drosophila* is determined by the balance between (ratio of) autosomal alleles that favor maleness and alleles on the X chromosomes that favor femaleness. He calculated a ratio of X chromosomes to autosomal sets to see if this ratio would predict the sex of a fly. An **autosomal set** (A) in *Drosophila* consists of one chromosome from each autosomal pair, or three chromosomes. (An autosomal set in human beings consists of twenty-two chromosomes.) Table 5.1, which presents his results, shows that Bridges's **genic balance**



Calvin B. Bridges (1889–1938). (From *Genetics* 25 (1940): frontispiece. Courtesy of the Genetics Society of America.)

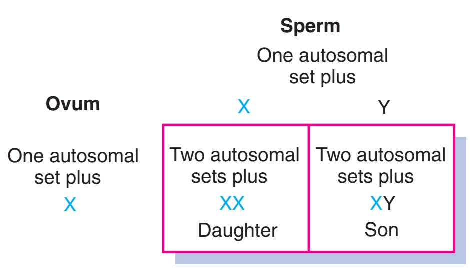


Figure 5.2 Segregation of human sex chromosomes during meiosis, with subsequent zygote formation.

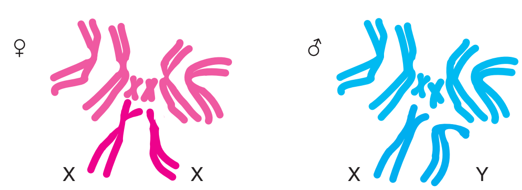


Figure 5.3 Chromosomes of *Drosophila melanogaster.*

Table 5.1 Data Supporting Bridges's Theory of Sex Determination by Genic Balance in *Drosophila*

Number of X Chromosomes	Number of Autosomal Sets (A)	Total Number of Chromosomes	$\frac{X}{A}$ Ratio	Sex
3	2	9	1.50	Metafemale
4	3	13	1.33	Female
4	4	16	1.00	Female
3	3	12	1.00	Female
2	2	8	1.00	Female
1	1	4	1.00	Female
2	3	11	0.67	Intersex
1	2	7	0.50	Male
1	3	10	0.33	Metamale

theory of sex determination was essentially correct. When the X:A ratio is 1.00, as in a normal female, or greater than 1.00, the organism is a female. When this ratio is 0.50, as in a normal male, or less than 0.50, the organism is a male. At 0.67, the organism is an **intersex. Metamales** (X/A = 0.33) and **metafemales** (X/A = 1.50) are usually very weak and sterile. The metafemales usually do not even emerge from their pupal cases.

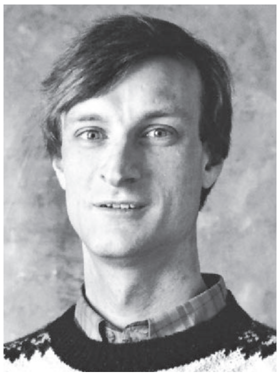
A **sex-switch** gene has been discovered that directs female development. This gene, **Sex-lethal** (*Sxl*), is located on the X chromosome. (It was originally called *female-lethal* because mutations of this gene killed female embryos.) Apparently, *Sxl* has two states of activity. When it is "on," it directs female development; when it is "off," maleness ensues. Other genes located on the X chromosome and the autosomes regulate this sex-switch gene. Genes on the X chromosome that act to regulate *Sxl* into the on state (female development) are called **numerator elements** because they act on the numerator of the X/A genic balance equation. Genes on the autosomes that act to regulate *Sxl* into the off state (male development) are called **denominator elements.** Geneticists have discovered four numerator elements—genes named *sisterless-a, sisterless-b, sisterless-c,* and *runt*. *Sxl* "counts" the number of X chromosomes; it turns on when two are present. It counts by measuring the level of the numerator genes' protein product. If the level is high, *Sxl* turns on, and the organism develops as a female. If the level is relatively low, *Sxl* does not turn on, and development proceeds as a male.

Sex Determination in Human Beings

Since the X0 genotype in human beings is a female (having Turner syndrome), it seems reasonable to conclude that the Y chromosome is male determining in human beings. The fact that persons with Klinefelter syndrome (XXY, XXXY, XXXXY) are all male, and XXX, XXXX, and other multiple-X karyotypes are all female, verifies this idea. (More details on these anomalies are presented in chapter 8.) For a long time, researchers have sought a single gene, a **testis-determining factor (TDF),** located on the Y chromosome that acts as a sex switch to initiate male development. Human embryologists had discovered that during the first month of embryonic development, the gonads that develop are neither testes nor ovaries, but instead are indeterminate. At about six or seven weeks of development, the indeterminate gonads become either ovaries or testes.

In the 1950s, Ernst Eichwald found that males had a protein on their cell surfaces not found in females; he discovered that female mice rejected skin grafts from genetically identical brothers, whereas the brothers accepted grafts from sisters. This implies that an antigen exists on the surface of male cells that is not found on female cells. This protein was called the *histocompatibility Y antigen* **(H-Y antigen).** The gene for this protein was found on the Y chromosome, near the centromere. At first, scientists believed it to be the sex switch: if the gene were present, the gonads would begin development as testes. Further male development, as in male secondary sexual characteristics, came about through the testosterone the functional testes produced. If the gene were absent, the gonads would develop into ovaries. Recently, however, by studying "sex-reversed" individuals, biologists refuted this theory.

Sex-reversed individuals are XX males or XY females. David Page, at the Whitehead Institute for Biomedical Research, found twenty XX males who had a small piece of the short arm of the Y chromosome attached to one of their X chromosomes. He found six XY females in whom the Y chromosome was missing the same small piece at the end of its short arm. This region, which did not contain the *HYA* gene, must carry the testis-determining factor. The first candidate gene from this region believed to code for the testis-determining factor was named the



David Page (1956–).
(Courtesy of Dr. David Page.)

***ZFY* gene,** for zinc finger on the Y chromosome. Zinc fingers are protein configurations known to interact with DNA (discussed in detail in chapter 16). Thus, researchers believed that the *ZFY* gene, coding for the testis-determining factor, worked by directly interacting with DNA. (Later in the book we look at the way regulatory genes, whose proteins interact with DNA, work.) However, men who lack the *ZFY* gene have been found, suggesting that the testis-determining factor is very close to, but not, the *ZFY* gene. From work in mice, it has been suggested that the *ZFY* gene controls the initiation of sperm cell development, but not maleness.

In 1991, Robin Lovell-Badge and Peter Goodfellow and their colleagues in England isolated a gene called **Sex-determining region Y (*SRY*)**—*Sry* in mice—adjacent to the *ZFY* gene. *Sry* has been positively identified as the testis-determining factor because, when injected into normal (XX) female mice, it caused them to develop as males (fig. 5.4). Although these XX males are sterile, they appear as normal males in every other way. (We discuss in chapter 13 how scientists introduce new genes into an organism.) Note also that the mouse and human systems are very similar genetically, and the homologous genes have been isolated from both. However, at present,

Robin Lovell-Badge (1953–).
(Courtesy of Robin Lovell-Badge.)

Peter Goodfellow (1951–).
(Courtesy of Peter Goodfellow.)

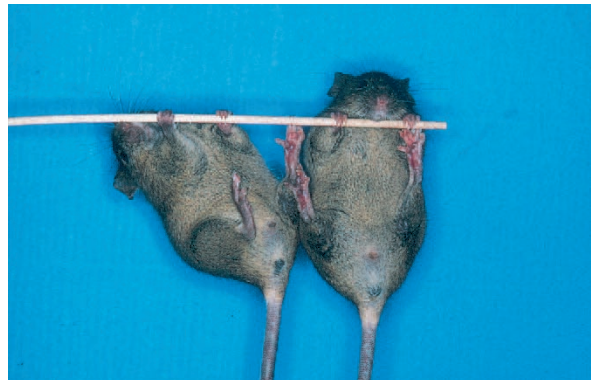


Figure 5.4 Normal male mouse (*left*) and female littermate given the *Sry* gene (*right*). Both mice are indistinguishably male.
(Courtesy of Robin Lovell-Badge.)

the human *SRY* gene does not convert XX female mice into males. Like the *ZFY* gene product, Sry protein (the protein the *SRY* gene produces) also binds to DNA.

The Sry protein appears to bind to at least two genes. One, the p450 aromatase gene, has a protein product that converts the male hormone testosterone to the female hormone estradiol; the Sry protein inhibits production of p450 aromatase. The second gene the Sry protein affects is the gene for the Müllerian-inhibiting substance, which induces testicular development and the digression of female reproductive ducts; the Sry protein enhances this gene's activity. Thus, the Sry protein points an indifferent embryo toward maleness and the maintenance of testosterone production. The sex switch initiates a developmental sequence involving numerous genes. Eva Eicher and Linda Washburn have developed a model in which two pathways of coordinated gene action help determine sex, one pathway for each sex. The first gene in the ovary-determining pathway is termed *ovary determining* (*Od*). The first gene in the testis-determining pathway must function before the *Od* gene begins, in order to allow XY individuals to develop as males. Once the steps of a pathway are initiated, the other pathway is inhibited (fig. 5.5).

Other Chromosomal Systems

The X0 system, sometimes referred to as an X0-XX system, occurs in many species of insects. It functions just as the XY chromosomal mechanism does, except that instead of a Y chromosome, the heterogametic sex (male) has only one X chromosome. Males produce gametes that contain either an X chromosome or no sex chromosome, whereas all the gametes from a female contain the X chromosome. The result of this arrangement is that females have an even number of chromosomes (all in homomorphic pairs) and males have an odd number of chromosomes.

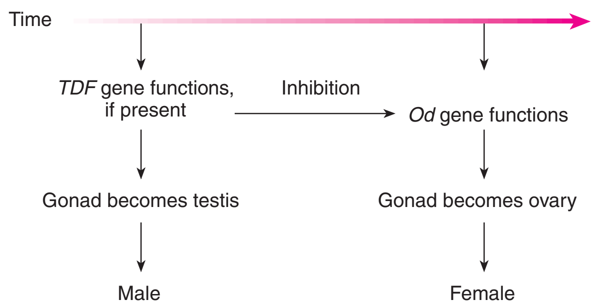


Figure 5.5 A model for the initiation of gonad determination in mammals.

The ZW system is identical to the XY system except that males are homogametic and females are heterogametic. This situation occurs in birds, some fishes, and moths.

Compound chromosomal systems tend to be complex. For example, *Ascaris incurva,* a nematode, has eight X chromosomes and one Y. The species has twenty-six autosomes. Males have thirty-five chromosomes (26A + 8X + Y), and females have forty-two chromosomes (26A + 16X). During meiosis, the X chromosomes unite end to end and so behave as one unit.

The Y Chromosome

In both human beings and fruit flies, the Y chromosome has very few functioning genes. In human beings, two homologous regions exist, one at either end of the X and Y chromosomes, allowing the chromosomes to pair during meiosis. These regions are termed **pseudoautosomal.** Mapping the Y chromosome (see chapters 6 and 13) has shown us the existence of about thirty-five genes (fig. 5.6). Other, nonfunctioning genes are present, too, remnants of a time in the evolutionary past when those genes were probably active (box 5.1). The *Drosophila* Y chromosome is known to carry genes for at least six fertility factors, two on the short arm (*ks-1* and *ks-2*) and four on the long arm (*kl-1, kl-2, kl-3,* and *kl-5*). The Y chromosome carries two other known genes: *bobbed,* which is a locus of ribosomal RNA genes (the nucleolar organizer), and *Suppressor of Stellate* or *Su(Ste),* a gene required for RNA splicing (see chapter 10). The fertility factors code for proteins needed during spermatogenesis. For example, *kl-5* codes for part of the dynein motor needed for sperm flagellar movement.

Sex Determination in Flowering Plants

Flowering plant species (angiosperms) generally have three kinds of flowers: hermaphroditic, male, and female. Hermaphroditic flowers have both male and female parts. The male parts are the anthers and filaments, making up the *stamen,* and the female parts are the stigma, style, and ovary, making up the *pistil* (see fig. 2.2). Ninety percent of angiosperms have hermaphroditic flowers. Of the 10% of the species that have unisexual flowers, some are *monoecious* (Greek, one house), bearing both male and female flowers on the same plant (e.g., walnut); and some are *dioecious* (Greek, two houses), having plants with just male or just female flowers (e.g., date palm).

Within the group of plant species with unisexual flowers, sex-determining mechanisms vary. Some species have a single locus determining sex, some have two or more loci involved in sex determination, and some have X and Y chromosomes. In most of the species with X and Y chromosomes, the sex chromosomes are indistinguishable. Among these species, most have heterogametic males, although in some species, such as the strawberry, females are heterogametic. In the very few species that have distinguishable X and Y chromosomes—only thirteen are known—two sex-determination mechanisms are found. One is similar to the system in mammals, in which the Y chromosome has a gene or genes present

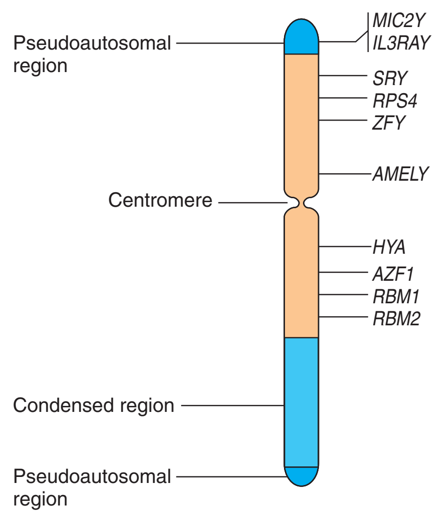


Figure 5.6 The human Y chromosome. In addition to the genes shown, the Y chromosome carries other genes, homologous to X chromosome genes, that do not function because of accumulated mutations. Some of these are in multiple copies. Note the two pseudoautosomal regions that allow synapsis between the Y and X chromosomes. The gene symbols shown include *MIC2Y,* T cell adhesion antigen; *IL3RAY,* interleukin-3 receptor; *RPS4,* a ribosomal protein; *AMELY,* amelogenin; *HYA,* histocompatibility Y antigen; *AZF1,* azoospermia factor 1 (mutants result in tailless sperm); and *RBM1, RBM2,* RNA binding proteins 1 and 2. (Adapted from Online Mendelian Inheritance in Man website. http://www3.ncbi.nlm.nih.gov/omim/. Reprinted with permission.)

BOX 5.1

Experimental Methods

Why Sex and Why Y?

Evolutionary biologists have asked, Why does sex exist? A haploid, asexual way of life seems like a very efficient form of existence. Haploid fungi can produce thousands of haploid spores, each of which can grow into a new colony. What evolutionary benefit do organisms gain by developing diploidy and sexual processes? Although this may not seem like a serious question, evolutionary biologists look for compelling answers.

In chapter 21, we discuss evolutionary thinking in some detail. For the moment, accept that evolutionary biologists look for an adaptive advantage in most evolutionary outcomes. Thus they ask, What is better about the combining of gametes to produce a new generation of offspring? Why would a diploid organism take a random sample of its genome and combine it with a random sample of someone else's genome to produce offspring? Why not simply produce offspring by mitosis? If offspring are produced by mitosis, all of an individual's genes pass into the next generation with every offspring. Not only does just half the genome of an individual pass into the next generation with every offspring produced sexually, but that half is a random jumble of what might be a very highly adapted genome. In addition, males are doubly expensive to produce because males themselves do not produce offspring: males fertilize females who produce offspring. Thus, on the surface, evolutionary biologists need to find very strong reasons for an organism to turn to sexual reproduction when an individual might be at an advantage evolutionarily to reproduce asexually.

There have been numerous suggestions as to the advantage of sex, nicely summarized in a 1994 article by James Crow, of the University of Wisconsin, in *Developmental Genetics,* and more recently in a special section of the 25 September 1998 issue of *Science* magazine. We aren't really sure what the true evolutionary reasons for sex are, but at least three explanations seem reasonable to evolutionary biologists:

- *Adjusting to a changing environment.* Sexual reproduction allows for much more variation in organisms. A haploid, asexual organism collects variation over time by mutation. A sexual organism, on the other hand, can achieve a tremendous amount of variation by recombination and fertilization. Remember that a human being can produce potentially $2^{100,000}$ different gametes. In a changing environment, a sexually reproduced organism is much more likely than an asexual organism to produce offspring that will be adapted to the changes.
- *Combining beneficial mutations.* As mentioned, a haploid, asexual organism accrues mutations as they happen over time in a given individual. A sexual organism can combine beneficial mutations each generation by recombination and fertilization. Thus, sexually reproducing organisms can adapt at a much more rapid rate than asexual organisms.
- *Removing deleterious mutations.* Mutation is more likely to produce deleterious changes than beneficial ones. An asexual organism gathers more and more deleterious mutations as time goes by (a process referred to as *Muller's ratchet,* in honor of Nobel Prize-winning geneticist H. J. Muller and referring to a ratchet wheel that can only go forward). Sexually reproducing organisms can eliminate deleterious mutations each generation by forming recombined offspring that are relatively free of mutation.

Hence, this list provides three of the generally assumed advantages of sexual reproduction that offset its disadvantages.

Another subtle question about sexual reproduction that evolutionary biologists ask is, Why is there a Y chromosome? In other words, why do we have, in some species (e.g., people), a heteromorphic pair of chromosomes involved in sex determination, with one of the chromosomes having the gene for that sex and very few other loci? In people, the Y chromosome is basically a degenerate chromosome with very few loci. This morphological difference between the members of the sex chromosome pair is puzzling. After all, chromosome pairs that do not carry sex-determining loci do not tend to be morphologically heterogeneous. Consider the following possible scenario that Virginia Morell presented in the 14 January 1994 issue of *Science.*

In a particular species in the past—evolutionarily speaking—a sex-determining gene arises on a particular chromosome. One allele at this locus confers maleness on its bearer. The absence of this allele causes the carrier to be female. At this point, millions of years ago, the sex chromosomes are not morphologically heterogeneous: the X and Y chromosomes are identical. In time, however, the Y chromosome comes to carry a gene that is beneficial to the male but not the female. For example, there might be a gene with an allele for a colorful marking; this allele confers a reproductive advantage for the male but also confers a predatory risk on the bearer, whether male or female. Males have a reproductive advantage to outweigh the predation risk, whereas females have none. Thus, the allele is favored in males and selected against in females.

An evolutionary solution to this situation is to isolate the gene for this marking on the Y chromosome and keep it off the X chromosome so that males have it but females do not. This can take place if the two chromosomes do not recombine over most of their lengths. Assume then, that some mechanism evolves to prevent recombination of the X and Y chromosomes. Thereafter, the Y chromosome degenerates, losing most of its genes but retaining the sex-determining locus and the loci conferring an advantage on males but a disadvantage on females.

What evidence do we have that any of these links in this complex line of logic are true? To begin with, when we look at evolutionary lineages, we usually see a spectrum of species with sex chromosomes in all stages of differentiation. Evolutionary biologists generally accept the notion that the similar sex chromosomes are the original condition and the morphologically heterogeneous sex chromosomes are the more evolved condition. In addition, as reported in the same issue of *Science,* William Rice of the University of California at Santa Cruz has shown experimentally with fruit flies that if recombination is prevented between sex chromosomes, the Y chromosome degenerates; it loses the function of many loci that are also found on the X chromosome. Rice showed this with an ingenious set of experiments that successfully prevented a nascent Y chromosome from recombining with the X. The results confirmed the prediction that the Y chromosome degenerates (fig. 1).

More recently, in an October 1999 article in *Science,* Bruce Lahn and David Page, at the Massachusetts Institute of Technology, reported research findings indicating that degeneration of the human Y chromosome has taken place in four stages, starting as long as 320 million years ago in our mammalian ancestors. Using DNA sequence data and methods discussed in chapter 21, they showed that the 19 genes known from both the X and Y chromosomes are arranged as if the Y chromosome has undergone four rearrangements, each preventing further recombination of the X and Y. According to their calculations, this process began shortly after the mammals split from the birds, which themselves went on to evolve a ZW pair of sex chromosomes.

Clearly, much more work is needed to validate all the steps in this logical, evolutionary argument. However, at this point, enough empirical support exists to make the idea attractive to evolutionary biologists.

Although we have gotten a bit ahead of ourselves by talking about subtle evolutionary arguments before reaching that material in the book, it is a good idea to keep an evolutionary perspective on processes and structures. Presumably, evolution has shaped us and the biological world in which we live. If that is so, we should be able to figure out how evolution was working. That thinking should hold from the level of the molecule (e.g., enzymes and DNA) to that of the whole organism. Behind every process and structure should be a hint of the evolutionary pressures that caused that structure or process to evolve.

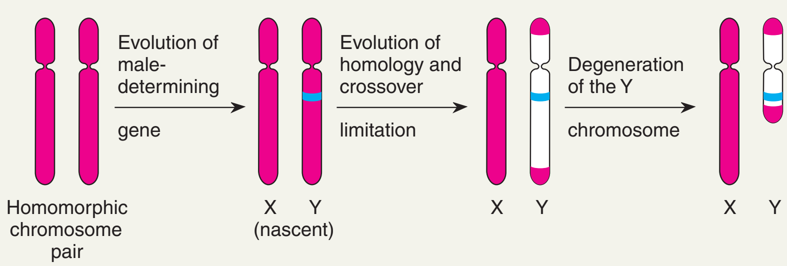


Figure 1 Evolution of a hypothetical Y chromosome. *Red* represents homologous regions, *blue* shows the male-determining gene, and *white* marks evolved areas of the Y chromosome that no longer recombine with the X chromosome.

that actively determine male-flowering plants. The other system is similar to that found in fruit flies, in which the X:A ratio determines sex.

In the mammalian-type system, the Y chromosome carries genes needed for the development of male flower parts while suppressing the development of female parts. An example of this is in the white campion *(Silene latifolia).* In the *Drosophila*-type system, found in the sorrel *(Rumex acetosa),* the ratios determine sex exactly as in the flies. That is, an X:A ratio of 0.5 or lower results in a male; a ratio of 1.0 or higher results in a female; and an intermediate ratio results in a plant with hermaphroditic flowers. It seems that all flowers have the potential to be hermaphroditic. That is, flower primordia for hermaphroditic, male, and female flowers look identical during early development. The simplest mechanism of sex determination would involve repressing the development of the female flower parts in male flowers and repressing the male flower parts in female flowers. Current research indicates that this repression of one component or another is probably involved in most flower sex determination and is under genetic and hormonal control. (We discuss further the genetic control of flower development in chapter 16.)

DOSAGE COMPENSATION

In the XY chromosomal system of sex determination, males have only one X chromosome, whereas females have two. Thus, disregarding pseudoautosomal regions, males have half the number of X-linked alleles as females for genes that are not primarily related to gender. A question arises: How does the organism compensate for this dosage difference between the sexes, given the potential for serious abnormality? In general, an incorrect number of autosomes is usually highly deleterious to an organism (see chapter 8). In human beings and other mammals, the necessary **dosage compensation** is accomplished by the inactivation of one of the X chromosomes in females so that both males and females have only one functional X chromosome per cell.

In 1949, M. Barr and E. Bertram first observed a condensed body in the nucleus that was not the nucleolus. Noting that normal female cats show a single condensed body, while males show none, these researchers referred to the body as sex chromatin, since known as a **Barr body** (fig 5.7). Mary Lyon then suggested that this Barr body represented an inactive X chromosome, which in females becomes tightly coiled into heterochromatin, a condensed, and therefore visible, form of chromatin.

Various lines of evidence support the **Lyon hypothesis** that only one X chromosome is active in any cell. First, XXY males have a Barr body, whereas X0 females have none. Second, persons with abnormal numbers of X

Mary F. Lyon (1925–). (Courtesy of Dr. Mary F. Lyon.)

chromosomes have one fewer Barr body than they have X chromosomes per cell: XXX females have two Barr bodies and XXXX females have three.

Proof of the Lyon Hypothesis

Direct proof of the Lyon hypothesis came when cytologists identified the Barr body in normal females as an X chromosome. Genetic evidence also supports the Lyon hypothesis: Females heterozygous for a locus on the X chromosome show a unique pattern of phenotypic expression. We now know that in human females, an X chromosome is inactivated in each cell on about the twelfth day of embryonic life; we also know that the inactivated X is randomly determined in a given cell. From that point on, the same X remains a Barr body for future cell generations. Thus, heterozygous females show **mosaicism** at the cellular level for X-linked traits. Instead of being typically heterozygous, they express only one or the other of the X chromosomal alleles in each cell.

Glucose-6-phosphate dehydrogenase (G-6-PD) is an enzyme that a locus on the X chromosome controls. The

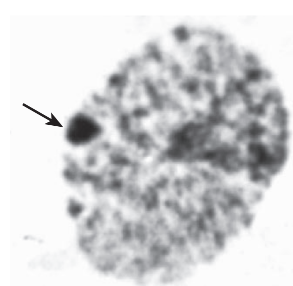


Figure 5.7 Barr body (*arrow*) in the nucleus of a cheek mucosal cell of a normal woman. This visible mass of heterochromatin is an inactivated X chromosome. (Thomas G. Brewster and Park S. Gerald, "Chromosome disorders associated with mental retardation," *Pediatric Annals,* 7, no. 2, 1978. Reproduced courtesy of Dr. Thomas G. Brewster, Foundation for Blood Research, Scarborough, Maine.)

enzyme occurs in several different allelic forms that differ by single amino acids. Thus, both forms (A and B) will dehydrogenate glucose-6-phosphate—both are fully functional enzymes—but because they differ by an amino acid, they can be distinguished by their rate of migration in an electrical field (one form moves faster than another). This electrical separation, termed **electrophoresis,** is carried out by placing samples of the enzymes in a supporting gel, usually starch, polyacrylamide, agarose, or cellulose acetate (fig. 5.8 and box 5.2). After the electric current is applied for several hours, the enzymes move in the gel as bands, revealing the distance each enzyme traveled. Since blood serum is a conglomerate of proteins from many cells, the serum of a female heterozygote (fig. 5.8, lane 3) has both A and B forms (bands), whereas any single cell (lanes 4–10) has only one or the other. Since the gene for glucose-6-phosphate dehydrogenase is carried on the X chromosome, this electrophoretic display indicates that only one X is active in any particular cell.

Another aspect of the glucose-6-phosphate dehydrogenase system provides further proof of the Lyon hypothesis. If a cell has both alleles functioning, both A and B proteins should be present. Since the functioning glucose-6-phosphate dehydrogenase enzyme is a dimer (made up of two protein subunits), 50% of the enzymes should be heterodimers (AB). These would form a third, intermediate band between the A form (AA dimer) and the B form (BB dimer; fig. 5.9). The lack of heterodimers in the blood of heterozygotes is further proof that both G-6-PD alleles are not active within the same cells. That is, in any one cell, only AA or BB dimers can form, because no single cell has both the A and B forms.

The Lyon hypothesis has been demonstrated with many X-linked loci, but the most striking examples are those for color phenotypes in some mammals. For example, the tortoiseshell pattern of cats is due to the inactivation of X chromosomes (fig. 5.10). Tortoiseshell cats are normally females heterozygous for the *yellow* and *black* alleles of the X-linked color locus. They exhibit patches of these two colors, indicating that at a certain stage in development, one or the other of the X chromosomes was inactivated and all of the ensuing daughter cells in that line kept the same X chromosome inactive. The result is patches of coat color.

The X chromosome is inactivated starting at a point called the **X inactivation center (XIC).** That region contains a gene called *XIST* (for *X i*nactive-*s*pecific *t*ranscripts, referring to the transcriptional activity of this gene in the inactivated X chromosome). The *XIST* gene has been putatively identified as the gene that initiates the inactivation of the X chromosome. This gene is known to be active only in the inactive X chromosome in a normal XX female. Another aspect of "Lyonization" is that several other loci are known to be active on the inactivated X chromosome; they are active in both X chromosomes, even though one is heterochromatic (inactivated). Although several of these loci are in the pseudoautosomal region of the short arm of the X chromosome, several other of the thirty or more genes known to be active are on other places on the mammalian X chromosome. Active genes on the inactive X include the gene for the enzyme steroid sulphatase; the red-cell antigen Xg^a; *MIC2;* a *ZFY*-like gene termed *ZFX;* the gene for Kallmann syndrome; and several others.

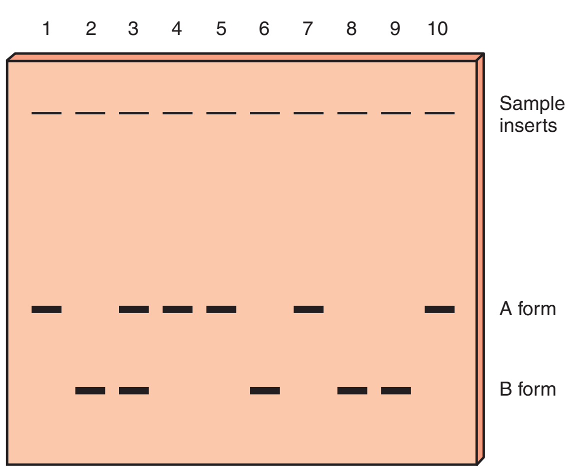


Figure 5.8 Electrophoretic gel stained for glucose-6-phosphate dehydrogenase. Lanes *1–3* contain blood from an *AA* homozygote, a *BB* homozygote, and an *AB* heterozygote, respectively. Lanes *4–10* contain homogenates of individual cells of an *AB* heterozygote.

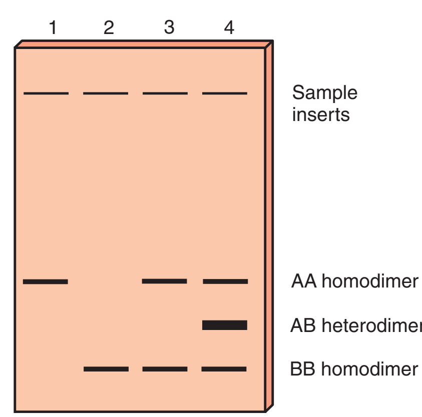


Figure 5.9 Electrophoretic gel stained for glucose-6-phosphate dehydrogenase. Lanes *1* and *2* contain blood serum from *AA* and *BB* women, respectively, and lane *3* contains serum from an *AB* heterozygote. Lane *4* shows the pattern expected if both the *A* and *B* alleles were active within the same cell.

BOX 5.2

Experimental Methods

Electrophoresis

Electrophoresis, a technique for separating relatively similar types of molecules (for example, proteins and nucleic acids), has opened up new and exciting areas of research in population, biochemical, and molecular genetics. It has allowed us to see variations in large numbers of loci, previously difficult or impossible to sample. In biochemical genetics, electrophoretic techniques can be used to study enzyme pathways. In molecular genetics, electrophoresis is used to sequence nucleotides (see chapter 13) and to assign various loci to particular chromosomes. In population genetics (see chapter 21), electrophoresis has made it possible to estimate the amount of variability that occurs in natural populations.

Here we discuss protein electrophoresis, a process that entails placing a sample—often blood serum or a cell homogenate—at the top of a gel prepared from a suitable substrate (e.g., hydrolyzed starch, polyacrylamide, or cellulose acetate) and a buffer. An electrical current is

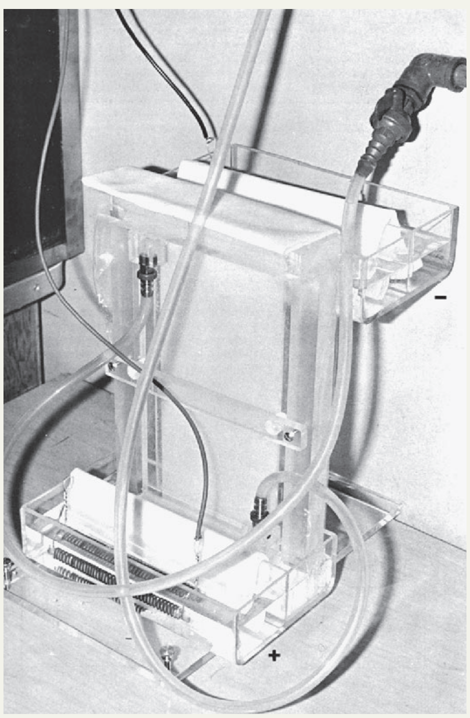


Figure 1 Vertical starch gel apparatus. Current flows from the upper buffer chamber to the lower one by way of the paper wicks and the starch gel. Cooling water flows around the system. (R. P. Canham, "Serum protein variations and selection in fluctuating populations of cricetid rodents," Ph.D. thesis, University of Alberta, 1969. Reproduced by permission.)

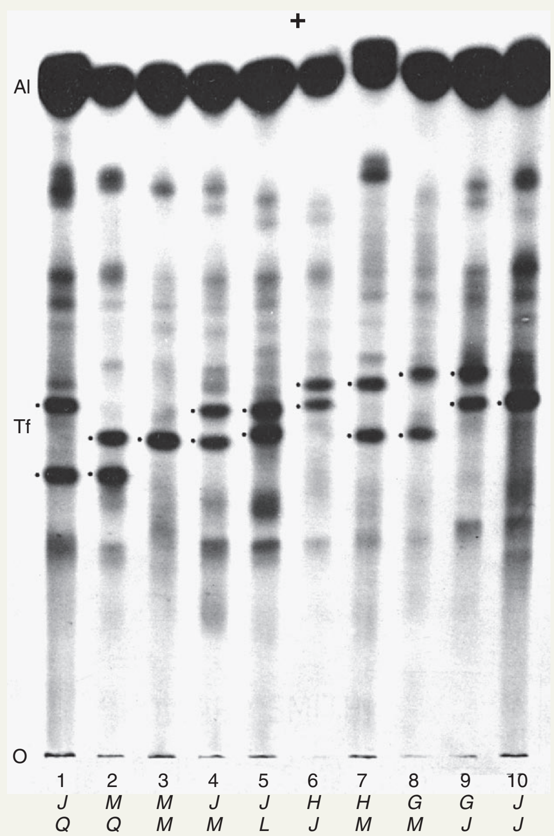


Figure 2 Ten samples of deer mouse *(Peromyscus maniculatus)* blood studied for general protein. *Al* is albumin and *Tf* is transferrin, the two most abundant proteins in mammalian blood. The six *Tf* allozymes are labeled G, H, J, L, M, and Q. (R. P. Canham, "Serum protein variations and selection in fluctuating populations of cricetid rodents," Ph.D. thesis, University of Alberta, 1969. Reproduced by permission.)

passed through the gel to cause charged molecules to move (fig. 1), and the gel is then treated with a dye that stains the protein. In the simplest case, if a protein is homogeneous (usually the product of a homozygote), it forms a single band on the gel. If it is heterogeneous (usually the product of a heterozygote), it forms two bands. This is because the two allelic protein products differ by an amino acid; they have different electrical charges and therefore travel through the gel at different rates (see fig. 5.8). The term **allozyme** refers to different electrophoretic forms of an enzyme controlled by alleles at the same locus.

Figure 2 shows samples of mouse blood serum that have been stained for protein. Most of the staining reveals albumins and β-globulins (transferrin). Because they are present in very small concentrations, many enzymes present in the serum are not visible, but a stain that is specific for a particular enzyme can make that enzyme visible on the gel. For example, lactate dehydrogenase (LDH) can be located because it catalyzes this reaction:

$$\text{lactic acid} + \text{NAD}^+ \overset{\text{LDH}}{\leftrightarrow} \text{pyruvic acid} + \text{NADH}$$

Thus, we can stain specifically for the lactate dehydrogenase enzyme by adding the substrates of the enzyme (lactic acid and nicotinamide adenine dinucleotide, NAD^+) and a suitable stain specific for a product of the enzyme reaction (pyruvic acid or nicotinamide adenine dinucleotide, reduced form, NADH). That is, if lactic acid and NAD^+ are poured on the gel, only lactate dehydrogenase converts them to pyruvic acid and NADH. We can then test for the presence of NADH by having it reduce the dye, nitro blue tetrazolium, to the blue precipitate, formazan, an electron carrier. We then add all the preceding reagents and look for blue bands on the gel (fig. 3).

In addition to its uses in population genetics and chromosome mapping, electrophoresis has been extremely useful in determining the structure of many proteins and for studying developmental pathways. As we can see from the lactate dehydrogenase gel in figure 3, five bands can occur. In some tissues of a homozygote, these bands occur roughly in a ratio of 1:4:6:4:1. This pattern can come about if the enzyme is a tetramer whose four subunits are random mixtures of two gene products (from the *A* and *B* loci). Thus we would get

AAAA (1/16)
AAAB (4/16)
AABB (6/16)
ABBB (4/16)
BBBB (1/16)

(Note that the ratio 1:4:6:4:1 is the expansion of $[A + B]^4$, and the relative "intensity" of each band—the number of protein doses—is calculated from the rule of unordered events described in chapter 4.)

continued

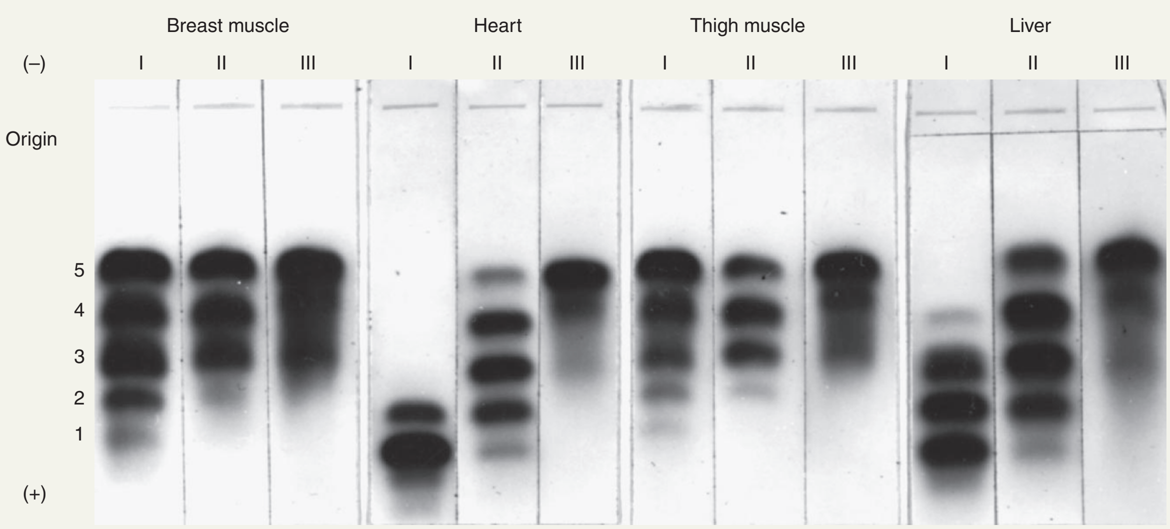


Figure 3 Lactate dehydrogenase isozyme patterns in pigeons. Note the five bands for some individual samples. Lanes *I, II,* and *III* under each tissue type indicate the range of individual variation. (W. H. Zinkham, et al., "A Variant of Lactate Dehydrogenase in Somatic Tissues of Pigeons" in *Journal of Experimental Zoology* 162, no. 1 (June):45–46, 1966. Reproduced by permission of the Wistar Institute.)

BOX 5.2 (CONTINUED)

Protein chemists have verified this tetramer model. In this way, electrophoresis has helped us determine the structure of several enzymes. (The term **isozymes** refers to multiple electrophoretic forms of an enzyme due to subunit interaction rather than allelic differences.) Chemists have also discovered that the five forms differ in concentration in different tissues of the body (fig. 4). This has led to various hypotheses as to how the production of enzymes is controlled developmentally.

Electrophoresis is also valuable in clinical diagnosis. In various diseases, cell destruction causes the release of proteins into the bloodstream. Thus, the lactate dehydrogenase pattern is found in the blood in certain disease states (fig. 5). This is why examination of the blood LDH is often a diagnostic test used to pick up early signs of heart and liver disease (among others).

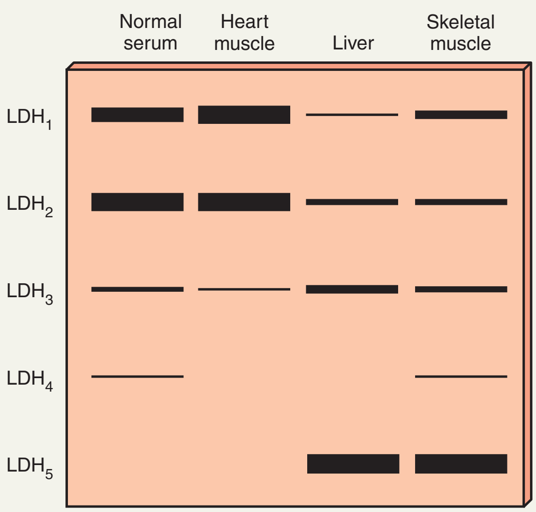


Figure 4 LDH patterns found in different tissues in human beings.

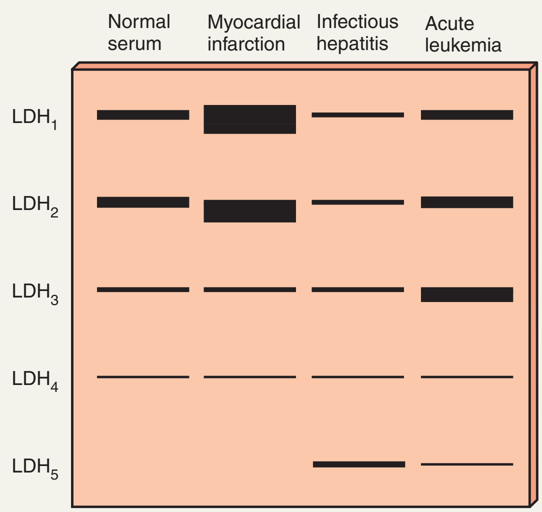


Figure 5 LDH patterns from normal human serum and from serum affected by various disease states.

The gene product of *XIST* is an RNA that does not seem to be translated into a protein. Rather, using localization techniques, geneticists have found this RNA is associated with Barr bodies, coating the inactive chromosome. Current research is aimed at determining the details of this interaction.

Dosage Compensation for Drosophila

Dosage compensation also occurs in fruit flies, and it appears that the gene activity of X chromosome loci is also about equal in males and females. The mechanism is different from that in mammals since no Barr bodies are found in fruit flies. Instead, the male's single X chromosome is hyperactive, approaching the level of transcriptional activity of both of the female's X chromosomes combined. Researchers have discovered a multisubunit protein complex called MSL (for male-specific lethal) that binds to hundreds of sites on the single X chromosome in males. Presumably, the binding mediates the hyperactivity of the genes on the X chromosome. (We discuss control of transcription later in the book.) At least five genes contribute products to this protein complex: *msl1, msl2, msl3, mle,* and *mof.* (*Mle* comes from *male*less, and *mof* comes from *m*ales absent *o*n the *f*irst.) Along with this protein complex are RNAs that also bind to the male X chromosome. These RNAs, also implicated in dosage compensation, are the products of the *rox1* and *rox2* genes (for *R*NA *o*n the *X*). Together, the MSL protein complex and the RNAs comprise a **compensasome.**

Mutant alleles of the male-specific lethal (*msl*) genes disrupt dosage compensation in males and are, as their names imply, lethal. However, they appear to have no effect in females. Expression of at least one of these genes, *msl2,* is repressed by the protein product of the *Sxl* gene. Thus, sex determination and dosage compensation are

Figure 5.10 Tortoiseshell cat. A female heterozygous for the X-linked *yellow* and *black* alleles. (Courtesy of Donna Bass.)

ultimately under the control of the same master switch gene, *Sxl.* This should not be surprising since the ability of *Sxl* to count the number of X chromosomes in a cell makes it the most efficient initiator of both sexual development and dosage compensation.

SEX LINKAGE

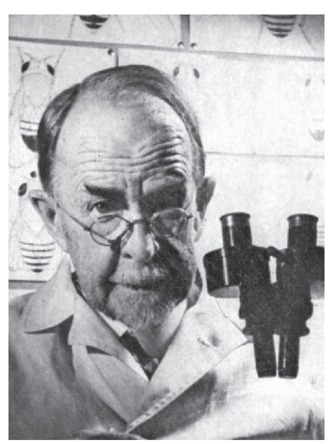
In an XY chromosomal system of sex determination, the pattern of inheritance for loci on the heteromorphic sex chromosomes differs from the pattern for loci on the homomorphic autosomal chromosomes because alleles of the sex chromosome are inherited in association with the sex of the offspring. Alleles on a male's X chromosome go to his daughters but not to his sons, because the presence of his X chromosome normally determines that his offspring is a daughter. For example, the inheritance pattern of hemophilia (failure of blood to clot), the common form of which is caused by an allele located on the X chromosome, has been known since the end of the eighteenth century. It was known that mostly men had the disease, whereas women could pass on the disease without actually having it. (In fact, the general nature of the inheritance of this trait was known in biblical times. The Talmud—the Jewish book of laws and traditions—specified exemptions to circumcision on the basis of hemophilia among relatives consistent with an understanding of who was at risk.)

Before we continue, we need to make a small distinction. Since both X and Y are sex chromosomes, three different patterns of inheritance are possible, all sex linked (for loci found only on the X chromosome, only on the Y chromosome, or on both). However, the term **sex-linked** usually refers to loci found only on the X chromosome; the term **Y-linked** is used to refer to loci found only on the Y chromosome, which control **holandric traits** (traits found only in males). Loci found on both the X and Y chromosomes are called *pseudoautosomal.* In human beings, at least four hundred loci are known to be on the X chromosome; only a few are known to be on the Y chromosome.

X Linkage in Drosophila

T. H. Morgan demonstrated the **X-linked** pattern of inheritance in *Drosophila* in 1910, when a white-eyed male appeared in a culture of wild-type (red-eyed) flies (fig. 5.11). This male was crossed with a wild-type female. All of the offspring were wild-type. However, when these F_1 individuals were crossed with each other, their offspring fell into two categories (fig. 5.12). All the females and half the males were wild-type, whereas the remaining half of the males were white-eyed. Ultimately, Morgan and others interpreted this to mean that the white-eye locus was on the X chromosome. We can redraw figure 5.12 to include the sex chromosomes of Morgan's flies (fig. 5.13). We denote the X chromosome with the white-eye allele as X^w. Similarly X^+ is the X chromosome with the wild-type allele, and Y is the Y chromosome, which does not have this locus.

Another property of sex linkage appears in figure 5.13. Since females have two X chromosomes, they can have normal homozygous and heterozygous allelic combinations. But males, with only one copy of the X chromosome, can be neither homozygous nor heterozygous.



Thomas Hunt Morgan (1866–1945). (From *Genetics* 32 (1947): frontispiece. Courtesy of the Genetics Society of America.)

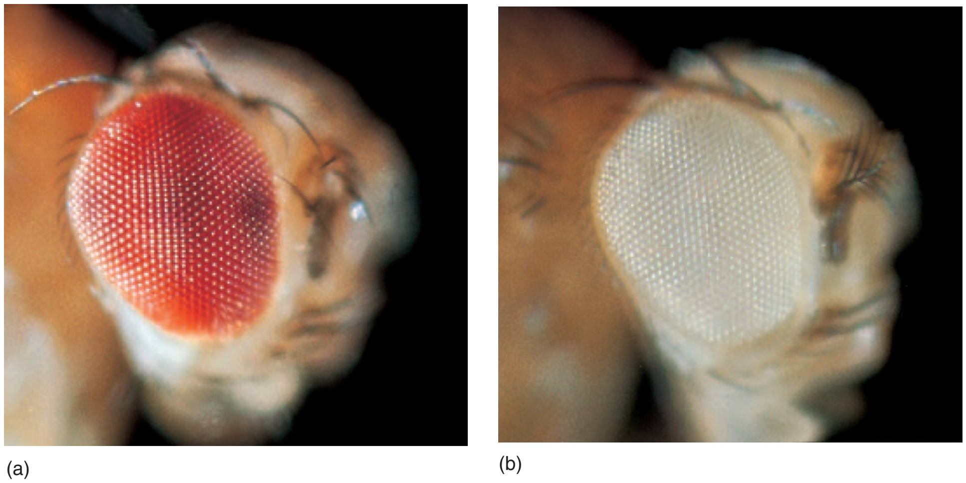


Figure 5.11 (*a*) Wild-type (red-eyed) and (*b*) white-eyed fruit flies. (Carolina Biological Supply Company.)

Instead, the term **hemizygous** describes the presence of X-linked genes (and other genes present in only one copy) in males. Since only one allele is present, a single copy of a recessive allele determines the phenotype in a phenomenon called **pseudodominance.** Thus, a male with one *w* allele is white-eyed, the allele acting in a dominant fashion. This is the same way one copy of a dominant autosomal allele would determine the phenotype of a normal diploid organism. Hence the term *pseudodominance.*

Nonreciprocity

The X-linked pattern has long been known as the **crisscross pattern of inheritance** because the father passes a trait to his daughters, who pass it to their sons. Figure 5.14 shows why this analysis is correct and the inheritance pattern is not reciprocal through a cross between a white-eyed female and a wild-type male. Here the F_1 males are white-eyed, the F_1 females are wild-type, and 50% of each sex in the F_2 generation are white-eyed. Such nonreciprocity and different ratios in the two sexes suggest sex linkage, which the crisscross pattern confirms.

Figure 5.15 shows the inheritance pattern of a sex-linked trait in chickens, in which the male is the homogametic sex (ZZ). The gene for barred plumage is Z linked, and barred plumage is dominant to nonbarred plumage. If we substitute white-eyed for nonbarred and male for female, we get the same pattern as in fruit flies (fig. 5.13)—in which, of course, females are homogametic.

The Y chromosome in fruit flies carries the pseudoautosomal *bobbed* locus (*bb*), the nucleolar organizer. In the homozygous recessive state, it causes bristles to shorten. Figures 5.16 and 5.17 show the results of reciprocal crosses involving *bobbed.* In both cases, one quarter of the F_2 individuals are *bobbed.* In one cross it is males, and in the other it is females.

P_1	♀ Wild-type	×	♂ White eye
F_1		♂ and ♀ Wild-type	
F_2	♀ Wild-type		♂ 1/2 Wild-type 1/2 White eye

Figure 5.12 Pattern of inheritance of the white-eye trait in *Drosophila.*

Sex-Limited and Sex-Influenced Traits

Aside from X-linked, holandric, and pseudoautosomal inheritance, two inheritance patterns show nonreciprocity without necessarily being under the control of loci on the sex chromosomes. **Sex-limited traits** are traits expressed in only one sex, although the genes are present in both. In women, breast and ovary formation are sex-limited traits, as are facial hair distribution and sperm production in men. Nonhuman examples are plumage patterns in birds—in many species, the male is brightly colored—and horns found only in males of certain sheep species. Milk yield in mammals is expressed phenotypically only in females. **Sex-influenced,** or **sex-conditioned, traits** appear in both sexes but occur in one sex more than the other. Pattern, or premature, baldness in human beings is an example of a sex-influenced trait. In women, it is usually expressed as a thinning of hair rather than as balding. Apparently testosterone, the male hormone, is required for the full expression of the allele.

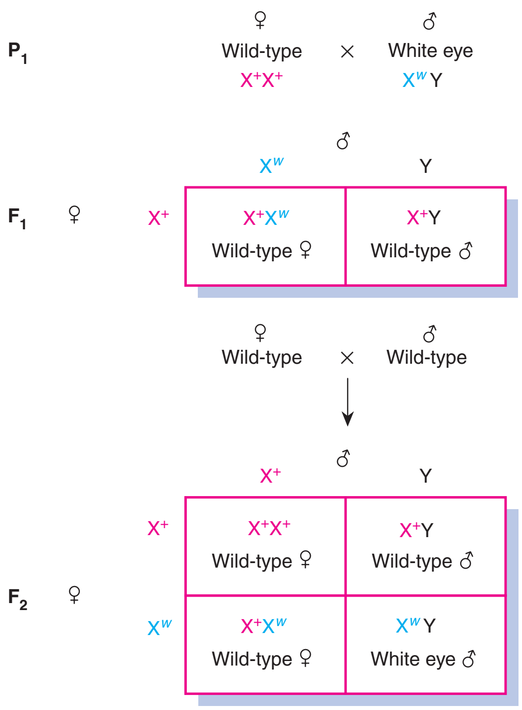


Figure 5.13 Crosses of figure 5.12 redrawn to include the sex chromosomes.

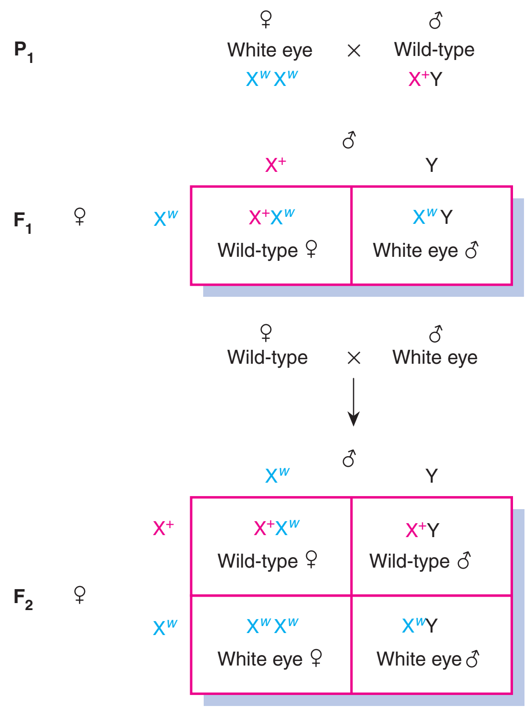


Figure 5.14 Reciprocal cross to that in figure 5.13.

PEDIGREE ANALYSIS

Inheritance patterns in many organisms are relatively easy to determine, because crucial crosses can test hypotheses about the genetic control of a particular trait. Many of these same organisms produce an abundance of offspring so that investigators can gather numbers large enough to compute ratios. Recall Mendel's work with garden peas; his 3:1 ratio in the F_2 generation led him to suggest the rule of segregation. If Mendel's sample sizes had been smaller, he might not have seen the ratio. Think of the difficulties Mendel would have faced had he decided to work with human beings instead of pea plants. Human geneticists face the same problems today. The occurrence of a trait in one of four children does not necessarily indicate a true 3:1 ratio.

To determine the inheritance pattern of many human traits, human geneticists often have little more to go on than a **pedigree** that many times does not include critical mating combinations. Frequently uncertainties and ambiguities plague pedigree analysis, a procedure whereby conclusions are often a product of the process of elimination. Other difficulties human geneticists encounter are the lack of **penetrance** and different degrees of **expressivity** in many traits. Both are aspects of the expression of a phenotype.

Penetrance and Expressivity

Penetrance refers to the appearance in the phenotype of genotypically determined traits. Unfortunately for geneticists, not all genotypes "penetrate" the phenotype. For example, a person could have the genotype that specifies vitamin-D-resistant rickets and yet not have rickets (a bone disease). This disease is caused by a sex-linked dominant allele and is distinguished from normal vitamin D deficiency by its failure to respond to low levels of vitamin D. It does, however, respond to very high levels of vitamin D and is thus treatable. In any case, in some family trees, affected children are born to unaffected parents. This would violate the rules of dominant inheritance because one of the parents must have had the allele yet did not express it. The fact that the parent actually had the allele is demonstrated by the occurrence of low levels of phosphorus in the blood, a pleiotropic effect of the same allele. The low-phosphorus aspect of the phenotype is always fully penetrant.

Thus, certain genotypes, often those for developmental traits, are not always fully penetrant. Most genotypes, however, are fully penetrant. For example, no known cases exist of individuals homozygous for albinism who do not actually lack pigment. Vitamin-D-resistant rickets illustrates another case in which a phenotype that is not genetically determined mimics a phenotype that is. This

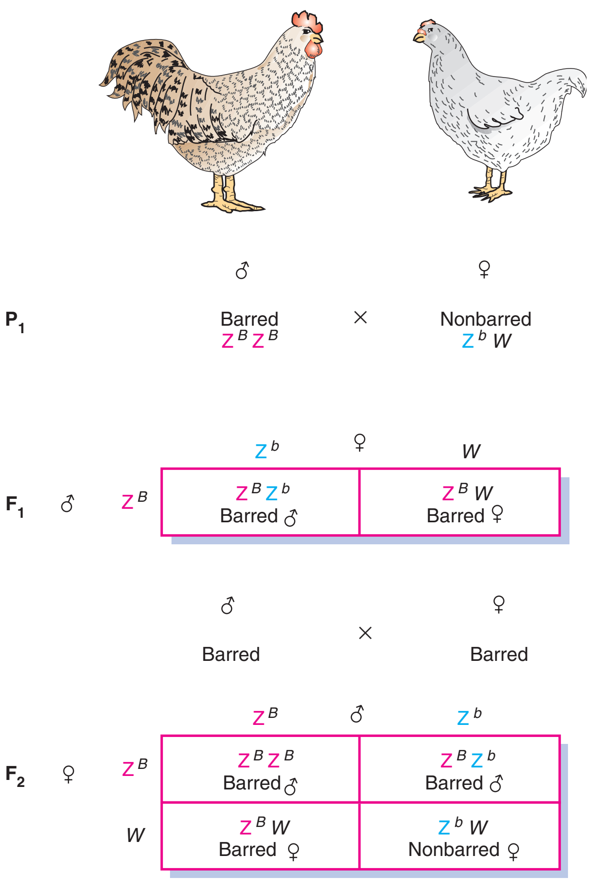


Figure 5.15 Inheritance pattern of barred plumage in chickens in which males are homogametic (ZZ) and females are heterogametic (ZW).

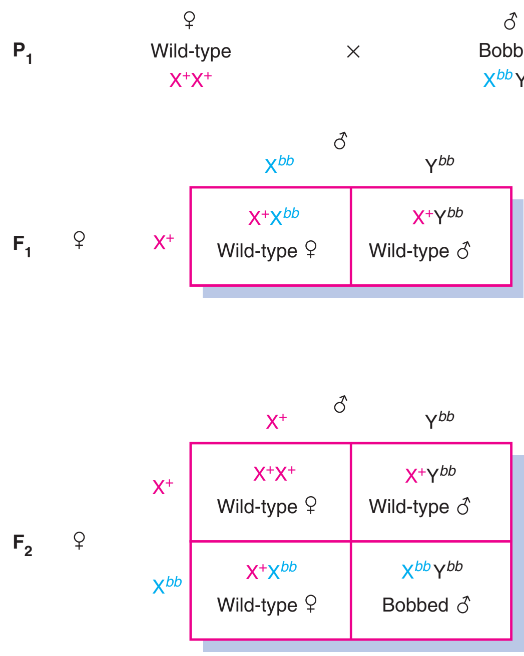


Figure 5.16 Inheritance pattern of the *bobbed* locus in *Drosophila*.

phenocopy is the result of dietary deficiency or environmental trauma. A dietary deficiency of vitamin D, for example, produces rickets that is virtually indistinguishable from genetically caused rickets.

Many developmental traits not only sometimes fail to penetrate, but also show a variable pattern of expression, from very mild to very extreme, when they do. For example, cleft palate is a trait that shows both variable penetrance and variable expressivity. Once the genotype penetrates, the severity of the impairment varies considerably, from a very mild external cleft to a very severe clefting of the hard and soft palates. Failure to penetrate and variable expressivity are not unique to human traits but are characteristic of developmental traits in many organisms.

Family Tree

One way to examine a pattern of inheritance is to draw a family tree. Figure 5.18 defines the symbols used in constructing a family tree, or pedigree. The circles represent females, and the squares represent males. Symbols that are filled in represent individuals who have the trait under study; these individuals are said to be **affected.** The open symbols represent those who do not have the trait. Direct horizontal lines between two individuals (one male, one female) are called marriage lines. Children are attached to a marriage line by a vertical line. All the brothers and sisters **(siblings** or **sibs)** from the same parents are connected by a horizontal line above their symbols. Siblings are numbered below their symbols according to birth order (fig. 5.19), and generations are numbered on the right in Roman numerals. When the sex of a child is unknown, the symbol is diamond-shaped (e.g., the children of III-1 and III-2 in fig. 5.19). A number within a symbol represents the number of siblings not separately listed. Individuals IV-7 and IV-8 in figure 5.19 are fraternal (dizygotic or nonidentical) twins: they originate from the same point. Individuals III-3 and III-4 are identical (monozygotic) twins: they originate from the same short vertical line.

When other symbols occur in a pedigree, they are usually defined in the legend. Individual V-5 in figure 5.19 is called a **proband** or **propositus** (female, **proposita**). The arrow pointing to individual V-5 indicates that the pedigree was ascertained through this individual, usually by a physician or clinical investigator.

On the basis of the information in a pedigree, geneticists attempt to determine the mode of inheritance

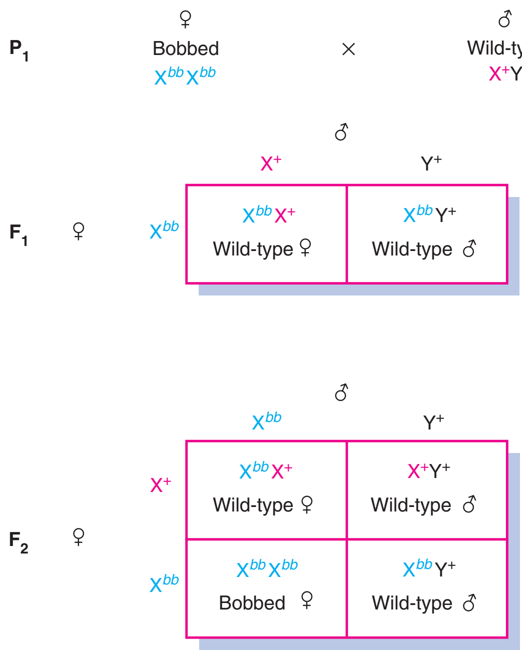


Figure 5.17 Reciprocal cross to that in figure 5.16.

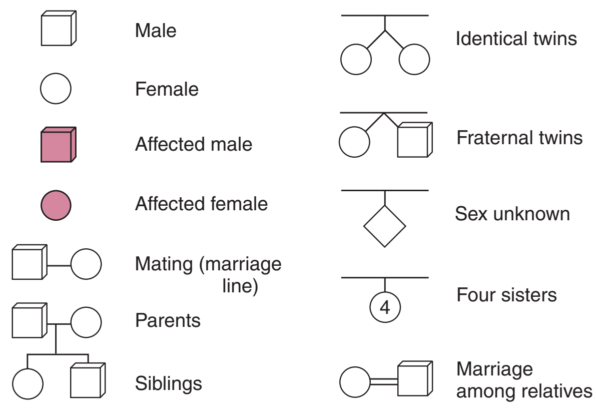


Figure 5.18 Symbols used in a pedigree.

of a trait. There are two types of questions the pedigree might be used to answer. First, are there patterns within the pedigree that are consistent with a particular mode of inheritance? Second, are there patterns within the pedigree that are inconsistent with a particular mode of inheritance? Often, it is not possible to determine the mode of inheritance of a particular trait with certainty. McKusick has reported that, as of 2001, the mode of inheritance of over nine thousand loci in human beings was known with some confidence, including autosomal dominant, autosomal recessive, and sex-linked genes.

Dominant Inheritance

If we look again at the pedigree in figure 5.19, several points emerge. First, polydactyly (fig. 5.20) occurs in every generation. Every affected child has an affected parent—no generations are skipped. This suggests dominant inheritance. Second, the trait occurs about equally in both sexes; there are seven affected males and six affected females in the pedigree. This indicates autosomal rather than sex-linked inheritance. Thus, so far, we would categorize polydactyly as an autosomal dominant trait. Note also that individual IV-11, a male, passed on the trait to two of his three sons. This would rule out sex linkage. (Remember that a male gives his X chromosome to all of his daughters but none of his sons. His sons receive his Y chromosome.) Consistency in many such pedigrees, has confirmed that an autosomal dominant gene causes polydactyly.

Polydactyly shows variable penetrance and expressivity. The most extreme manifestation of the trait is an extra digit on each hand (fig. 5.20) and one or two extra toes on each foot. However, some individuals have only extra toes, some have extra fingers, and some have an asymmetrical distribution of digits such as six toes on one foot and seven on the other.

Recessive Inheritance

Figure 5.21 is a pedigree with a different pattern of inheritance. Here affected individuals are not found in each generation. The affected daughters, identical triplets, come from unaffected parents. They represent, in fact, the first appearance of the trait in the pedigree. A telling point is that the parents of the triplets are first cousins; a mating between relatives is referred to as **consanguineous.** If the degree of relatedness is closer than law permits, the union is called **incestuous.** In all states, brother-sister and mother-son marriages are forbidden; and in all states except Georgia, father-daughter marriages are forbidden. Georgia did not intend to permit father-daughter marriages. However, the law was drafted using biblical terminology that inadvertently did not prohibit a man from marrying his daughter or his grandmother. Thirty states prohibit the marriage of first cousins.

Consanguineous matings often produce offspring that have rare recessive, and often deleterious, traits. The reason is that through common ancestry (e.g., when first cousins have a pair of grandparents in common), a rare allele can be passed on both sides of the pedigree and become homozygous in a child. The occurrence of a trait in a pedigree with common ancestry is often good evidence

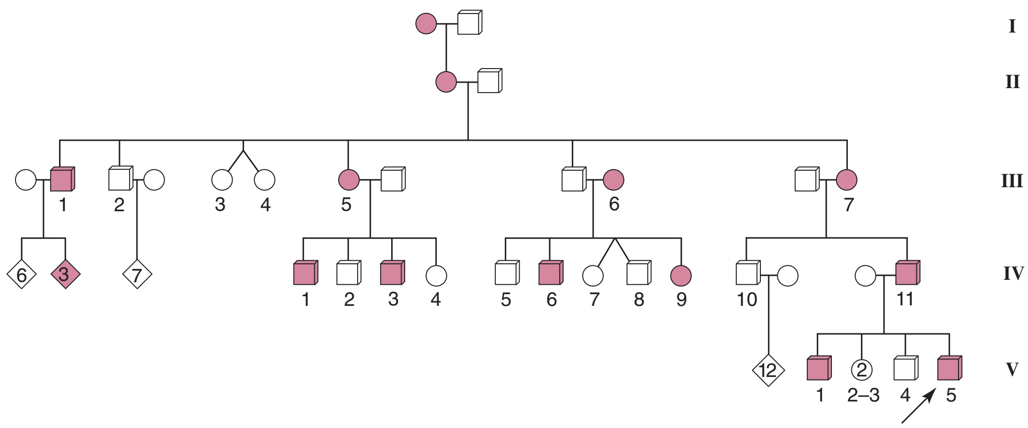


Figure 5.19 Part of a pedigree for polydactyly.

for an autosomal recessive mode of inheritance. Consanguinity by itself does not guarantee that a trait has an autosomal recessive mode of inheritance; all modes of inheritance appear in consanguineous pedigrees. Conversely, recessive inheritance is not confined to consanguineous pedigrees. Hundreds of recessive traits are known from pedigrees lacking consanguinity.

Sex-Linked Inheritance

Figure 5.22 is the pedigree of Queen Victoria of England. Through her children, hemophilia was passed on to many of the royal houses of Europe. Several interesting aspects of this pedigree help to confirm the method of inheritance. First, generations are skipped. Although Alexis (1904–18) was a hemophiliac, neither his parents nor his grandparents were. This pattern occurs in several other places in the pedigree and indicates a recessive mode of inheritance. From other pedigrees and from the biochemical nature of the defect, scientists have determined that hemophilia is a recessive trait.

Further inspection of the pedigree in figure 5.22 reveals that all the affected individuals are sons, strongly suggesting sex linkage. Since males are hemizygous for the X chromosome, more males than females should have the phenotype of a sex-linked recessive trait because males do not have a second X chromosome that might carry the normal allele. If this is correct, we can make several predictions. First, since all males get their X chromosomes from their mothers, affected males should be the offspring of carrier (heterozygous) females. A female is automatically a carrier if her father had the disease. She has a 50% chance of being a carrier if her brother, but not her father, has the disease. In that case, her mother was a carrier. The pedigree in figure 5.22 is consistent with these predictions.

Figure 5.20 Hands of a person with polydactyly. Manifestations range in severity from one extra finger or toe to one or more extra digits on each hand and foot. (© L.V. Bergman/ The Bergman Collection.)

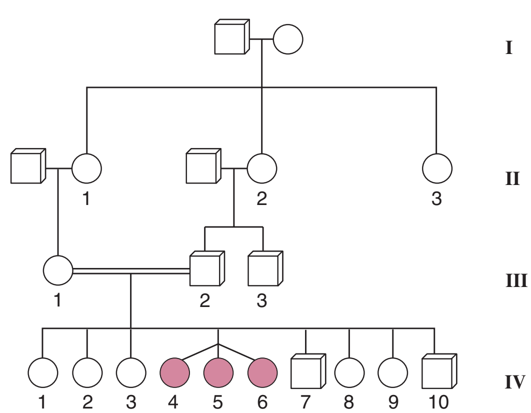


Figure 5.21 Part of a pedigree of hypotrichosis (early hair loss).

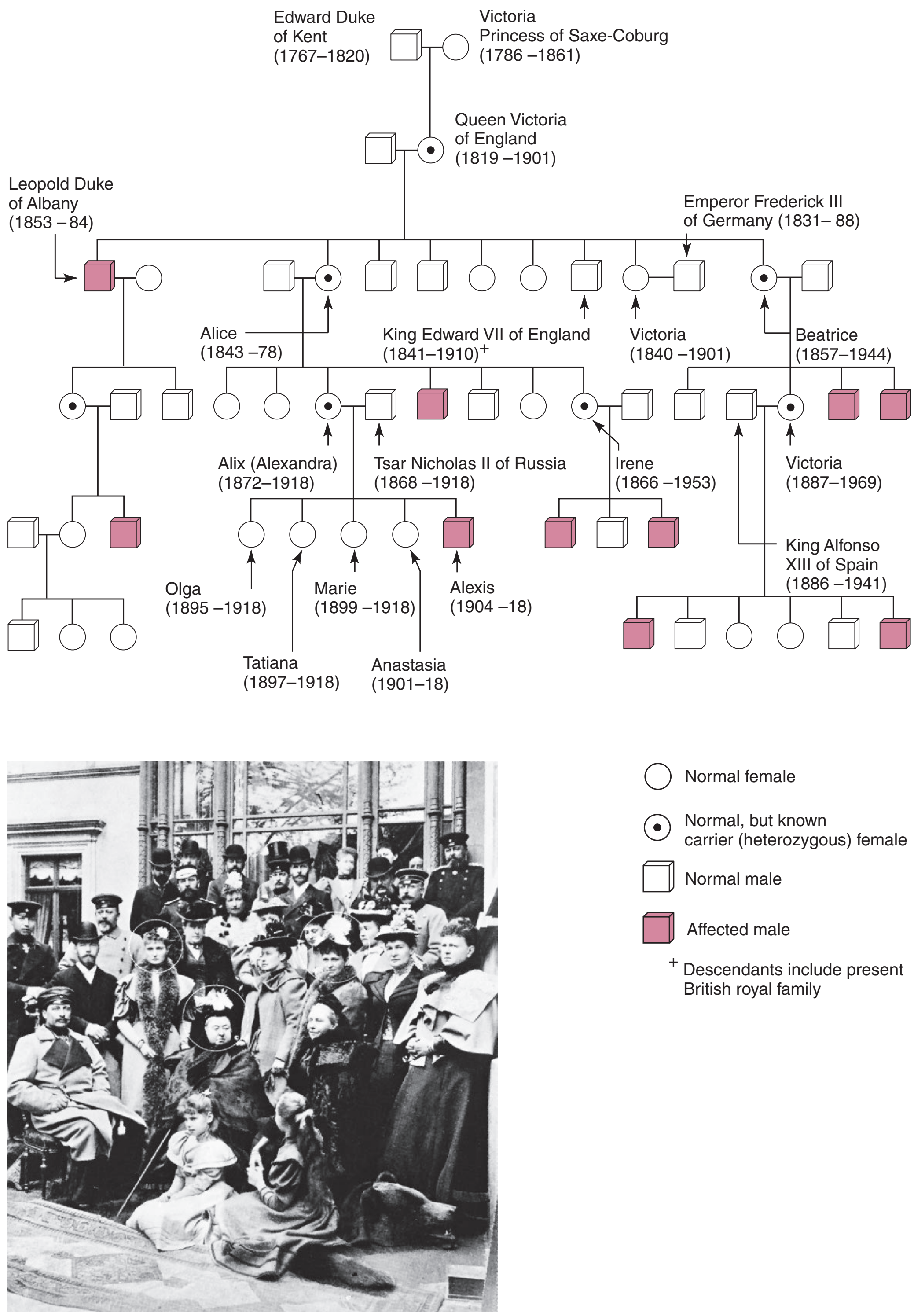


Figure 5.22 Hemophilia in the pedigree of Queen Victoria of England. In the photograph of the Queen and some of her descendants, three carriers—Queen Victoria (*center*), Princess Irene of Prussia (*right*), and Princess Alix (Alexandra) of Hesse (*left*)—are indicated. (Photo © Mary Evans Picture Library/Photo Researchers, Inc.)

In no place in the pedigree is the trait passed from father to son. This would defy the route of an affected X chromosome. We can conclude from the pedigree that hemophilia is a sex-linked recessive trait. (Several different inherited forms of hemophilia are known, each deficient in one of the steps in the pathway that forms fibrinogen, the blood clot protein. Two of these forms, "classic" hemophilia A and hemophilia B, also called Christmas disease, are sex linked. Other hemophilias are autosomal.)

One other interesting point about this pedigree is that there is no evidence of the disease in Queen Victoria's ancestors, yet she was obviously a heterozygote, having one affected son and two daughters who were known carriers. Thus, though she was born to what appears to be a homozygous normal mother and a hemizygous normal father, one of Queen Victoria's X chromosomes had the hemophilia allele. This could have happened if a change (mutation) had occurred in one of the gametes that formed Queen Victoria. (We explore the mechanisms of mutation in chapter 12.)

Figure 5.23 is another pedigree that points to dominant inheritance because the trait skips no generations. The pedigree shows the distribution of low blood-phosphorus levels, the fully penetrant aspect of vitamin-D-resistant rickets, among the sexes. Affected males pass on the trait to their daughters but not their sons. This pattern follows that of the X chromosome: a male passes it on to all of his daughters but to none of his sons. Although this pedigree accords with a sex-linked dominant mode of inheritance, it does not rule out autosomal inheritance. The pedigree shown is a small part of one involving hundreds of people, all with phenotypes consistent with the hypothesis of sex-linked dominant inheritance.

In figure 5.23, there is the slight possibility that the trait is recessive. This could be true if the male in generation I and the mates of II-5 and II-7 were all heterozygotes. Since this is a rare trait, the possibility that all these conditions occurred is small. For example, if one person in fifty (0.02) is a heterozygote, then the probability of three heterozygotes mating within the same pedigree is $(0.02)^3$, or eight in one million. The rareness of this event further supports the hypothesis of dominant inheritance. The expected patterns for the various types of inheritance in pedigrees can be summarized in the following four categories:

Autosomal Recessive Inheritance

1. Trait often skips generations.
2. An almost equal number of affected males and females occur.
3. Traits are often found in pedigrees with consanguineous matings.
4. If both parents are affected, all children should be affected.
5. In most cases when unaffected people mate with affected individuals, all children are unaffected. When at least one child is affected (indicating that the unaffected parent is heterozygous), approximately half the children should be affected.
6. Most affected individuals have unaffected parents.

Autosomal Dominant Inheritance

1. Trait should not skip generations (unless trait lacks full penetrance).
2. When an affected person mates with an unaffected person, approximately 50% of their offspring should be affected (indicating also that the affected individual is heterozygous).
3. The trait should appear in almost equal numbers among the sexes.

Sex-Linked Recessive Inheritance

1. Most affected individuals are male.
2. Affected males result from mothers who are affected or who are known to be carriers (heterozygotes) because they have affected brothers, fathers, or maternal uncles.
3. Affected females come from affected fathers and affected or carrier mothers.
4. The sons of affected females should be affected.
5. Approximately half the sons of carrier females should be affected.

Sex-Linked Dominant Inheritance

1. The trait does not skip generations.
2. Affected males must come from affected mothers.
3. Approximately half the children of an affected heterozygous female are affected.
4. Affected females come from affected mothers or fathers.
5. All the daughters, but none of the sons, of an affected father are affected.

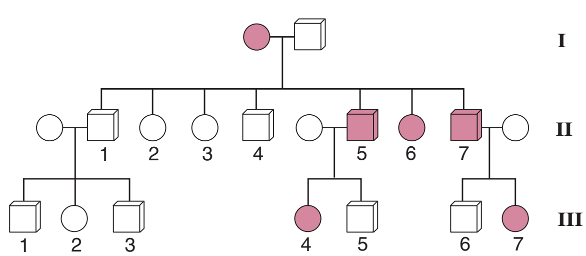


Figure 5.23 Part of a pedigree of vitamin-D-resistant rickets. Affected individuals have low blood-phosphorus levels. Although the sample is too small for certainty, dominance is indicated because every generation was affected, and sex linkage is suggested by the distribution of affected individuals.

SUMMARY

This chapter begins a four-chapter sequence that analyzes the relationship of genes to chromosomes. We begin with the study of sex determination.

STUDY OBJECTIVE 1: To analyze the causes of sex determination in various organisms 83–90

Sex determination in animals is often based on chromosomal differences. In human beings and fruit flies, females are homogametic (XX) and males are heterogametic (XY). In human beings, a locus on the Y chromosome, *SRY,* determines maleness; in *Drosophila,* sex is determined by the balance between genes on the X chromosome and genes on the autosomes that regulate the state of the sex-switch gene, *Sxl.*

STUDY OBJECTIVE 2: To understand methods of dosage compensation 90–95

Different organisms have different ways of solving problems of dosage compensation for loci on the X chromosome. In human beings, one of the X chromosomes in cells in a woman is Lyonized, or inactivated. Lyonization in women leads to cellular mosaicism for most loci on the X chromosome. In *Drosophila,* the X chromosome in males is hyperactive.

STUDY OBJECTIVE 3: To analyze the inheritance patterns of traits that loci on the sex chromosomes control 95–97

Since different chromosomes are normally associated with each sex, inheritance of loci located on these chromosomes shows specific, nonreciprocal patterns. The white-eye locus in *Drosophila* was the first case when a locus was assigned to the X chromosome. Over four hundred sex-linked loci are now known in human beings.

STUDY OBJECTIVE 4: To use pedigrees to infer inheritance patterns 97–102

Human genetic studies use pedigree analysis to determine inheritance patterns because it is impossible to carry out large-scale, controlled human crosses. However, not all traits determined by genotype are apparent in the phenotype, and this lack of penetrance can pose problems in genetic analysis.

SOLVED PROBLEMS

PROBLEM 1: A Female fruit fly with a yellow body is discovered in a wild-type culture. The female is crossed with a wild-type male. In the F_1 generation, the males are yellow-bodied and the females are wild-type. When these flies are crossed among themselves, the F_2 produced are both yellow-bodied and wild-type, equally split among males and females (see fig. 1). Explain the genetic control of this trait.

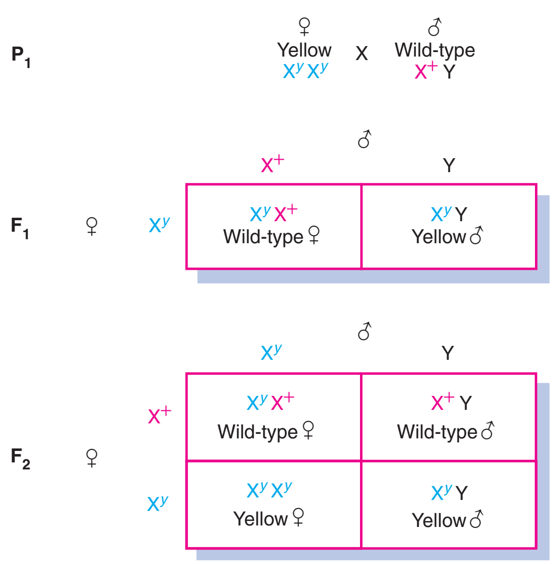


Figure 1 Cross between yellow-bodied and wild-type fruit flies.

Answer: Since the results in the F_1 generation differ between the two sexes, we suspect that a sex-linked locus is responsible for the control of body color. If we assume that it is a recessive trait, then the female parent must have been a recessive homozygote, and the male must have been a wild-type hemizygote. If we assign the wild-type allele as X^+, the yellow-body allele as X^y, and the Y chromosome as Y, then figure 1, showing the crosses into the F_2 generation, is consistent with the data. Thus, a recessive X-linked gene controls yellow body color in fruit flies.

PROBLEM 2: The affected individuals in the pedigree in figure 2 are chronic alcoholics (data from the National Institute of Alcohol Abuse and Alcoholism). What can you say about the inheritance of this trait?

Answer: We begin by assuming 100% penetrance. If that is the case, then we can rule out either sex-linked or autosomal recessive inheritance because both parents had the trait, yet they produced some unaffected children.

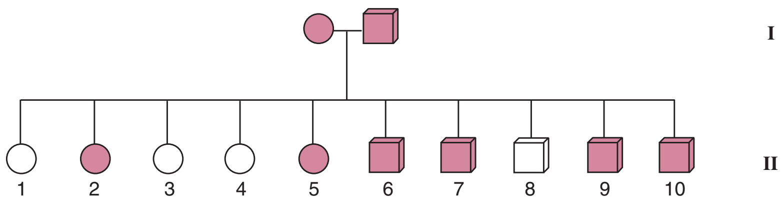


Figure 2 A pedigree for alcoholism.

Nor can the mode of inheritance be by a sex-linked dominant gene because an affected male would have only affected daughters, since his daughters get copies of his single X chromosome. We are thus left with autosomal dominance as the mode of inheritance. If that is the case, then both parents must be heterozygotes; otherwise, all the children would be affected. If both parents are heterozygotes, we expect a 3:1 ratio of affected to unaffected offspring (a cross of $Aa \times Aa$ produces offspring of A-:aa in a 3:1 ratio); here, the ratio is 6:4. If we did a chi-square test, the expected numbers would be 7.5:2.5 (3/4 and 1/4, respectively, of 10). Although the expected value of 2.5 makes it inappropriate to do a chi-square test (the expected value is too small), we can see that the observed and expected numbers are very close. Thus, from the pedigree we would conclude that an autosomal dominant allele controls chronic alcoholism. (Although the analysis is consistent, we actually cannot draw that conclusion about alcoholism because other pedigrees are not consistent with 100% penetrance, a one-gene model, or the lack of environmental influences. In fact, scientists are currently debating whether alcoholism is inherited at all. These types of problems related to complex human traits are discussed in chapter 18.)

PROBLEM 3: A female fly with orange eyes is crossed with a male fly with short wings. The F_1 females have wild-type (red) eyes and long wings; the F_1 males have orange eyes and long wings. The F_1 flies are crossed to yield

47 long wings, red eyes
45 long wings, orange eyes
17 short wings, red eyes
14 short wings, orange eyes

with no differences between the sexes. What is the genetic basis of each trait?

Answer: In the F_1 flies, we see a difference in eye color between the sexes, indicating some type of sex linkage. Since the females are wild-type, wild-type is probably dominant to orange. We can thus diagram the cross for eye color as

(female) X^oX^o orange			X^+Y (male) red	P_1
		↓		
X^+X^o red			X^oY orange	F_1
		↓		
X^+X^o red	X^oX^o orange		X^+Y red	X^oY orange	F_2

We would thus expect to see equal numbers of red-eyed and orange-eyed males and females, which is what we observe. Now look at long versus short wings. If we disregard eye color, wing length seems to be under autosomal control with short wings being recessive. Thus, the parents are homozygotes (ss and s^+s^+), the F_1 offspring are heterozygotes (s^+s), and the F_2 progeny have a phenotypic ratio of 3:1, wild-type (long) to short wings.

EXERCISES AND PROBLEMS*

SEX DETERMINATION

1. What is the difference between an X and a Z chromosome?
2. Transformer (*tra*) is an autosomal recessive gene that converts chromosomal females into sterile males. A female *Drosophila* heterozygous for the transformer allele (*tra*) is mated with a normal male homozygous for transformer. What is the sex ratio of their offspring? What is the sex ratio of their offspring's offspring?
3. The autosomal recessive doublesex (*dsx*) gene converts males and females into developmental intersexes. Two fruit flies, both heterozygous for the *doublesex* (*dsx*) allele, are mated. What are the sexes of their offspring?
4. What is a sex switch? What genes serve as sex switches in human beings and *Drosophila?*

*Answers to selected exercises and problems are on page A-4.

DOSAGE COMPENSATION

5. The diagram of an electrophoretic gel in figure 3 shows activity for a particular enzyme. Lane 1 is a sample from a "fast" homozygote. Lane 2 is a sample from a "slow" homozygote. In lane 3, the blood from the first two was mixed. Lane 4 comes from one of the children of the two homozygotes.

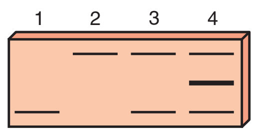


Figure 3 The activity of a particular enzyme as revealed on an electrophoretic gel.

Can you guess the structure of the enzyme? If this were an X-linked trait, what pattern would you expect from a heterozygous female's

a. whole blood?
b. individual cells?

6. How many different zones of activity (bands) would you see on a gel stained for lactate dehydrogenase (LDH) activity from a person homozygous for the A protein gene but heterozygous for the B protein gene? Are the bands due to the activity of allozymes or isozymes?

7. How many Barr bodies would you see in the nuclei of persons with the following sex chromosomes?

a. X0 **e.** XXX
b. XX **f.** XXXXX
c. XY **g.** XX/XY mosaic
d. XXY

What would the sex of each of these persons be? If these were the sex chromosomes of individual *Drosophila* that were diploid for all other chromosomes, what would their sexes be?

8. Calico cats have large patches of colored fur. What does this indicate about the age of onset of Lyonization (is it early or late)? Tortoiseshell cats have very small color patches. Explain the difference between the two phenotypes.

SEX LINKAGE

9. In *Drosophila,* the lozenge phenotype, caused by a sex-linked recessive allele (*lz*), is narrow eyes. Diagram to the F_2 generation a cross of a lozenge male and a homozygous normal female. Diagram the reciprocal cross.

10. Sex linkage was originally detected in 1906 in moths with a ZW sex-determining mechanism. In the currant moth, a pale color (*p*) is recessive to the wild-type and located on the Z chromosome. Diagram reciprocal crosses to the F_2 generation in these moths.

11. What family history of hemophilia would indicate to you that a newborn male baby should be exempted from circumcision?

12. What is the difference between *pseudodominance* and *phenocopy?*

13. In *Drosophila,* cut wings are controlled by a recessive sex-linked allele (*ct*), and fuzzy body is controlled by a recessive autosomal allele (*fy*). When a fuzzy female is mated with a cut male, all the members of the F_1 generation are wild-type. What are the proportions of F_2 phenotypes, by sex?

14. Consider the following crosses in canaries:

Parents	Progeny
a. pink-eyed female × pink-eyed male	all pink-eyed
b. pink-eyed female × black-eyed male	all black-eyed
c. black-eyed female × pink-eyed male	all females pink-eyed, all males black-eyed

Explain these results by determining which allele is dominant and how eye color is inherited.

15. Consider the following crosses involving yellow and gray true-breeding *Drosophila:*

Cross	F_1	F_2
gray female × yellow male	all males gray, all females gray	97 gray females, 42 yellow males, 48 gray males
yellow female × gray male	all females gray, all males yellow	?

a. Is color controlled by an autosomal or an X-linked gene?
b. Which allele, gray or yellow, is dominant?
c. Assume 100 F_2 offspring are produced in the second cross. What kinds and what numbers of progeny do you expect? List males and females separately.

16. A man with brown teeth mates with a woman with normal white teeth. They have four daughters, all with brown teeth, and three sons, all with white teeth. The sons all mate with women with white teeth, and all their children have white teeth. One of the daughters (A) mates with a man with white teeth (B), and they have two brown-toothed daughters, one white-toothed daughter, one brown-toothed son, and one white-toothed son.

a. Explain these observations.
b. Based on your answer to *a,* what is the chance that the next child of the A-B couple will have brown teeth?

17. In human beings, red-green color blindness is inherited as an X-linked recessive trait. A woman with normal vision whose father was color-blind marries a man with normal vision whose father was also color-blind. This couple has a color-blind daughter with a normal complement of chromosomes. Is infidelity suspected? Explain.

18. A white-eyed male fly is mated with a pink-eyed female. All the F_1 offspring have wild-type red eyes. F_1 individuals are mated among themselves to yield:

Females		**Males**	
red-eyed	450	red-eyed	231
pink-eyed	155	white-eyed	301
		pink-eyed	70

Provide a genetic explanation for the results.

19. In *Drosophila,* white eye is an X-linked recessive trait, and ebony body is an autosomal recessive trait. A homozygous white-eyed female is crossed with a homozygous ebony male.

a. What phenotypic ratio do you expect in the F_1 generation?

b. What phenotypic ratio do you expect in the F_2 generation?

c. Suppose the initial cross was reversed: ebony female × white-eyed male. What phenotypic ratio would you expect in the F_2 generation?

20. In *Drosophila,* abnormal eyes can result from mutations in many different genes. A true-breeding wild-type male is mated with three different females, each with abnormal eyes. The results of these crosses are as follows:

		Females	**Males**
male × abnormal-1	→	all normal	all normal
male × abnormal-2	→	1/2 normal, 1/2 abnormal	1/2 normal, 1/2 abnormal
male × abnormal-3	→	all abnormal	all abnormal

Explain the results by determining the mode of inheritance for each abnormal trait.

21. A black and orange female cat is crossed with a black male, and the progeny are as follows:

females: two black, three orange and black

males: two black, two orange

Explain the results.

22. Based on the following *Drosophila* crosses, explain the genetic basis for each trait and determine the genotypes of all individuals:

white-eyed, dark-bodied female × red-eyed, tan-bodied male

F_1: females are all red-eyed, tan-bodied; males are all white-eyed, tan-bodied

F_2: 27 red-eyed, tan-bodied
24 white-eyed, tan-bodied
9 red-eyed, dark-bodied
7 white-eyed, dark-bodied

(No differences between males and females in the F_2 generation.)

PEDIGREE ANALYSIS

23. What is the difference between penetrance and expressivity?

24. What are the possible modes of inheritance in pedigrees *a–c* in figure 4? What modes of inheritance are not possible for a given pedigree?

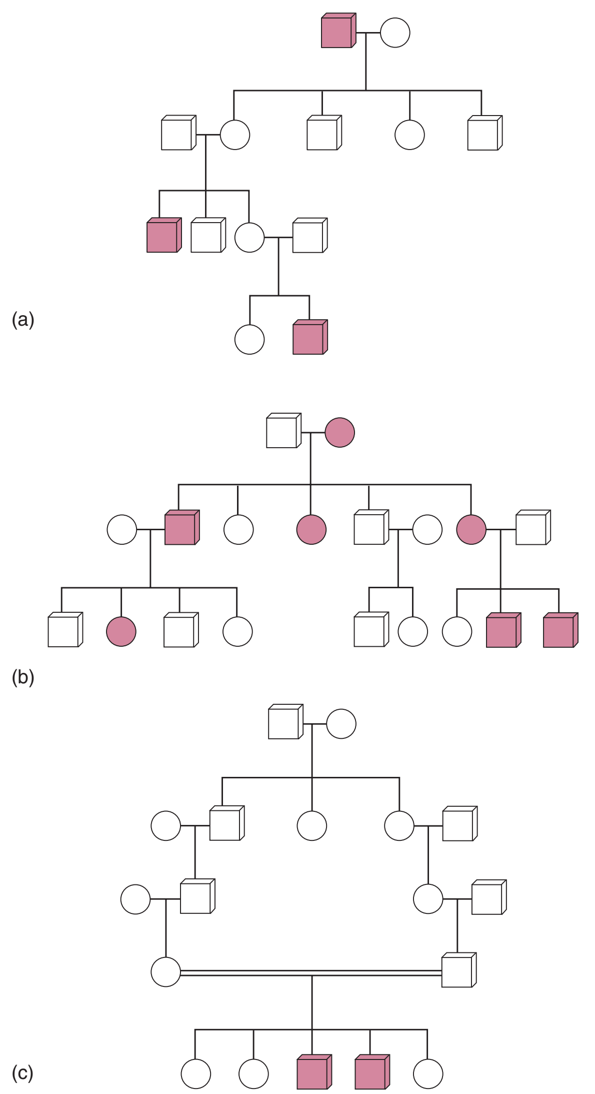


Figure 4 Three pedigrees showing different modes of inheritance.

25. In pedigrees *a–d* in figure 5, which show the inheritance of rare human traits, including twin production, determine which modes of inheritance are most probable, possible, or impossible.

26. Hairy ears, a human trait expressed as excessive hair on the rims of ears in men, shows reduced penetrance (less than 100% penetrant). Mechanisms proposed include Y linkage, autosomal dominance, and autosomal recessiveness. Construct a pedigree consistent with each of these mechanisms.

27. Construct pedigrees for traits that *could not be*
- **a.** autosomal recessive.
- **b.** autosomal dominant.
- **c.** sex-linked recessive.
- **d.** sex-linked dominant.

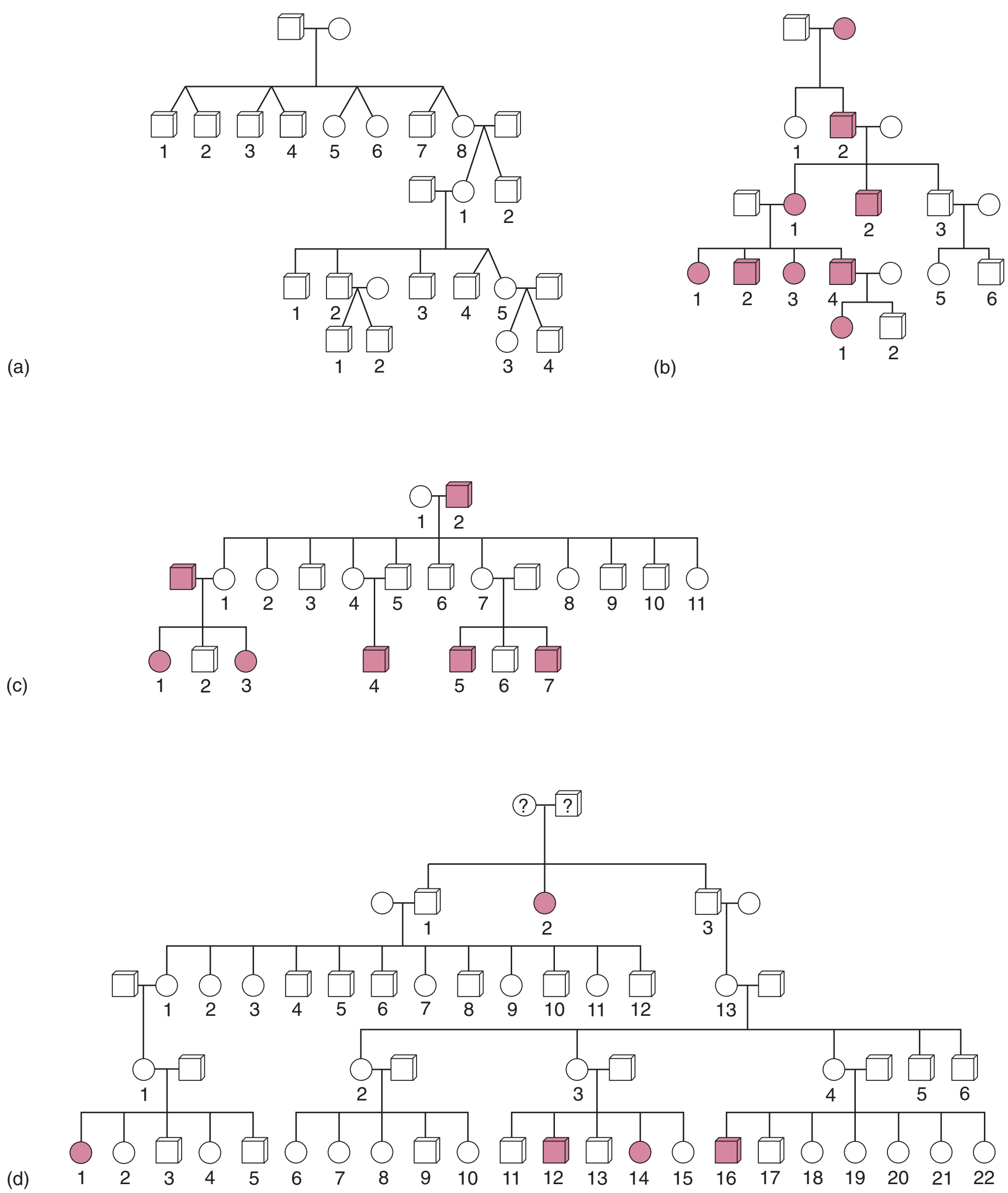


Figure 5 Pedigrees of rare human traits, including twin production (*a*).

28. Determine the possible modes of inheritance for each trait in pedigrees *a–c* in figure 6.

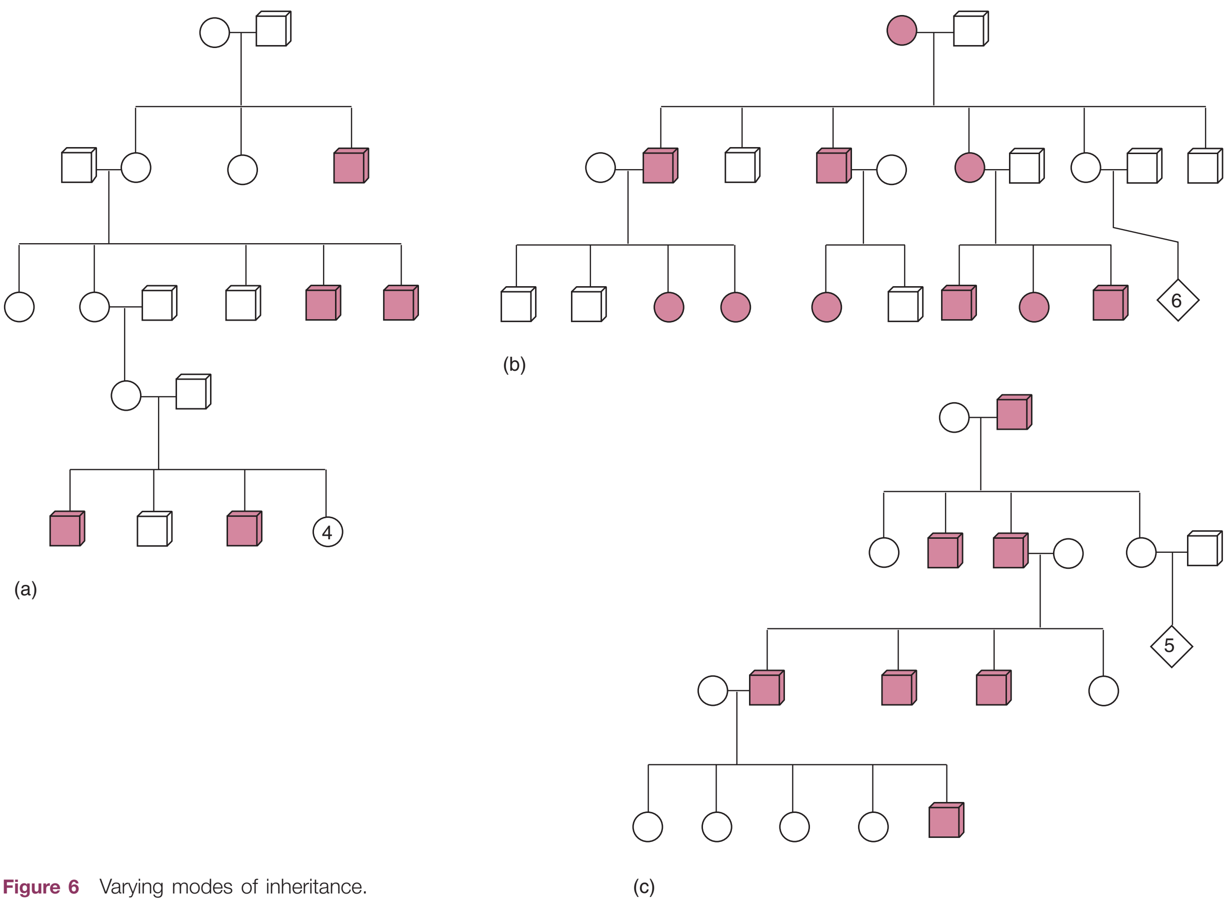


Figure 6 Varying modes of inheritance.

CRITICAL THINKING QUESTIONS

1. What effects do null alleles, alleles that produce no protein product, have in electrophoretic systems? How could you tell if a null allele were present?

2. In 1918, the Bolsheviks killed Tsar Nicholas II of Russia and his family (fig. 5.22). However, the remains of one daughter, Princess Anastasia, were never recovered. At one point, a woman appeared who claimed to be Anastasia. How could you validate her claim genetically?

Suggested Readings for chapter 5 are on page B-2.

6

LINKAGE AND MAPPING IN EUKARYOTES

STUDY OBJECTIVES

1. To learn about analytical techniques for locating the relative positions of genes on chromosomes in diploid eukaryotic organisms 110
2. To learn about analytical techniques for locating the relative positions of genes on chromosomes in ascomycete fungi 122
3. To learn about analytical techniques for locating the relative positions of genes on human chromosomes 132

STUDY OUTLINE

Scanning electron micrograph (false color) of a fruit fly, *Drosophila melanogaster*.
(© Dr. Jeremy Burgess/SPL/Photo Researchers.)

After Sutton suggested the chromosomal theory of inheritance in 1903, evidence accumulated that genes were located on chromosomes. For example, Morgan showed by an analysis of inheritance patterns that the white-eye locus in Drosophila is located on the X chromosome. Given that any organism has many more genes than chromosomes, it follows that each chromosome has many loci. Since chromosomes in eukaryotes are linear, it also follows that genes are arranged in a linear fashion on chromosomes, like beads on a string. Sturtevant first demonstrated this in 1913. In this chapter, we look at analytical techniques for mapping chromosomes—techniques for determining the relationship between different genes on the same chromosome. These techniques are powerful tools that allow us to find out about the physical relationships of genes on chromosomes without ever having to see a gene or a chromosome. We determine that genes are on the same chromosome when the genes fail to undergo independent assortment, and then we use recombination frequencies to determine the distance between genes.

If loci were locked together permanently on a chromosome, allelic combinations would always be the same. However, at meiosis, crossing over allows the alleles of associated loci to show some measure of independence. A geneticist can use crossing over between loci to determine how close one locus actually is to another on a chromosome and thus to map an entire chromosome and eventually the entire **genome** (genetic complement) of an organism.

Loci carried on the same chromosome are said to be linked to each other. There are as many **linkage groups** (l) as there are autosomes in the haploid set plus sex chromosomes. *Drosophila* has five linkage groups ($2n = 8$; $l = 3$ autosomes + X + Y), whereas human beings have twenty-four linkage groups ($2n = 46$; $l = 22$ autosomes + X + Y). Prokaryotes and viruses, which usually have a single chromosome, are discussed in chapter 7.

Historically, classical mapping techniques, as described in this chapter and the next, gave researchers their only tools to determine the relationships of particular genes and their chromosomes. When geneticists know the locations of specific genes, they can study them in relation to each other and begin to develop a comprehensive catalogue of the genome of an organism. Knowing the location of a gene also helps in isolating the gene and studying its function and structure. And mapping the genes of different types of organisms (diploid, haploid, eukaryotic, prokaryotic) gives geneticists insight into genetic processes. More recently, recombinant DNA technology has allowed researchers to sequence whole genomes, including the human and fruit fly genomes; this means they now know the exact locations of all the genes on all the chromosomes of these organisms (see chapter 13). Geneticists are now creating massive databases containing this information, much of it available for free or by subscription on the World Wide Web. Until investigators mine all this information for all organisms of interest, they will still use analytical techniques in the laboratory and field to locate genes on chromosomes.

DIPLOID MAPPING

Two-Point Cross

In *Drosophila,* the recessive band gene (*bn*) causes a dark transverse band on the thorax, and the detached gene (*det*) causes the crossveins of the wings to be either detached or absent (fig. 6.1). A banded fly was crossed with a detached fly to produce wild-type, dihybrid offspring in the F_1 generation. F_1 females were then testcrossed to banded, detached males (fig. 6.2). (There is no crossing over in male fruit flies; in experiments designed to detect linkage, heterozygous females—in which crossing over occurs—are usually crossed with homozygous recessive males.) If the loci were assorting independently, we would expect a 1:1:1:1 ratio of the four possible phenotypes. However, of the first one thousand offspring examined, experimenters recorded a ratio of 2:483: 512:3.

Several points emerge from the data in figure 6.2. First, no simple ratio is apparent. If we divide by two, we get a ratio of 1:241:256:1.5. Although the first and last categories seem about equal, as do the middle two, no simple numerical relation seems to exist between the middle and end categories. Second, the two cate-

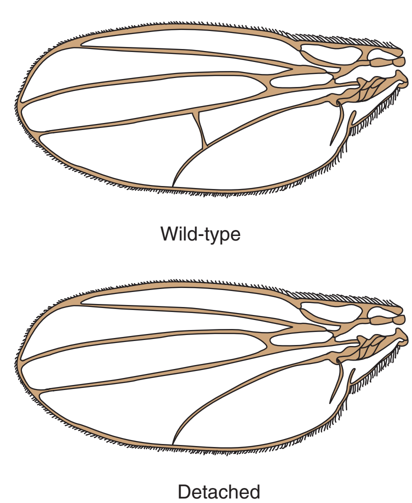


Figure 6.1 Wild-type (det^+) and detached (*det*) crossveins in *Drosophila*.

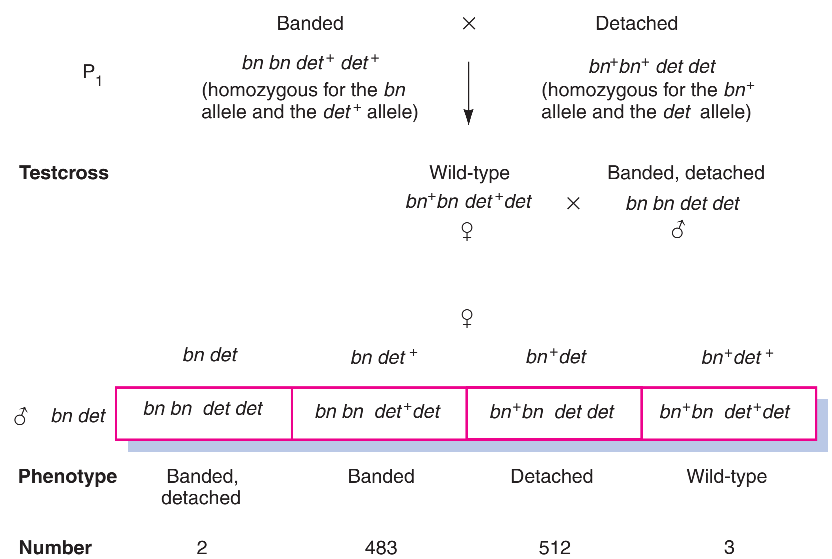


Figure 6.2 Testcrossing a dihybrid *Drosophila.*

gories in very high frequency have the same phenotypes as the original parents in the cross (P_1 of fig. 6.2). That is, banded flies and detached flies were the original parents as well as the great majority of the testcross offspring. We call these phenotypic categories **parentals,** or **nonrecombinants.** On the other hand, the testcross offspring in low frequency combine the phenotypes of the two original parents (P_1). These two categories are referred to as **nonparentals,** or **recombinants.** The simplest explanation for these results is that the banded and detached loci are located near each other on the same chromosome (they are a linkage group), and therefore they move together as associated alleles during meiosis.

We can analyze the original cross by drawing the loci as points on a chromosome (fig. 6.3). This shows that 99.5% of the testcross offspring (the nonrecombinants) come about through the simple linkage of the two loci. The remaining 0.5% (the recombinants) must have arisen through a crossover of homologues, from a chiasma at meiosis, between the two loci (fig. 6.4). Note that since it is not possible to tell from these crosses which chromosome the loci are actually on or where the centromere is in relation to the loci, the centromeres are not included in the figures. The crossover event is viewed as a breakage and reunion of two chromatids lying adjacent to each other during prophase I of meiosis. Later in this chapter, we find cytological proof for this; in chapter 12, we explore the molecular mechanisms of this breakage and reunion process.

From the testcross in figure 6.3, we see that 99.5% of the gametes produced by the dihybrid are nonrecombinant, whereas only 0.5% are recombinant. This very small frequency of recombinant offspring indicates that the two loci lie very close to each other on their particular chromosome. In fact, we can use the recombination percentages of gametes, and therefore of testcross offspring, as estimates of distance between loci on a chromosome: 1% recombinant offspring is referred to as one **map unit** (or one **centimorgan,** in honor of geneticist T. H. Morgan, the first geneticist to win the Nobel Prize; box 6.1). Although a map unit is not a physical distance, it is a relative distance that makes it possible to know the order of and relative separation between loci on a chromosome. In this case, the two loci are 0.5 map units apart. (From sequencing various chromosomal segments—see chapter 13—we have learned that the relationship between centimorgans and DNA base pairs is highly variable, depending on species, sex, and region of the chromosome. For example, in human beings, 1 centimorgan can vary between 100,000 and 10,000,000 base pairs. In the fission yeast, *Schizosaccharomyces pombe,* 1 centimorgan is only about 6,000 base pairs.)

The arrangement of the *bn* and *det* alleles in the dihybrid of figure 6.3 is termed the **trans** configuration, meaning "across," because the two mutants are across from each other, as are the two wild-type alleles. The alternative arrangement, in which one chromosome carries both mutants and the other chromosome carries both wild-type alleles (fig. 6.5), is referred to as the **cis** configuration. (Two other terms, **repulsion** and **coupling,** have the same meanings as *trans* and *cis,* respectively.)

A cross involving two loci is usually referred to as a **two-point cross;** it gives us a powerful tool for dissecting the makeup of a chromosome. The next step in our